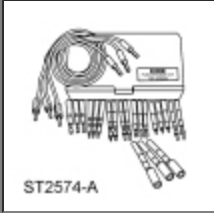


Airbag and Safety Belt Pretensioner Supplemental Restraint System (SRS)

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)	
	Flex Probe Kit NUD105-R025D or equivalent
	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

The Restraints Control Module (RCM) continuously receives/monitors all inputs from the following Supplemental Restraint System (SRS) components.

- LH and RH front impact severity sensors
- Driver and passenger front door side impact sensors
- Driver and passenger second row side impact sensors
- Safety belt buckle switches (driver and passenger)
- Driver seat track position sensor
- Occupant Classification System Module (OCSM)

If the RCM detects sudden vehicle deceleration and/or lateral deceleration based on all the information received from all sensors and switches and determines that deployment is required, the RCM sends voltage and current to deploy the appropriate SRS components.

The RCM contains an internal sensor that detects vehicle rollover angles. If the RCM determines that a rollover condition exists and deployment is required, the RCM deploys the applicable SRS components.

During a frontal or side crash, the RCM can deploy the following SRS components, based on crash severity and sensor input.

- Safety belt retractor pretensioner(s)
- Seat side air bag(s)
- Safety Canopy®
- Driver and passenger dual-stage air bag(s) (one or both stages)

The fact that the safety belt pretensioners, Safety Canopy® or air bags did not activate for both front seat occupants in a crash does not indicate that something is wrong with the system.

The **RCM** carries out a self-test of the **SRS** during each startup. If a **SRS** fault exists, the air bag warning indicator illuminates and remains illuminated for the rest of the ignition cycle. In addition to the self-test at start up, the **RCM** continuously monitors all of its **SRS** components and circuitry for correct operation.

Air Bag Warning Indicator

The air bag warning indicator:

- is located in the Instrument Panel Cluster (IPC) and proves out by lighting for 6 seconds and then turn off.
- flashes and/or illuminates based on the message the **IPC** receives from the **RCM** .
- illuminates if the **IPC** does not receive a message from the **RCM** .

Air Bag Module Second Stage Deployment Check

Because the driver and passenger front air bags each have 2 deployment stages, it is possible that Stage 1 has deployed and Stage 2 has not.

If a front air bag module has deployed, it is **mandatory** that the front air bag module be remotely deployed using the appropriate air bag disposal procedure.

- For information on driver air bag module and/or passenger air bag module remote deployment, refer to Pyrotechnic Device Disposal in this section.

Clockspring

The clockspring allows for continuous electrical connections between the driver air bag module and the **RCM** when the steering wheel is turned.

Driver Air Bag Module

The driver air bag module is a dual-stage air bag, upon receiving a flow of current from the **RCM** , deploying at 1 of 2 different rates depending upon vehicle impact severity and impact sensor input.

High Speed Controller Area Network (HS-CAN)

This vehicle utilizes a communication system called a High Speed Controller Area Network (HS-CAN) . The **RCM** communicates with various modules for required information about the vehicle. Refer to Section 418-00 for information about the **HS-CAN** .

Occupant Classification System (OCS)

The Occupant Classification System (OCS) is found only on the front passenger seat. The **OCS** classifies the size of the front passenger seat occupant and provides this information to the **RCM** . The **RCM** uses this information (and other inputs) in determining the deployment strategy of the passenger air bag and pretensioner. The **OCS** has 2 weight sensor bolts that are mounted to the inboard rail of the seat track and are only serviced with the seat track assembly. The **OCSM** is mounted to the electrical bracket underneath the seat cushion pan and can be serviced separately from the seat track assembly.

The **OCS** is also used for operation of the passenger Belt-Minder®. For information on the passenger Belt-Minder® feature, refer to Section 501-20A. To deactivate or reactivate the passenger Belt-Minder® feature, refer to Section 413-01 or the Owner's Literature.

Belt Tension Sensor (BTS)

The Belt Tension Sensor (BTS) :

- is a 3-wire Hall-effect sensor located at the safety belt anchor point and is part of the front passenger safety belt and retractor assembly.

- is used in conjunction with the **QCS** .

The **BTS** is used by the **QCS** to identify the presence of a child safety seat on the front passenger seat. The **BTS** senses the tension on the safety belt assembly then provides an output to the **QCSM** , indicating that the safety belt assembly is cinched. After sensing the weight applied to the seat by the occupant and using the **BTS** input, the **QCSM** determines how the occupant should be classified and communicates this information to the **RCM** . If the occupant is classified as a child, the **RCM** automatically deactivates the passenger air bag module and illuminates the Passenger Air Bag Deactivation (PAD) indicator.

Passenger Air Bag Deactivation (PAD) Indicator — Vehicles without 4.2" LCD IPC display

The **PAD** indicator is a visual indicator used to inform the front seat occupants of the front passenger air bag state. The **PAD** indicator is located in the instrument panel as part of the hazard/traction control switch and is visible to each front seat occupant.

For vehicles without a 4.2" **LCD IPC** display, the **RCM** controls the state of the **PAD** indicator through a direct hard-wired connection. The **PAD** indicator output is based on information provided by the **QCSM** and illuminates to indicate the front passenger air bag module is disabled. An exception to this is when the front passenger seat is determined to be empty, in which case it is not necessary to indicate that the passenger airbag is disabled. In all other cases, the **PAD** indicator is unlit when the passenger air bag module is enabled.

The **RCM** briefly activates the **PAD** indicator to prove-out the indicator function and verify to the front occupants the **PAD** indicator is functioning correctly

The following table indicates the passenger air bag status and the **PAD** indicator status based the size of the front passenger occupant.

Passenger Air Bag and Passenger Air Bag Deactivation (PAD) Indicator Status — Vehicles without 4.2" LCD IPC display

Occupant Size	Passenger Safety Belt Buckle Status	Passenger Air Bag Status	PAD Indicator Status
Empty	Unbuckled	Disabled	Unlit
Empty	Buckled	Disabled	Lit
Small	Buckled/ Unbuckled	Disabled	Lit
Large	Buckled/ Unbuckled	Enabled	Unlit

Passenger Air Bag Deactivation (PAD) Indicator — Vehicles with 4.2" LCD IPC display

The **PAD** indicator is a visual indicator used to inform the front seat occupants of the front passenger air bag state. The **PAD** indicator is located in the instrument panel as part of the **FCIM** and is visible to each front seat occupant.

For vehicles with a 4.2" **LCD IPC** display, the **RCM** controls the state of the **PAD** indicator via the **HS-CAN** and **LCAN** . The **PAD** indicator output is based on information provided by the **QCSM** . When the passenger airbag is activated, the **PAD** indicator displays ON. If the **RCM** has deactivated the passenger airbag, the **PAD** indicator displays OFF.

The **RCM** briefly activates the **PAD** indicator to prove-out the indicator function and verify to the front occupants the **PAD** indicator is functioning correctly.

The following table indicates the passenger air bag status and the **PAD** indicator status based the size of the front passenger occupant.

Passenger Air Bag and Passenger Air Bag Deactivation (PAD) Indicator Status — Vehicles with 4.2" LCD IPC display

Occupant Size	Passenger Safety Belt Buckle Status	Passenger Air Bag Status	PAD Indicator Status
Empty	Buckled/ Unbuckled	Disabled	OFF: Lit
			ON: Unlit
Small	Buckled/ Unbuckled	Disabled	OFF: Lit
			ON: Unlit
Large	Buckled/ Unbuckled	Enabled	OFF: Unlit
			ON: Lit

Passenger Air Bag Module

The passenger air bag module is a dual-stage air bag, deploying at 1 of 2 different rates depending upon vehicle impact severity and sensor input.

Restraints Control Module (RCM)

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

WARNING: If a vehicle has been in a crash, inspect the restraints control module (RCM) and the impact sensor (if equipped) mounting areas for deformation. If damaged, restore the mounting areas to the original production configuration. A new RCM and sensors must be installed whether or not the airbags have deployed. Failure to follow these instructions may result in serious personal injury or death in a crash.

NOTE: This vehicle is equipped with the SYNC® feature which contains the 911 assist option. Refer to the Owner's Literature.

The RCM carries out the following functions:

- Deploys the air bag(s)/Safety Canopy® in the event of a deployable crash
- Activates the driver and passenger retractor pretensioner(s) to control the tension of the driver and passenger safety belts in the event of a deployable crash
- Monitors the SRS for faults
- Sends a message to the IPC to illuminate the air bag warning indicator if a fault is detected
- Communicates through the Data Link Connector (DLC) the current or historical DTCs

The RCM monitors the SRS for possible faults. If a fault is detected, the RCM requests the IPC to illuminate the air bag warning indicator. When the ignition is turned OFF and then ON, the IPC proves out the air bag warning indicator by lighting for 6 seconds. If no faults are detected by the RCM, the IPC turns the air bag warning indicator off and it remains off. If the RCM detects a fault, it sends a message to the IPC to turn the air bag warning indicator on and it remains on for the rest of the ignition cycle. If the RCM requests illumination of the air bag warning indicator and the air bag warning indicator does not function, the IPC automatically activates an audible chime. The chime is a series of 5 sets of 5 tone bursts. If the chime is heard, the SRS and the air bag warning indicator require repair.

The RCM includes a backup power supply. This feature provides sufficient backup power to deploy the air bags in the event that the ignition circuit is lost or damaged during impact. The backup power supply depletes its stored energy approximately one minute after power and/or ground is removed from the RCM.

Post Crash Alert Function

The post crash alert is a function controlled by the Body Control Module (BCM) . If the RCM determines that an impact of a pre-determined severity (the air bags may or may not be deployed) has occurred, the RCM sends a message to the BCM over the High Speed Controller Area Network (HS-CAN) and the post crash alert function is activated. The post-crash alert function activates the hazard lamp function and cycles the horn on 3 times, then off for 4 seconds, and repeats this cycle until the battery is discharged or the function is turned off. The post crash alert function can be turned off by:

- pressing the hazard flasher lamp switch (which may need to be pressed more than once).
- pressing the Remote Keyless Entry (RKE) transmitter UNLOCK button.
- pressing the RKE transmitter PANIC button.

Safety Belt Buckle Switches

The safety belt buckle switches indicate to the RCM whether the safety belts are buckled or unbuckled.

The RCM also communicates the driver safety belt buckle switch status to the IPC , which monitors the information to control the safety belt warning indicator. For information on the safety belt warning indicator, refer to Section 413-01.

Safety Belt Retractor Pretensioner

Safety belt retractor pretensioners control the tension of the driver and passenger safety belts in the event of a deployable collision.

Safety Canopy® Module

WARNING: Anytime the Safety Canopy® or side air curtain module has deployed, a new headliner and new A-, B- and C-pillar upper trim panels and attaching hardware must be installed. Remove any other damaged components and hardware and install new components and hardware as needed. Failure to follow these instructions may result in the Safety Canopy® or side air curtain module deploying incorrectly and increases the risk of serious personal injury or death in a crash.

NOTE: The Safety Canopy® modules are NOT interchangeable. If installing a new Safety Canopy® module, the correct Safety Canopy® module MUST be installed for the vehicle application.

The Safety Canopy® module:

- is mounted above the headliner.
- attaches from the A-pillar frame to the B-pillar or C-pillar frame.

Seat Track Position Sensor

The seat track position sensor indicates the position of the driver seat along the seat track to the RCM . The sensor detects the presence of a shunt bracket on the track, indicating that the seat has moved forward beyond a certain point of adjustment. The RCM uses seat track position sensor information to determine the deployment rate of the dual-stage driver air bag module.

Secondary Air Bag Warning (Chime)

The secondary air bag warning chime is an audible chime located in and controlled by the IPC . If a fault is detected with the air bag warning indicator, a DTC is stored in memory of the IPC . Upon receiving the message from the RCM that a SRS fault has been detected, the IPC sounds the secondary air bag warning chime in a pattern of 5 sets of 5 beeps.

Impact Sensor

For this vehicle line, the SRS uses up to 6 satellite sensors in addition to the RCM . The RCM is mounted to the center tunnel beneath the console (if equipped) or front center seat. All vehicles have 2 front impact severity sensors located in the front of the vehicle, mounted on the radiator support. The first row impact sensors (side restraints sensor 1) are mounted in each of the front doors behind the trim panel, the second row impact sensors (side restraints sensor 2, Crew Cab and SuperCab) are located on each C-pillar. The driver/passenger front door side impact sensor are pressure sensors and are **not**

interchangeable with other sensors on the vehicle. Mounting orientation is critical for correct operation of all impact sensors.

Seat Side Air Bag Module

The side air bag module:

- deploys upon receiving a flow of current from the **RCM** , initiated by the side restraints sensor and internal **RCM** circuitry.
- is mounted in the seat backrest.
- is used in conjunction with a Safety Canopy® module.

Inspection and Verification

1. Verify the customer concern by checking the air bag warning indicator in the Instrument Panel Cluster (IPC) . Refer to Prove Out the System in this section.
2. Visually inspect for obvious signs of mechanical or electrical damage. Do not disconnect electrical connectors until directed to do so within the pinpoint test.

Visual Inspection Chart

Mechanical	Electrical
• Damaged Restraints Control Module (RCM) or loose mounting • Damaged LH or RH front impact severity sensor(s) • Damaged side impact sensor front door and/or C-pillar • Damaged or disconnected Passenger Air Bag Deactivation (PAD) indicator • Damaged front driver/passenger seat assembly	• Open Body Control Module (BCM) fuse(s): ▪ 36 (10A) ▪ 41 (7.5A) • Open Battery Junction Box (BJB) fuse: ▪ 27 (20A) • Damaged wiring harness • Loose, damaged or corroded connectors • Circuitry open/shorted • Damaged shorting bars • Loose, damaged or pinched passenger seat wire harness

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

4. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

5. **NOTE:** *The Vehicle Communication Module (VCM) LED prove out confirms power and ground from the **DLC** are provided to the **VCM** .*

If the scan tool does not communicate with the **VCM** :

- check the **VCM** connection to the vehicle.
- check the scan tool connection to the **VCM** .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

6. If the scan tool does not communicate with the vehicle:
 - verify the ignition is ON.

- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00 to diagnose no response from the PCM.

7. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, retrieve and record on-demand and Continuous Memory Diagnostic Trouble Codes (CMDTCs) from the RCM and Occupant Classification System Module (OCSM) .

8. If the DTCs retrieved are related to the concern, go to the DTC Charts. For all other DTCs, refer to Section 419-10.

9. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Charts

The DTCs can be retrieved from the RCM and the Occupant Classification System Module (OCSM) with a scan tool via the Data Link Connector (DLC) .

Restraints Control Module (RCM) DTC Chart

NOTE: Always make sure the correct Supplemental Restraint System (SRS) component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. Continuous Memory Diagnostic Trouble Codes (CMDTCs) have an additional 2-character DTC status code suffix to assist in determining DTC history.

DTC	Description	Action To Take
Continuous lamp	The Air Bag Warning Indicator is Illuminated Continuously	GO to Pinpoint Test A.
B0001:11	Driver Frontal Stage 1 Deployment Control: Circuit Short to Ground	GO to Pinpoint Test B.
B0001:12	Driver Frontal Stage 1 Deployment Control: Circuit Short to Battery	GO to Pinpoint Test B.
B0001:13	Driver Frontal Stage 1 Deployment Control: Circuit Open	GO to Pinpoint Test B.
B0001:1A	Driver Frontal Stage 1 Deployment Control: Circuit Resistance Below Threshold	GO to Pinpoint Test B.
B0001:2B	Driver Frontal Stage 1 Deployment Control: Signal Cross Coupled	GO to Pinpoint Test W.
B0001:4A	Driver Frontal Stage 1 Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
B0002:11	Driver Frontal Stage 2 Deployment Control: Circuit Short to Ground	GO to Pinpoint Test C.
B0002:12	Driver Frontal Stage 2 Deployment Control: Circuit Short to Battery	GO to Pinpoint Test C.
B0002:13	Driver Frontal Stage 2 Deployment Control: Circuit Open	GO to Pinpoint Test C.

DTC	Description	Action To Take
B0002:1A	Driver Frontal Stage 2 Deployment Control: Circuit Resistance Below Threshold	GO to Pinpoint Test C.
B0002:2B	Driver Frontal Stage 2 Deployment Control: Signal Cross Coupled	GO to Pinpoint Test W.
B0002:4A	Driver Frontal Stage 2 Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
B0004:11	Driver Knee Bolster Deployment Control: Circuit Short to Ground	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT Programmable Module Installation (PMI) . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0004:12	Driver Knee Bolster Deployment Control: Circuit Short to Battery	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0004:13	Driver Knee Bolster Deployment Control: Circuit Open	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0004:1A	Driver Knee Bolster Deployment Control: Circuit Resistance Below Threshold	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0004:2B	Driver Knee Bolster Deployment Control: Signal Cross Coupled	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0004:4A	Driver Knee Bolster Deployment Control: Incorrect Component Installed	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0005:11	Collapsible Steering Column Deployment Control: Circuit Short	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is</i>

DTC	Description	Action To Take
	to Ground	<i>installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0005:12	Collapsible Steering Column Deployment Control: Circuit Short to Battery	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0005:13	Collapsible Steering Column Deployment Control: Circuit Open	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0005:1A	Collapsible Steering Column Deployment Control: Circuit Resistance Below Threshold	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0005:2B	Collapsible Steering Column Deployment Control: Signal Cross Coupled	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0005:4A	Collapsible Steering Column Deployment Control: Incorrect Component Installed	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0010:11	Passenger Frontal Stage 1 Deployment Control: Circuit Short to Ground	GO to Pinpoint Test D.
B0010:12	Passenger Frontal Stage 1 Deployment Control: Circuit Short to Battery	GO to Pinpoint Test D.
B0010:13	Passenger Frontal Stage 1 Deployment Control: Circuit Open	GO to Pinpoint Test D.
B0010:1A	Passenger Frontal Stage 1 Deployment Control: Circuit Resistance Below Threshold	GO to Pinpoint Test D.
B0010:2B	Passenger Frontal Stage 1 Deployment Control: Signal Cross	GO to Pinpoint Test W.

DTC	Description	Action To Take
	Coupled	
B0010:4A	Passenger Frontal Stage 1 Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
B0011:11	Passenger Frontal Stage 2 Deployment Control: Circuit Short to Ground	GO to Pinpoint Test E.
B0011:12	Passenger Frontal Stage 2 Deployment Control: Circuit Short to Battery	GO to Pinpoint Test E.
B0011:13	Passenger Frontal Stage 2 Deployment Control: Circuit Open	GO to Pinpoint Test E.
B0011:1A	Passenger Frontal Stage 2 Deployment Control: Circuit Resistance Below Threshold	GO to Pinpoint Test E.
B0011:2B	Passenger Frontal Stage 2 Deployment Control: Signal Cross Coupled	GO to Pinpoint Test W.
B0011:4A	Passenger Frontal Stage 2 Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
B0020:11	Left Side Airbag Deployment Control: Circuit Short to Ground	GO to Pinpoint Test F.
B0020:12	Left Side Airbag Deployment Control: Circuit Short to Battery	GO to Pinpoint Test F.
B0020:13	Left Side Airbag Deployment Control: Circuit Open	GO to Pinpoint Test F.
B0020:1A	Left Side Airbag Deployment Control 1: Circuit Resistance Below Threshold	GO to Pinpoint Test F.
B0020:2B	Left Side Airbag Deployment Control: Signal Cross Coupled	GO to Pinpoint Test W.
B0020:4A	Left Side Airbag Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
B0021:11	Left Curtain Deployment Control 1: Circuit Short to Ground	GO to Pinpoint Test G.
B0021:12	Left Curtain Deployment Control 1: Circuit Short to Battery	GO to Pinpoint Test G.
B0021:13	Left Curtain Deployment Control 1: Circuit Open	GO to Pinpoint Test G.
B0021:1A	Left Curtain Deployment Control 1: Circuit Resistance Below Threshold	GO to Pinpoint Test G.
B0021:2B	Left Curtain Deployment Control 1: Signal Cross Coupled	GO to Pinpoint Test W.
B0021:4A	Left Curtain Deployment Control 1: Incorrect Component Installed	GO to Pinpoint Test X.

DTC	Description	Action To Take
B0028:11	Right Side Airbag Deployment Control: Circuit Short to Ground	GO to Pinpoint Test H.
B0028:12	Right Side Airbag Deployment Control: Circuit Short to Battery	GO to Pinpoint Test H.
B0028:13	Right Side Airbag Deployment Control: Circuit Open	GO to Pinpoint Test H.
B0028:1A	Right Side Airbag Deployment Control 1: Circuit Resistance Below Threshold	GO to Pinpoint Test H.
B0028:2B	Right Side Airbag Deployment Control: Signal Cross Coupled	GO to Pinpoint Test W.
B0028:4A	Right Side Airbag Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
B0029:11	Right Curtain Deployment Control 1: Circuit Short to Ground	GO to Pinpoint Test I.
B0029:12	Right Curtain Deployment Control 1: Circuit Short to Battery	GO to Pinpoint Test I.
B0029:13	Right Curtain Deployment Control 1: Circuit Open	GO to Pinpoint Test I.
B0029:1A	Right Curtain Deployment Control 1: Circuit Resistance Below Threshold	GO to Pinpoint Test I.
B0029:2B	Right Curtain Deployment Control 1: Signal Cross Coupled	GO to Pinpoint Test W.
B0029:4A	Right Curtain Deployment Control 1: Incorrect Component Installed	GO to Pinpoint Test X.
B0050:11	Driver Seatbelt Sensor: Circuit Short to Ground	GO to Pinpoint Test J.
B0050:12	Driver Seatbelt Sensor: Circuit Short to Battery	GO to Pinpoint Test J.
B0050:13	Driver Seatbelt Sensor: Circuit Open	GO to Pinpoint Test J.
B0050:1D	Driver Seatbelt Sensor: Circuit Current Out of Range	GO to Pinpoint Test J.
B0050:2B	Driver Seatbelt Sensor: Signal Cross Coupled	GO to Pinpoint Test W.
B0050:4A	Driver Seatbelt Sensor: Incorrect Component Installed	GO to Pinpoint Test X.
B0052:11	Passenger Seatbelt Sensor: Circuit Short to Ground	GO to Pinpoint Test K.
B0052:12	Passenger Seatbelt Sensor: Circuit Short to Battery	GO to Pinpoint Test K.
B0052:13	Passenger Seatbelt Sensor: Circuit Open	GO to Pinpoint Test K.
B0052:1D	Passenger Seatbelt Sensor: Circuit Current Out of Range	GO to Pinpoint Test K.

DTC	Description	Action To Take
B0052:2B	Passenger Seatbelt Sensor: Signal Cross Coupled	GO to Pinpoint Test W.
B0052:4A	Passenger Seatbelt Sensor: Incorrect Component Installed	GO to Pinpoint Test X.
B0080:11	Driver Seatbelt Load Limiter Deployment Control: Circuit Short to Ground	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0080:12	Driver Seatbelt Load Limiter Deployment Control: Circuit Short to Battery	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0080:13	Driver Seatbelt Load Limiter Deployment Control: Circuit Open	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0080:1A	Driver Seatbelt Load Limiter Deployment Control: Circuit Resistance Below Threshold	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT Programmable Module Installation (PMI) . Use As-Built data when CARRYING OUT Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0080:2B	Driver Seatbelt Load Limiter Deployment Control: Signal Cross Coupled	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0080:4A	Driver Seatbelt Load Limiter Deployment Control: Incorrect Component Installed	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0082:11	Passenger Seatbelt Load Limiter Deployment Control: Circuit Short to Ground	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.

DTC	Description	Action To Take
B0082:12	Passenger Seatbelt Load Limiter Deployment Control: Circuit Short to Battery	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0082:13	Passenger Seatbelt Load Limiter Deployment Control: Circuit Open	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0082:1A	Passenger Seatbelt Load Limiter Deployment Control: Circuit Resistance Below Threshold	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0082:2B	Passenger Seatbelt Load Limiter Deployment Control: Signal Cross Coupled	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0082:4A	Passenger Seatbelt Load Limiter Deployment Control: Incorrect Component Installed	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B0090:11	Left Frontal Restraints Sensor: Circuit Short to Ground	GO to Pinpoint Test L.
B0090:4A	Left Frontal Restraints Sensor: Incorrect Component Installed	GO to Pinpoint Test X.
B0090:81	Left Frontal Restraints Sensor: Invalid Serial Data Received	GO to Pinpoint Test L.
B0090:93	Left Frontal Restraints Sensor: No Operation	GO to Pinpoint Test L.
B0090:96	Left Frontal Restraints Sensor: Component Internal Failure	GO to Pinpoint Test L.
B0091:11	Left Side Restraints Sensor 1: Circuit Short to Ground	GO to Pinpoint Test M.
B0091:4A	Left Side Restraints Sensor 1: Incorrect Component Installed	GO to Pinpoint Test X.
B0091:81	Left Side Restraints Sensor 1: Invalid Serial Data Received	GO to Pinpoint Test M.

DTC	Description	Action To Take
B0091:93	Left Side Restraints Sensor 1: No Operation	GO to Pinpoint Test M.
B0091:96	Left Side Restraints Sensor 1: Component Internal Failure	GO to Pinpoint Test M.
B0092:11	Left Side Restraints Sensor 2: Circuit Short to Ground	GO to Pinpoint Test N.
B0092:4A	Left Side Restraints Sensor 2: Incorrect Component Installed	GO to Pinpoint Test X.
B0092:81	Left Side Restraints Sensor 2: Invalid Serial Data Received	GO to Pinpoint Test N.
B0092:93	Left Side Restraints Sensor 2: No Operation	GO to Pinpoint Test N.
B0092:96	Left Side Restraints Sensor 2: Component Internal Failure	GO to Pinpoint Test N.
B0095:11	Right Frontal Restraints Sensor: Circuit Short to Ground	GO to Pinpoint Test O.
B0095:4A	Right Frontal Restraints Sensor: Incorrect Component Installed	GO to Pinpoint Test X.
B0095:81	Right Frontal Restraints Sensor: Invalid Serial Data Received	GO to Pinpoint Test O.
B0095:93	Right Frontal Restraints Sensor: No Operation	GO to Pinpoint Test O.
B0095:96	Right Frontal Restraints Sensor: Component Internal Failure	GO to Pinpoint Test O.
B0096:11	Right Side Restraints Sensor 1: Circuit Short to Ground	GO to Pinpoint Test N.
B0096:4A	Right Side Restraints Sensor 1: Incorrect Component Installed	GO to Pinpoint Test X.
B0096:81	Right Side Restraints Sensor 1: Invalid Serial Data Received	GO to Pinpoint Test N.
B0096:93	Right Side Restraints Sensor 1: No Operation	GO to Pinpoint Test N.
B0096:96	Right Side Restraints Sensor 1: Component Internal Failure	GO to Pinpoint Test N.
B0097:11	Right Side Restraints Sensor 2: Circuit Short to Ground	GO to Pinpoint Test M.
B0097:4A	Right Side Restraints Sensor 2: Incorrect Component Installed	GO to Pinpoint Test X.
B0097:81	Right Side Restraints Sensor 2: Invalid Serial Data Received	GO to Pinpoint Test M.
B0097:93	Right Side Restraints Sensor 2: No Operation	GO to Pinpoint Test M.
B0097:96	Right Side Restraints Sensor 2: Component Internal Failure	GO to Pinpoint Test M.
B00A0:09	Occupant Classification System: Component Failure	GO to Pinpoint Test R.

DTC	Description	Action To Take
B00A0:4A	Occupant Classification System: Incorrect Component Installed	GO to Pinpoint Test R.
B00A0:63	Occupant Classification System: Circuit / Component Protection Time-Out	GO to Pinpoint Test R.
B00A0:64	Occupant Classification System: Signal Plausibility Failure	GO to Pinpoint Test R.
B00A0:68	Occupant Classification System: Event Information	GO to Pinpoint Test R.
B00B5:11	Driver Seat Track Position Restraints Sensor: Circuit Short to Ground	GO to Pinpoint Test S.
B00B5:12	Driver Seat Track Position Restraints Sensor: Circuit Short to Battery	GO to Pinpoint Test S.
B00B5:13	Driver Seat Track Position Restraints Sensor: Circuit Open	GO to Pinpoint Test S.
B00B5:1D	Driver Seat Track Position Restraints Sensor: Circuit Current Out of Range	GO to Pinpoint Test S.
B00B5:2B	Driver Seat Track Position Restraints Sensor: Signal Cross Coupled	GO to Pinpoint Test W.
B00B5:4A	Driver Seat Track Position Restraints Sensor: Incorrect Component Installed	GO to Pinpoint Test X.
B00D5:01	Restraint System Passenger Disable Indicator: General Electrical Failure	CONFIGURE the RCM using As-Built data. CLEAR all RCM DTCs and REPEAT the self-test. If DTC B00D5:01 returns, the RCM has received a fault message from the FCIM indicating a fault in the PAD indicator. CARRY OUT a self-test of the FCIM and REPAIR any DTCs. Do not replace the RCM for this DTC.
B00D5:11	Restraint System Passenger Disable Indicator: Circuit Short to Ground	GO to Pinpoint Test T.
B00D5:12	Restraint System Passenger Disable Indicator: Circuit Short to Battery	GO to Pinpoint Test T.
B00D5:13	Restraint System Passenger Disable Indicator: Circuit Open	GO to Pinpoint Test T.
B00D5:4A	Restraint System Passenger Disable Indicator: Incorrect Component Installed	GO to Pinpoint Test X.
B00DF:13	Passenger Restraints Disable Switch: Incorrect Component Installed	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER

DTC	Description	Action To Take
		to Restraints Control Module (RCM) in this section.
B1193:00	Crash Event Storage Full and Locked: No Sub Type Information	This DTC indicates the vehicle has been involved in a collision. INSTALL a new RCM and impact sensors. REFER to Inspection and Repair After a Supplemental Restraint System (SRS) Deployment in this section.
B11CD:11	Driver Frontal Airbag Canister Vent: Circuit Short to Ground	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B11CD:12	Driver Frontal Airbag Canister Vent: Circuit Short to Battery	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B11CD:13	Driver Frontal Airbag Canister Vent: Circuit Open	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B11CD:1A	Driver Frontal Airbag Canister Vent: Circuit Resistance Below Threshold	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B11CD:2B	Driver Frontal Airbag Canister Vent: Signal Cross Coupled	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B11CD:4A	Driver Frontal Airbag Canister Vent: Incorrect Component Installed	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B11D8:12	Restraint Event Notification: Circuit Short to Battery	GO to Pinpoint Test AI.
B11D8:14	Restraint Event Notification: Circuit Short to Ground or Open	GO to Pinpoint Test AI.
B1202:01	Restraint System Passenger Enable Indicator: General Electrical	The RCM has received a fault message from the ECIM indicating a fault in the PAD indicator. CARRY OUT a self-test of the ECIM and

DTC	Description	Action To Take
	Failure	REPAIR any DTCs. Do not replace the RCM for this DTC.
B1211:11	Driver Seatbelt Retractor Pretensioner Deployment Control: Circuit Short to Ground	GO to Pinpoint Test U.
B1211:12	Driver Seatbelt Retractor Pretensioner Deployment Control: Circuit Short to Battery	GO to Pinpoint Test U.
B1211:13	Driver Seatbelt Retractor Pretensioner Deployment Control: Circuit Open	GO to Pinpoint Test U.
B1211:1A	Driver Seatbelt Retractor Pretensioner Deployment Control: Circuit Resistance Below Threshold	GO to Pinpoint Test U.
B1211:2B	Driver Seatbelt Retractor Pretensioner Deployment Control: Signal Cross Coupled	GO to Pinpoint Test W.
B1211:4A	Driver Seatbelt Retractor Pretensioner Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
B1212:11	Driver Seatbelt Buckle Pretensioner Deployment Control: Circuit Short to Ground	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B1212:12	Driver Seatbelt Buckle Pretensioner Deployment Control: Circuit Short to Battery	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B1212:13	Driver Seatbelt Buckle Pretensioner Deployment Control: Circuit Open	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B1212:1A	Driver Seatbelt Buckle Pretensioner Deployment Control: Circuit Resistance Below Threshold	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B1212:2B	Driver Seatbelt Buckle Pretensioner Deployment Control: Signal Cross Coupled	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i>

DTC	Description	Action To Take
		INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B1212:4A	Driver Seatbelt Buckle Pretensioner Deployment Control: Incorrect Component Installed	NOTE: <i>This vehicle is not equipped with this component. This DTC indicates the RCM is incorrectly configured or the incorrect RCM is installed to the vehicle.</i> INSTALL a new RCM and CARRY OUT PMI . Use As-Built data when carrying out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
B1214:11	Passenger Seatbelt Retractor Pretensioner Deployment Control: Circuit Short to Ground	GO to Pinpoint Test V.
B1214:12	Passenger Seatbelt Retractor Pretensioner Deployment Control: Circuit Short to Battery	GO to Pinpoint Test V.
B1214:13	Passenger Seatbelt Retractor Pretensioner Deployment Control: Circuit Open	GO to Pinpoint Test V.
B1214:1A	Passenger Seatbelt Retractor Pretensioner Deployment Control: Circuit Resistance Below Threshold	GO to Pinpoint Test V.
B1214:2B	Passenger Seatbelt Retractor Pretensioner Deployment Control: Signal Cross Coupled	GO to Pinpoint Test W.
B1214:4A	Passenger Seatbelt Retractor Pretensioner Deployment Control: Incorrect Component Installed	GO to Pinpoint Test X.
U0028:08	Vehicle Communication Bus A: Bus Signal / Message Failure	GO to Pinpoint Test Y.
U0028:88	Vehicle Communication Bus A: Bus Off	GO to Pinpoint Test Y.
U0140:00	Lost Communication With Body Control Module: No Sub Type Information	NOTE: <i>The DTC sets in a module that is reporting a communication fault from another module on the data bus. The module that reports the fault is not the problem module. Do not install a new RCM as part of repair.</i> REFER to Section 418-00 to diagnose no communication with the Body Control Module (BCM) .
U0154:00	Lost Communication With Restraints Occupant Classification System Module: No Sub Type Information	NOTE: <i>The DTC sets in a module that is reporting a communication fault from another module on the data bus. The module that reports the fault is not the problem module. Do not install a new RCM as part of repair.</i> REFER to Section 418-00 to diagnose no communication with the Occupant Classification System Module (OCSM) .
U0155:00	Lost Communication With Instrument Panel Cluster (IPC) Control Module: No Sub Type Information	NOTE: <i>The DTC sets in a module that is reporting a communication fault from another module on the data bus. The module that reports the fault is not the problem module. Do not install a new RCM as part of repair.</i>

DTC	Description	Action To Take
		REFER to Section 418-00 to diagnose the communication concern.
U0253:00	Lost Communication With Accessory Protocol Interface Module: No Sub Type Information	NOTE: This DTC sets in the RCM if a message is not received from the Accessory Protocol Interface Module (APIM) . CHECK for additional RCM and APIM DTCs. If the APIM does not respond to the scan tool, REFER to Section 418-00 to diagnose the communication concern. REFER to the appropriate section to diagnose the DTCs. If no other DTCs are present, REFER to Section 418-00 to diagnose the APIM communication concern.
U0300:00	Internal Control Module Software Incompatibility: No Sub Type Information	CARRY OUT Programmable Module Installation (PMI) . RETRIEVE RCM DTCs. If RCM DTC U0300:00 is retrieved on-demand during self-test, INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section.
U0422:00	Invalid Data Received From Body Control Module: No Sub Type Information	CARRY OUT a self-test of the BCM . REFER to Section 419-10 to diagnose any DTCs that are retrieved.
U0554:00	Invalid Data Received From Accessory Protocol Interface Module: No Sub Type Information	CHECK for additional RCM and APIM DTCs. REFER to the appropriate section to diagnose the DTCs.
U2100:00	Initial Configuration Not Complete: No Sub Type Information	CARRY OUT PMI . RETRIEVE RCM DTCs. If RCM DTC U2100:00 is retrieved on-demand, INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section.
U2200:00	Control Module Configuration Memory Corrupt: No Sub Type Information	INSTALL a new RCM and CARRY OUT Programmable Module Installation (PMI) . Use As-Built data when CARRYING out Programmable Module Installation (PMI) for this DTC. REFER to Restraints Control Module (RCM) in this section.
U3000:46	Control Module: Calibration / Parameter Memory Failure	INSTALL a new RCM and CARRY OUT Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
U3000:49	Control Module: Internal Electronic Failure	INSTALL a new RCM and CARRY OUT Programmable Module Installation (PMI) . If DTC U2200:00 is also present, use As-Built data when CARRYING out Programmable Module Installation (PMI) . REFER to Restraints Control Module (RCM) in this section.
U3000:4A	Control Module: Incorrect Component Installed	GO to Pinpoint Test X.
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test Z.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test Z.

Occupant Classification System Module (OCSM) DTC Chart

NOTE: This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. Continuous Memory Diagnostic Trouble Codes (CMDTCs) have an additional 2-character DTC status code suffix to assist in determining DTC history.

DTC	Description	Action To Take
B0061:11	Passenger Seatbelt Tension Sensor: Circuit Short to Ground	GO to Pinpoint Test AA.
B0061:12	Passenger Seatbelt Tension Sensor: Circuit Short to Battery	GO to Pinpoint Test AA.
B0061:13	Passenger Seatbelt Tension Sensor: Circuit Open	GO to Pinpoint Test AA.
B0061:1C	Passenger Seatbelt Tension Sensor: Circuit Voltage Out of Range	GO to Pinpoint Test AA.
B00A0:54	Occupant Classification System: Missing Calibration	GO to Pinpoint Test AB.
B00A0:67	Occupant Classification System: Signal Incorrect After Event	CARRY OUT the Occupant Classification System (OCS) zero seat weight test. REFER to Occupant Classification Sensor (OCS) System Zero Seat Weight Test in this section.
B00A0:76	Occupant Classification System: Wrong Mounting Position	GO to Pinpoint Test AB.
B00A0:97	Occupant Classification System: Component or System Operation Obstructed or Blocked	GO to Pinpoint Test AC.
B00C2:12	Passenger Seat Occupant Classification Sensor "C": Circuit Short To Battery	GO to Pinpoint Test AD.
B00C2:13	Passenger Seat Occupant Classification Sensor "C": Circuit Open	GO to Pinpoint Test AD.
B00C2:1A	Passenger Seat Occupant Classification Sensor "C": Circuit Resistance Blow Threshold	GO to Pinpoint Test AD.
B00C2:49	Passenger Seat Occupant Classification Sensor "C": Internal Electronic Failure	GO to Pinpoint Test AB.
B00C2:81	Passenger Seat Occupant Classification Sensor "C": Invalid Serial Data	REPAIR all other Occupant Classification System Module (OCSM) DTCs first. If no other OCSM DTCs are present, INSTALL a new seat track (includes OCS weight sensor bolts). REFER to Section 501-10.
B00C3:12	Passenger Seat Occupant Classification Sensor "D": Circuit Short To Battery	GO to Pinpoint Test AE.
B00C3:13	Passenger Seat Occupant Classification Sensor "D": Circuit Open	GO to Pinpoint Test AE.
B00C3:1A	Passenger Seat Occupant Classification Sensor "D": Circuit Resistance Blow Threshold	GO to Pinpoint Test AE.
B00C3:49	Passenger Seat Occupant Classification Sensor "D":	GO to Pinpoint Test AB.

DTC	Description	Action To Take
	Internal Electronic Failure	
B00C3:81	Passenger Seat Occupant Classification Sensor "D": Invalid Serial Data	REPAIR all other Occupant Classification System Module (OCSM) DTCs first. If no other OCSM DTCs are present, INSTALL a new seat track (includes OCS weight sensor bolts). REFER to Section 501-10.
B1193:00	Crash Event Storage Full and Locked: No Sub Type Information	INSTALL new passenger seat track assembly with weight sensor bolts. REFER to Section 501-10. CLEAR all DTCs before carrying out the System Reset procedure. REFER to Inspection and Repair After a Supplemental Restraint System (SRS) Deployment in this section.
B1A59:01	Sensor 5 Volt Supply: General Electrical Failure	GO to Pinpoint Test AF.
B1A59:11	Sensor 5 Volt Supply: Circuit Short to Ground	GO to Pinpoint Test AF.
U0100:00	Lost Communication With ECM/PCM "A": No Sub Type Information	U0100:00 sets in a module that is reporting a communication fault from another module on the data bus. The module that reports the fault is not the problem module. Do not install a new OCSM as part of repair for DTC U0100:00. REFER to Section 418-00 to diagnose lost communication with the PCM.
U0140:00	Lost Communication With Body Control Module: No Sub Type Information	U0140:00 sets in a module that is reporting a communication fault from another module on the data bus. The module that reports the fault is not the problem module. Do not install a new OCSM as part of repair for DTC U0140:00. REFER to Section 418-00 to diagnose lost communication with the BCM .
U0151:00	Lost Communication With Restraints Control Module: No Sub Type Information	U0151:00 sets in a module that is reporting a communication fault from another module on the data bus. The module that reports the fault is not the problem module. Do not install a new OCSM as part of repair for DTC U0151:00. REFER to Section 418-00 to diagnose lost communication with the RCM .
U0401:00	Invalid Data Received From ECM/PCM "A": No Sub Type Information	U0401:00 sets in a module that is reporting a communication fault from another module on the data bus. The module that reports the fault is not the problem module. Do not install a new OCSM as part of repair for DTC U0401:00. CARRY OUT PCM Self Test. REPAIR any vehicle speed or communication-related DTCs. REFER to DTC Chart in Powertrain Control/Emissions Diagnosis (PC/ED) manual.
U0401:81	Invalid Data Received From ECM/PCM "A": Invalid Serial Data Received	This DTC sets when the OCSM is not receiving the vehicle speed data that it expects from the PCM. The OCSM is not the problem module. Do not install a new OCSM as part of repair for DTC U0401:81. CARRY OUT the PCM Self Test. REPAIR any DTCs. REFER to DTC Chart in Section 303-14 or the Powertrain Control/Emissions Diagnosis (PC/ED) manual.
U0422:00	Invalid Data Received From Body Control Module: No Sub Type Information	U0422:00 sets in a module that is reporting a communication fault from another module on the data bus. The OCSM is not the problem module. Do not install a new OCSM as part of repair for DTC U0422:00.

DTC	Description	Action To Take
		CARRY OUT Body Control Module (BCM) Self Test. REPAIR any ignition switch or communication-related DTCs. REFER to the DTC Chart in Section 419-10.
U0422:81	Invalid Data Received From Body Control Module: Invalid Serial Data Received	U0422:81 sets when the <u>OCSM</u> is not receiving the ignition status data that it expects from the <u>BCM</u>. The <u>OCSM</u> is not the problem module. Do not install a new <u>OCSM</u> as part of repair for DTC U0422:81. CARRY OUT Body Control Module (BCM) Self Test. REPAIR any ignition switch or communication-related DTCs. REFER to the DTC Chart in Section 419-10.
U0452:00	Invalid Data Received From Restraints Control Module: No Sub Type Information	U0452:00 sets in a module that is reporting a communication fault from another module on the data bus. The <u>OCSM</u> is not the problem module. Do not install a new <u>OCSM</u> as part of repair for DTC U0452:00. CARRY OUT <u>RCM</u> Self Test. REPAIR any <u>SRS</u> -related DTCs. REFER to the Restraints Control Module (RCM) DTC Chart in this section.
U0452:81	Invalid Data Received From Restraints Control Module: No Sub Type Information	U0452:00 sets when the <u>OCSM</u> is not receiving the <u>SRS</u> data that it expects from the <u>RCM</u>. The <u>OCSM</u> is not the problem module. Do not install a new <u>OCSM</u> as part of repair for DTC U0452:81. CARRY OUT <u>RCM</u> Self Test. REPAIR any <u>SRS</u> -related DTCs. REFER to the Restraints Control Module (RCM) DTC Chart in this section.
U200D:12	Control Module Output Power A: Circuit Short To Battery	GO to Pinpoint Test AG.
U200E:11	Control Module Output Power B: Circuit Short To Ground	GO to Pinpoint Test AH.
U200E:12	Control Module Output Power B: Circuit Short To Battery	GO to Pinpoint Test AH.
U200E:1A	Control Module Output Power B: Circuit Resistance Below Threshold	GO to Pinpoint Test AH.
U3000:49	Control Module: Internal Electronic Failure	REPAIR all other <u>OCSM</u> DTCs first. REPEAT the self-test. If U3000:49 returns, INSTALL a new <u>OCSM</u> . REFER to Occupant Classification System Module (OCSM) in this section.
U3000:53	Control Module: Deactivated	CLEAR all <u>OCSM</u> DTCs and REPEAT the self-test. If U3000:53 returns, INSTALL a new <u>OCSM</u> . REFER to Occupant Classification System Module (OCSM) in this section.
U3000:54	Control Module: Missing Calibration	CARRY OUT the <u>OCS</u> reset. REFER to Occupant Classification Sensor (OCS) System Reset in this section.
U3003:16	Battery Voltage: Circuit Voltage Below Threshold	GO to Pinpoint Test Z.
U3003:17	Battery Voltage: Circuit Voltage Above Threshold	GO to Pinpoint Test Z.

FCIM DTC Chart

NOTE: This module utilizes a 5-character DTC followed by a 2-character failure type code. The failure type code provides information about specific fault conditions such as opens or shorts to ground. Continuous Memory Diagnostic Trouble Codes (CMDTCs) have an additional 2-character DTC status code suffix to assist in determining DTC history.

DTC	Description	Action To Take
B00D5:01	Restraint System Passenger Disable Indicator: General Electrical Failure	Refer to the appropriate section in Group 415 for the procedure..
B1202:01	Restraint System Passenger Enable Indicator: General Electrical Failure	Refer to the appropriate section in Group 415 for the procedure..
All other DTCs	—	Refer to the appropriate section in Group 415 for the procedure..

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
The air bag warning indicator is illuminated continuously	- Fuse to Restraints Control Module (RCM) - Wiring, terminals or connectors - Damaged Data Link Connector (DLC) - Instrument Panel Cluster (IPC) - RCM	GO to Pinpoint Test A.
Air bag indicator flashing	Air bag/pretensioner Supplemental Restraint System (SRS) fault	CARRY OUT RCM and Occupant Classification System Module (OCSM) self-tests. REFER to DTC Charts in this section.
Audible tone — DTCs retrieved	Air bag SRS fault and air bag warning indicator fault	CARRY OUT RCM and IPC self-tests. REFER to Restraints Control Module (RCM) DTC Chart in this section and the IPC DTC Chart in Section 413-01.

• The RCM does not respond to the scan tool	• Fuse • Scan tool • DLC • Wiring, terminals or connectors • RCM	• REFER to Section 418-00 to diagnose the no communication concern.
• The Occupant Classification System Module (OCSM) does not respond to the scan tool	• Fuse • Scan tool • DLC • Wiring, terminals or connectors • OCSM	• REFER to Section 418-00 to diagnose the no communication concern.

Pinpoint Tests

Pinpoint Test A: The Air Bag Warning Indicator is Illuminated Continuously

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Instrument Panel Cluster (IPC) illuminates the air bag indicator continuously for 6 seconds. If the Supplemental Restraint System (SRS) is fault free, the air bag warning indicator turns off and remains off. If a fault is detected in the SRS , the air bag warning indicator illuminates and remains illuminated for the rest of the ignition cycle until the fault is no longer detected. The Restraints Control Module (RCM) communicates to the IPC module via the High Speed Controller Area Network (HS-CAN) . The IPC illuminates the air bag warning indicator based on messaging from the RCM or if there is no communication between the RCM and IPC .

This pinpoint test is intended to diagnose the following:

- IPC
- RCM

PINPOINT TEST A : THE AIR BAG WARNING INDICATOR IS ILLUMINATED CONTINUOUSLY

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

WARNING: Always tighten the fasteners of the restraints control module (RCM) and impact sensor (if equipped) to the specified torque. Failure to do so may result in incorrect restraint system operation, which increases the risk of personal injury or death in a crash.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

A1 RETRIEVE RCM DTCS

- Ignition ON.

- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Were any DTCs retrieved on-demand during self-test?

Yes	If DTCs were retrieved, GO to the DTC Charts in this section for diagnostic direction. Do not clear any DTCs until all DTCs have been resolved.
No	If the RCM does communicate with the scan tool, REFER to Section 413-01 to diagnose the IPC concern. If the RCM does not communicate with the scan tool, REFER to Section 418-00 to diagnose no power to the scan tool. Do not clear any DTCs until all DTCs have been resolved.

Pinpoint Test B: DTCs B0001:11, B0001:12, B0001:13 and B0001:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

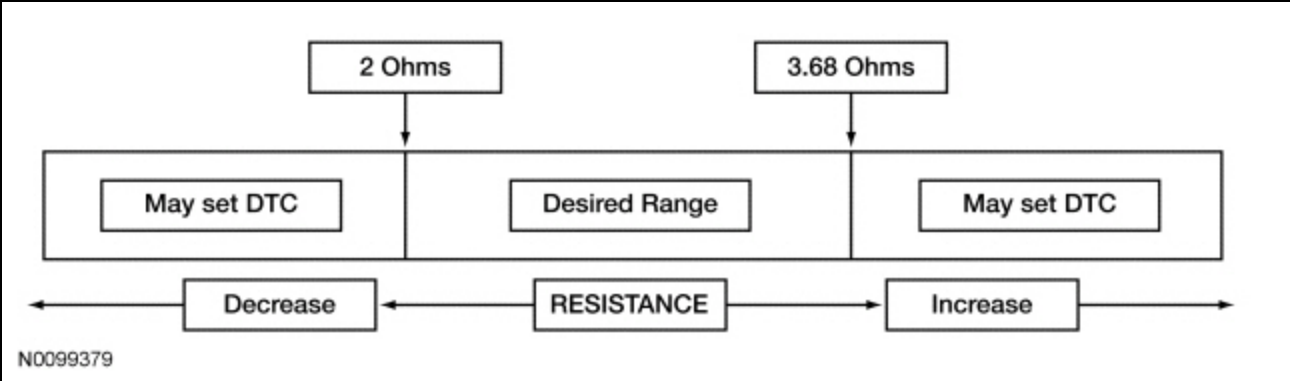
The Restraints Control Module (RCM) continuously monitors the driver air bag module stage 1 circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted driver air bag module

If a fault is detected, the **RCM** stores DTC B0001:11, B0001:12, B0001:13 or B0001:1A in memory and sends a message to the Instrument Panel Cluster (IPC) module to illuminate the air bag warning indicator.

The **RCM** analyzes the deployment loop resistance and uses this measurement, in combination with an age correction factor, to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the **RCM** . If the value displayed is lower or higher than the desired range (refer to diagram below), the **RCM** can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The **RCM** fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Clockspring
- Driver air bag module
- **RCM**



DTC Description	Fault Trigger Condition
B0001:11 — Driver Frontal Stage 1 Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either driver air bag stage 1 circuit.
B0001:12 — Driver Frontal Stage 1 Deployment Control: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either driver air bag stage 1 circuit.
B0001:13 — Driver Frontal Stage 1 Deployment Control: Circuit Open	A fault is indicated when the RCM measures more than the desired resistance range between driver air bag stage 1 circuits.
B0001:1A — Driver Frontal Stage 1 Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between driver air bag stage 1 circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Clockspring
- Driver air bag module
- RCM

PINPOINT TEST B : DTCS B0001:11, B0001:12, B0001:13 AND B0001:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

B1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0001:11, B0001:12, B0001:13 or B0001:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0001:13 or B0001:1A, GO to B2 . For DTC B0001:11, GO to B12 . For DTC B0001:12, GO to B15 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0001:13 or B0001:1A, GO to B20 . For DTC B0001:11, GO to B21 . For DTC B0001:12, GO to B22 .

B2 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_00_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_00_R.

Does the PID value read between 2 and 3.68 ohms?

Yes	GO to B19 .
No	GO to B3 .

B3 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_00_R) PID WHILE CARRYING OUT THE HARNESS TEST

- Remove the lower steering column shroud to access the clockspring electrical connector.
- While monitoring the DEPLOY_00_R PID, carry out the harness test of the driver air bag circuits and accessible connectors (including any in-line connectors) by wiggling and flexing the wire harness, connectors, tilting and rotating the steering wheel frequently.

Does the PID value read between 2 and 3.68 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness or INSTALL a new clockspring as needed. REFER to Clockspring in this section. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to B23 .
No	For PID value less than 2 ohms, GO to B4 . For PID value greater than 3.68 ohms, GO to B8 .

B4 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0001:1A to B0001:13?

Yes	GO to B17 .
No	GO to B5 .

B5 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED) (CLOCKSPRING DISCONNECTED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

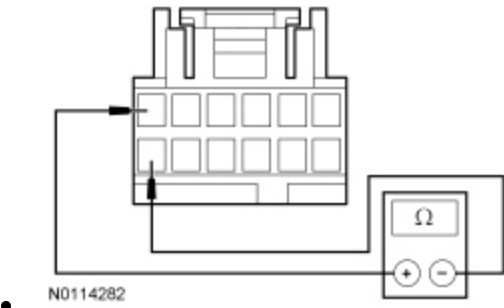
- Ignition OFF.
- Disconnect: Clockspring C218A .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module/clockspring disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0001:1A to B0001:13?

Yes	GO to B18 .
No	GO to B6 .

B6 CHECK FOR A SHORT BETWEEN DRIVER AIR BAG STAGE 1 CIRCUITS

- Measure the resistance between [C218A](#) Pin 6, circuit CR101 (VT/BN), harness side and [C218A](#) Pin 12, circuit RR101 (YE/GN), harness side.

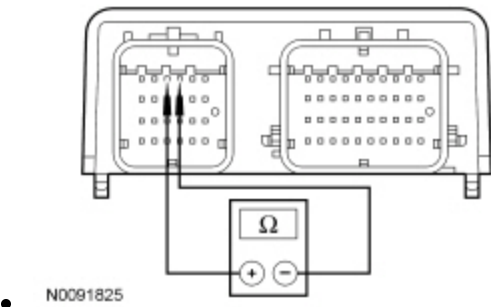


Is the resistance greater than 10,000 ohms?

Yes	GO to B7 .
No	REPAIR circuits CR101 (VT/BN) and RR101 (YE/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to B24 .

B7 CHECK THE RCM FOR LOW RESISTANCE

- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310A pins 3 and 4, component side.



Is the resistance greater than 10,000 ohms?

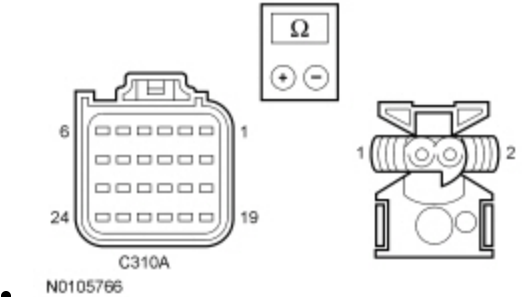
Yes	REPAIR circuits CR101 (VT/BN) and RR101 (YE/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to B24 .
No	GO to B19 .

B8 CHECK THE DRIVER AIR BAG STAGE 1 CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.

- Disconnect: RCM C310A and C310B .
- Measure the resistance between RCM C310A, harness side and driver air bag module stage 1 [C216A](#), harness side, using the following chart.

RCM	Circuit	Driver Air Bag Module
C310A Pin 4	CR101 (VT/BN)/(WH)	C216A Pin 1
C310A Pin 3	RR101 (YE/GN)/(BU)	C216A Pin 2



• N0105765

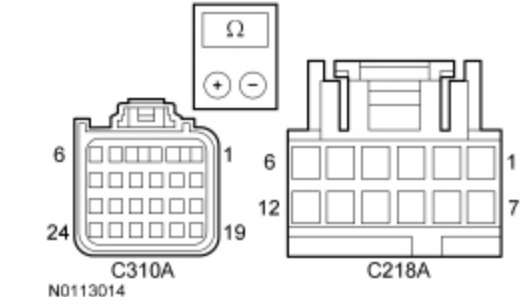
Are the resistances less than 1 ohm?

Yes	GO to B10 .
No	GO to B9 .

B9 CHECK THE DRIVER AIR BAG STAGE 1 CIRCUITS FOR AN OPEN (CLOCKSPRING DISCONNECTED)

- Disconnect: Clockspring C218A .
- Measure the resistance between RCM C310A, harness side and clockspring C218A, harness side using the following chart.

RCM	Circuit	Clockspring Electrical Connector
C310A Pin 4	CR101 (VT/BN)	C218A Pin 6
C310A Pin 3	RR101 (YE/GN)	C218A Pin 12



• N0113014

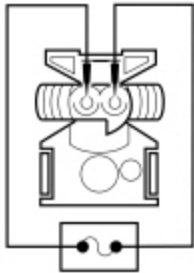
Are the resistances less than 0.5 ohm?

Yes	GO to B18 .
No	REPAIR circuit CR101 (VT/BN) or RR101 (YE/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to B24 .

B10 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Connect: *RCM* C310A and C310B .
- Connect a fused jumper wire between driver air bag module stage 1 [C216A](#) Pin 1, circuit CR101 (WH), harness side, and [C216A](#) Pin 2, circuit RR101 (BU), harness side.



- N0114275
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag stage 1 circuits shorted together, a low resistance fault is normally retrieved on stage 1. Stage 2 shows an open circuit fault due to the driver air bag being disconnected.

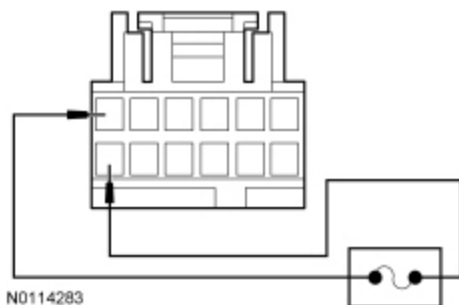
Was DTC B0001:1A retrieved on-demand during self-test?

Yes	REMOVE the fused jumper wire and GO to B17 .
No	REMOVE the fused jumper wire and GO to B11 .

B11 CHECK THE DRIVER AIR BAG STAGE 1 CIRCUITS FOR AN OPEN (CLOCKSPRING DISCONNECTED)

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Clockspring C218A .

- Connect a fused jumper wire between clockspring [C218A](#) Pin 6, circuit CR101 (VT/BN), harness side and [C218A](#) Pin 12, circuit RR101 (YE/GN), harness side.



- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the clockspring stage 1 circuits shorted together, a low resistance fault is normally retrieved on stage 1. Stage 2 shows an open circuit fault due to the driver air bag being disconnected.

Was DTC B0001:1A retrieved on-demand during self-test?

Yes	REMOVE the fused jumper wire and GO to B18 .
No	REMOVE the fused jumper wire and GO to B19 .

B12 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0001:11 to B0001:13?

Yes	GO to B17 .
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No	GO to B13 .
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B13 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED) (CLOCKSPRING DISCONNECTED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

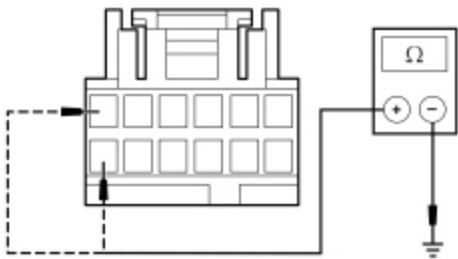
- Ignition OFF.
- Disconnect: Clockspring C218A .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the clockspring disconnected, open circuit faults are normally retrieved on driver air bag stages 1 and 2.

Did the on-demand DTC change from B0001:11 to B0001:13?

Yes	GO to B18 .
No	GO to B14 .

B14 CHECK THE DRIVER AIR BAG STAGE 1 CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: *RCM* C310A and C310B .
- Measure the resistance between clockspring:
 - [C218A](#) Pin 6, circuit CR101 (VT/BN), harness side and ground.
 - [C218A](#) Pin 12, circuit RR101 (YE/GN), harness side and ground.



N0118519

Are the resistances greater than 10,000 ohms?

Yes	GO to B19 .
No	Due to the shorting bar feature in the <i>RCM</i> electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar. REPAIR circuit CR101 (VT/BN) or RR101 (YE/GN).

	Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to B24 .
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B15 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO BATTERY INDICATED) (CLOCKSPRING DISCONNECTED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

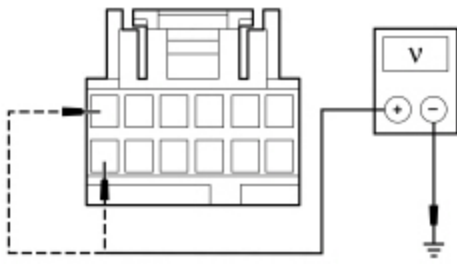
- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Disconnect: Clockspring C218A .
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module/clockspring disconnected, open circuit faults are normally retrieved on driver air bag stages 1 and 2.

Did the on-demand DTC change from B0001:12 to B0001:13?

Yes	GO to B18 .
No	GO to B16 .

B16 CHECK THE DRIVER AIR BAG STAGE 1 CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: *RCM* C310A and C310B .
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between clockspring:
 - [C218A](#) Pin 6, circuit CR101 (VT/BN), harness side and ground.
 - [C218A](#) Pin 12, circuit RR101 (YE/GN), harness side and ground.



N0117231

Is voltage present on either circuit?

Yes	Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar. REPAIR circuit CR101 (VT/BN) or RR101 (YE/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to B24 .
No	GO to B19 .

B17 CONFIRM THE DRIVER AIR BAG MODULE FAULT

NOTE: Make sure all **SRS** components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Install the driver air bag module. Refer to Driver Airbag in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new driver air bag module. REFER to Driver Airbag in this section. GO to B24 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0001:13 or B0001:1A, GO to B20 . For DTC B0001:11, GO to B21 . For DTC B0001:12, GO to B22 .

B18 CONFIRM THE CLOCKSPRING FAULT

NOTE: Make sure all **SRS** components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.

- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Clockspring C218A .
- Install the driver air bag module. Refer to Driver Airbag in this section.
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new clockspring. REFER to Clockspring in this section. GO to B24 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0001:13 or B0001:1A, GO to B20. For DTC B0001:11, GO to B21. For DTC B0001:12, GO to B22.

B19 CONFIRM THE RCM FAULT

NOTE: Make sure all ~~SRS~~ components and the ~~RCM~~ electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damaged, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- If previously removed, install the driver air bag module. Refer to Driver Airbag in this section.
- Connect: Clockspring C218A (if previously disconnected) .
- Connect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to B24 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0001:13 or B0001:1A, GO to B20. For DTC B0001:11, GO to B21. For DTC B0001:12, GO to B22.

B20 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_00_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_00_R.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.

Does the PID value read between 2 and 3.68 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to B23 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to B24.

B21 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0001:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to B24.
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to B23 .

B22 CHECK THE DRIVER FRONTAL STAGE 1 DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO BATTERY FAULT

- Ignition OFF.

- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0001:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to B24 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to B23 .

B23 CHECK THE HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module:
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
- Inspect **RCM** C310A and C310B **CPA** lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to B24 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to B24 .

B24 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the **RCM** and **OCSM** .*

Are any **RCM and/or **OCSM** DTCs retrieved on-demand during self-test?**

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test C: DTCs B0002:11, B0002:12, B0002:13 and B0002:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

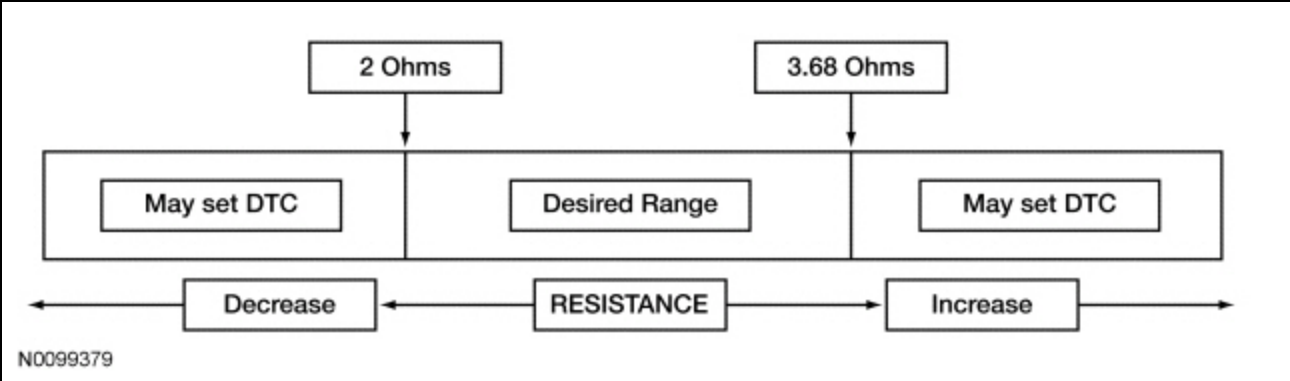
The Restraints Control Module (RCM) continuously monitors the driver air bag module stage 2 circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted driver air bag module

If a fault is detected, the **RCM** stores DTC B0002:11 , B0002:12, B0002:13 or B0002:1A in memory and sends a message to the Instrument Panel Cluster (IPC) module to illuminate the air bag warning indicator.

The **RCM** analyzes the deployment loop resistance and uses this measurement, in combination with an age correction factor, to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the **RCM** . If the value displayed is lower or higher than the desired range (refer to diagram below), the **RCM** can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The **RCM** fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Clockspring
- Driver air bag module
- **RCM**



DTC Description	Fault Trigger Condition
B0002:11 — Driver Frontal Stage 2 Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either driver air bag stage 2 circuit.
B0002:12 — Driver Frontal Stage 2 Deployment Control: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either driver air bag stage 2 circuit.
B0002:13 — Driver Frontal Stage 2 Deployment Control: Circuit Open	A fault is indicated when the RCM measures more than the desired resistance range between driver air bag stage 2 circuits.
B0002:1A — Driver Frontal Stage 2 Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between driver air bag stage 2 circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Clockspring
- Driver air bag module
- RCM

PINPOINT TEST C : DTCS B0002:11, B0002:12, B0002:13 AND B0002:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

C1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0002:11, B0002:12, B0002:13 or B0002:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0002:13 or B0002:1A, GO to C2 . For DTC B0002:11, GO to C11 . For DTC B0002:12, GO to C14 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0002:13 or B0002:1A, GO to C19 . For DTC B0002:11, GO to C20 . For DTC B0002:12, GO to C21 .

C2 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_01_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_01_R.

Does the PID value read between 2 and 3.68 ohms?

Yes	GO to C18 .
No	GO to C3 .

C3 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_01_R) PID WHILE CARRYING OUT THE HARNESS TEST

- Remove the lower steering column shroud to access the clockspring electrical connector.
- While monitoring the DEPLOY_01_R PID, carry out the harness test of the driver air bag circuits and accessible connectors (including any in-line connectors), by wiggling and flexing the wire harness, connectors, tilting and rotating the steering wheel frequently.

Does the PID value read between 2 and 3.68 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness or INSTALL a new clockspring as needed. REFER to Clockspring in this section. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to C23 .
No	For PID value less than 2 ohms, GO to C4 . For PID value greater than 3.68 ohms, GO to C7 .

C4 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0002:1A to B0002:13?

Yes	GO to C16 .
No	GO to C5 .

C5 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED) (CLOCKSPRING DISCONNECTED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Disconnect: Clockspring C218A .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module/clockspring disconnected, open circuit faults would normally be retrieved on stage 1 and 2.

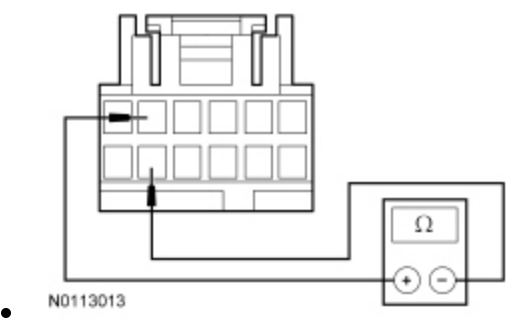
Did the on-demand DTC change from B0002:1A to B0002:13?

Yes	GO to C17 .
No	GO to C6 .

C6 CHECK FOR A SHORT BETWEEN DRIVER AIR BAG STAGE 2 CIRCUITS

- Ignition OFF.

- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between [C218A](#) Pin 5, circuit CR102 (BU), harness side and [C218A](#) Pin 11, circuit RR102 (WH), harness side.



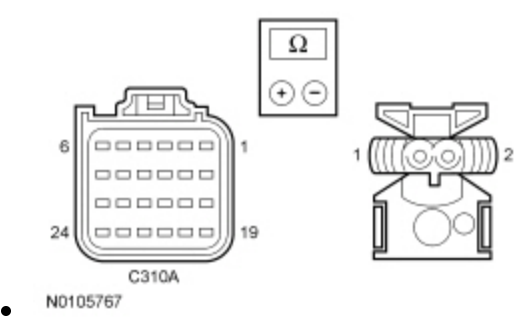
Is the resistance greater than 10,000 ohms?

Yes	GO to C18 .
No	REPAIR circuits CR102 (BU) and RR102 (WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to C23 .

C7 CHECK THE DRIVER AIR BAG STAGE 2 CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310A, harness side and driver air bag module stage 2 [C216B](#), harness side using the following chart.

RCM	Circuit	Driver Air Bag Module
C310A Pin 7	CR102 (BU)/(YE/GN)	C216B Pin 1
C310A Pin 8	RR102 (WH)/(VT/BN)	C216B Pin 2



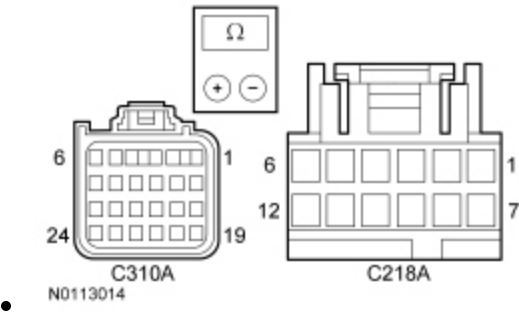
Are the resistances less than 1 ohm?

Yes	GO to C9 .
No	GO to C8 .

C8 CHECK THE DRIVER AIR BAG STAGE 2 CIRCUITS FOR AN OPEN (CLOCKSPRING DISCONNECTED)

- Disconnect: Clockspring C218A .
- Measure the resistance between ~~RCM~~ C310A, harness side and clockspring C218A, harness side using the following chart.

RCM	Circuit	Clockspring Electrical Connector
C310A Pin 7	CR102 (BU)	C218A Pin 5
C310A Pin 8	RR102 (WH)	C218A Pin 11



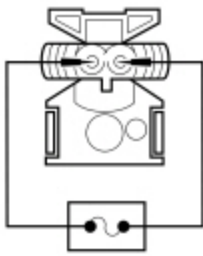
Are the resistances less than 0.5 ohm?

Yes	GO to C17 .
No	REPAIR circuit CR102 (BU) or RR102 (WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to C23 .

C9 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Connect: ~~RCM~~ C310A and C310B .
- Connect a fused jumper wire between driver air bag module stage 2 [C216B](#) Pin 1, circuit CR102 (YE/GN), harness side, and driver air bag module stage 2 [C216B](#) Pin 2, circuit RR102 (VT/BN), harness side.



• N0113511

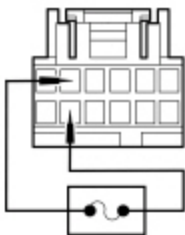
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with driver air bag module stage 2 circuits shorted together, a low resistance fault is normally retrieved on stage 2. Stage 1 shows an open circuit fault due to the driver air bag being disconnected.

Was DTC B0002:1A retrieved on-demand during self-test?

Yes	GO to C16 .
No	GO to C10 .

C10 CHECK THE DRIVER AIR BAG STAGE 2 CIRCUITS FOR AN OPEN (CLOCKSPRING DISCONNECTED)

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Clockspring C218A .
- Connect a fused jumper wire between clockspring [C218A](#) Pin 5, circuit CR102 (BU), harness side, and [C218A](#) Pin 11, circuit RR102 (WH), harness side.



• N0114274

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

- **DIAGNOSTIC TIP:** When viewing DTCs with the clockspring stage 2 circuits shorted together, a low resistance fault is normally retrieved on stage 2. Stage 1 shows an open circuit fault due to the driver air bag being disconnected.

Was DTC B0002:1A retrieved on-demand during self-test?

Yes	GO to C17 .
No	GO to C18 .

C11 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0002:11 to B0002:13?

Yes	GO to C16 .
No	GO to C12 .

C12 CHECK DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED) (CLOCKSPRING DISCONNECTED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

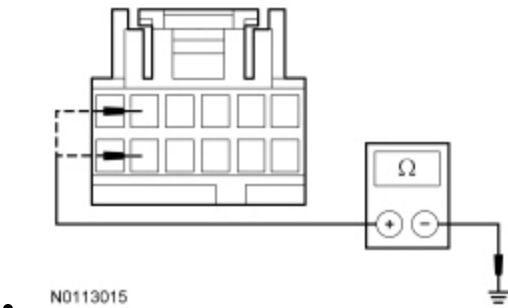
- Ignition OFF.
- Disconnect: Clockspring C218A .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module/clockspring disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0002:11 to B0002:13?

Yes	GO to C17 .
No	GO to C13 .

C13 CHECK THE DRIVER AIR BAG STAGE 2 CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between clockspring:
 - [C218A](#) Pin 5, circuit CR102 (BU), harness side and ground.
 - [C218A](#) Pin 11, circuit RR102 (WH), harness side and ground.



Are the resistances greater than 10,000 ohms?

Yes	GO to C18 .
No	REPAIR circuit CR102 (BU) or RR102 (WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to C23 .

C14 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO BATTERY INDICATED) (CLOCKSPRING DISCONNECTED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Disconnect: Clockspring C218A .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.

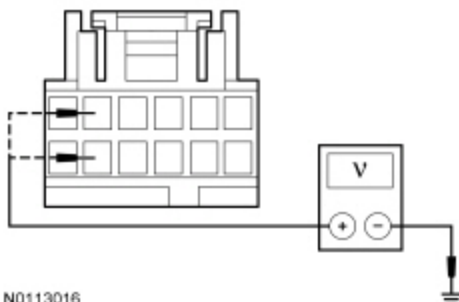
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver air bag module/clockspring disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0002:12 to B0002:13?

Yes	GO to C17 .
No	GO to C15 .

C15 CHECK THE DRIVER AIR BAG STAGE 2 CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between clockspring:
 - [C218A](#) Pin 5, circuit CR102 (BU), harness side and ground.
 - [C218A](#) Pin 11, circuit RR102 (WH), harness side and ground.



Is voltage present on either circuit?

Yes	REPAIR circuit CR102 (BU) or RR102 (WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to C23 .
No	GO to C18 .

C16 CONFIRM THE DRIVER AIR BAG MODULE FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Install the driver air bag module. Refer to Driver Airbag in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new driver air bag module. REFER to Driver Airbag in this section. GO to C23 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0002:13 or B0002:1A, GO to C19 . For DTC B0002:11, GO to C20 . For DTC B0002:12, GO to C21 .

C17 CONFIRM THE CLOCKSPrING FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Clockspring C218A .
- Install the driver air bag module. Refer to Driver Airbag in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new clockspring. REFER to Clockspring in this section. GO to C23 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0002:13 or B0002:1A, GO to C19 . For DTC B0002:11, GO to C20 . For DTC B0002:12, GO to C21 .

C18 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- If previously removed, install the driver air bag module. Refer to Driver Airbag in this section.
- Connect: Clockspring C218A (if previously disconnected) .
- Connect: **RCM** C310A and C310B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to C23 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0002:13 or B0002:1A, GO to C19. For DTC B0002:11, GO to C20. For DTC B0002:12, GO to C21.

C19 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_01_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_01_R.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.

Does the PID value read between 2 and 3.68 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to C22 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to C23.

C20 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0002:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to C23 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to C22 .

C21 CHECK THE DRIVER FRONTAL STAGE 2 DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO BATTERY FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module. Refer to Driver Airbag in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness, tilting and rotating the steering wheel frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0002:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to C23 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to C22 .

C22 CHECK HARNESS AND CONNECTORS

- Ignition OFF.

- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Remove the driver air bag module.
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
- Inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to C23 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to C23 .

C23 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test D: DTCs B0010:11, B0010:12, B0010:13 and B0010:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) continuously monitors the passenger air bag module stage 1 circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted passenger air bag module

If a fault is detected, the RCM stores DTC B0010:11, B0010:12, B0010:13 or B0010:1A in memory and sends a message to the Instrument Panel Cluster (IPC) module to illuminate the air bag warning indicator.

The RCM analyzes the deployment loop resistance and uses this measurement, in combination with an age correction factor, to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the RCM. If the value displayed is lower or higher than the desired range (refer to diagram below), the RCM can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The RCM fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Passenger air bag module
- RCM



DTC Description	Fault Trigger Condition
• B0010:11 — Passenger Frontal Stage 1 Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either passenger air bag stage 1 circuit.
• B0010:12 — Passenger Frontal Stage 1 Deployment Control: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either passenger air bag stage 1 circuit.
• B0010:13 — Passenger Frontal Stage 1 Deployment Control: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between passenger air bag stage 1 circuits.
• B0010:1A — Passenger Frontal Stage 1 Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between passenger air bag stage 1 circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger air bag module
- RCM

PINPOINT TEST D : DTCS B0010:11, B0010:12, B0010:13 AND B0010:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

D1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM.

Was DTC B0010:11, B0010:12, B0010:13 or B0010:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0010:13 or B0010:1A, GO to D2 . For DTC B0010:11, GO to D9 . For DTC B0010:12, GO to D11 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0010:13 or B0010:1A, GO to D14 . For DTC B0010:11, GO to D15 . For DTC B0010:12, GO to D16 .

D2 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_05_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — RCM.
 - Monitor and record PID DEPLOY_05_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to D13 .
No	GO to D3 .

D3 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_05_R) PID WHILE CARRYING OUT THE HARNESS TEST

- While monitoring the DEPLOY_05_R PID, carry out the harness test of the passenger air bag module circuits and accessible connectors (including any in-line connectors), by wiggling connectors and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to D18 .
No	For PID value less than 1.7 ohms, GO to D4 . For PID value greater than 2.78 ohms, GO to D7 .

D4 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

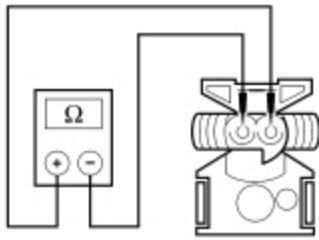
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- DIAGNOSTIC TIP:** When viewing DTCs with the passenger air bag module disconnected, open circuit faults are normally retrieved on passenger air bag stages 1 and 2.

Did the on-demand DTC change from B0010:1A to B0010:13?

Yes	GO to D12 .
No	GO to D5 .

D5 CHECK FOR A SHORT BETWEEN PASSENGER AIR BAG STAGE 1 CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Measure the resistance between passenger air bag [C256A](#) Pin 2, circuit CR103 (GY/BU), harness side and [C256A](#) Pin 1, circuit RR103 (VT/GN), harness side.



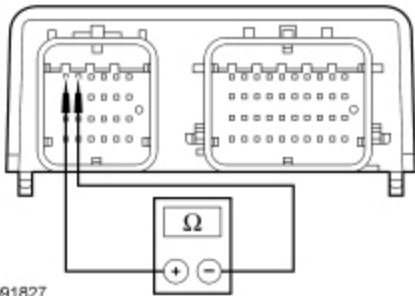
• N0093109

Is the resistance greater than 10,000 ohms?

Yes	GO to D13 .
No	GO to D6 .

D6 CHECK THE RCM FOR LOW RESISTANCE

- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310A pins 1 and 2, component side.



• N0091827

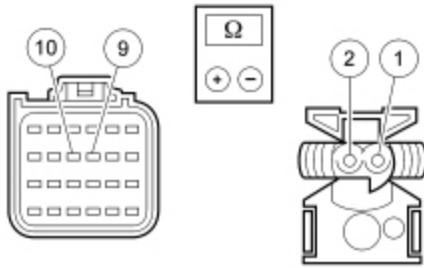
Is the resistance greater than 10,000 ohms?

Yes	REPAIR circuits CR103 (GY/BU) and RR103 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to D18 .
No	GO to D13 .

D7 CHECK THE PASSENGER AIR BAG STAGE 1 CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310A, harness side and passenger air bag module C256A, harness side using the following chart.

RCM	Circuit	Passenger Air Bag Module Stage 1 Electrical Connector
C310A Pin 1	CR103 (GY/BU)	C256A Pin 2
C310A Pin 2	RR103 (VT/GN)	C256A Pin 1



N0093290

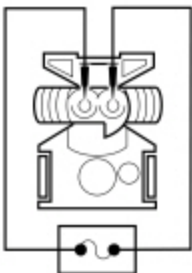
Are the resistances less than 0.5 ohm?

Yes	GO to D8 .
No	REPAIR circuit CR103 (GY/BU) or RR103 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to D18 .

D8 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Connect: **RCM** C310A and C310B .
- Connect a fused jumper wire between passenger air bag module stage 1 [C256A](#) Pin 2, circuit CR103 (GY/BU), harness side, and [C256A](#) Pin 1, circuit RR103 (VT/GN), harness side.



N0114275

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- DIAGNOSTIC TIP:** When viewing DTCs with the passenger air bag module stage 1 circuits shorted together, a low resistance fault is normally retrieved on stage 1. Stage 2 shows an open circuit fault due to the passenger

air bag being disconnected.

Did the on-demand DTC change from B0010:13 to B0010:1A?

Yes	REMOVE the fused jumper wire and GO to D12 .
No	REMOVE the fused jumper wire and GO to D13 .

D9 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

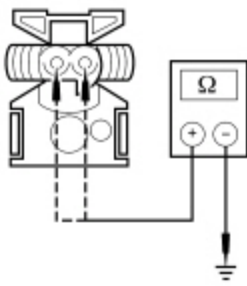
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger air bag module disconnected, open circuit faults are normally retrieved on passenger air bag stages 1 and 2.

Did the on-demand DTC change from B0010:11 to B0010:13?

Yes	GO to D12 .
No	GO to D10 .

D10 CHECK THE PASSENGER AIR BAG STAGE 1 CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between passenger air bag module:
 - [C256A](#) Pin 2, circuit CR103 (GY/BU), harness side and ground.
 - [C256A](#) Pin 1, circuit RR103 (VT/GN), harness side and ground.



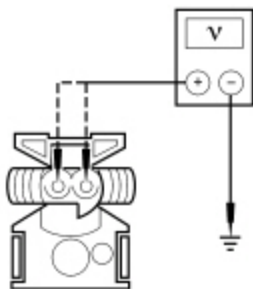
N0061730

Are the resistances greater than 10,000 ohms?

Yes	GO to D13 .
No	Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar. REPAIR circuit CR103 (GY/BU) or RR103 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to D18 .

D11 CHECK THE PASSENGER AIR BAG STAGE 1 CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Disconnect: **RCM** C310A and C310B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between passenger air bag module:
 - [C256A](#) Pin 2, circuit CR103 (GY/BU), harness side and ground.
 - [C256A](#) Pin 1, circuit RR103 (VT/GN), harness side and ground.



N0062797

Is voltage present on either circuit?

Yes	Due to the shorting bar feature in the RCM electrical connector, the fault can exist in either circuit. Do not remove or defeat the shorting bar. REPAIR circuit CR103 (GY/BU) or RR103 (VT/GN).
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	Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to D18 .
No	GO to D13 .

D12 CONFIRM THE PASSENGER AIR BAG MODULE FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Passenger Air Bag Module C256A and C256B (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new passenger air bag module. REFER to Passenger Airbag in this section. GO to D18 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0010:13 or B0010:1A, GO to D14 . For DTC B0010:11, GO to D15 . For DTC B0010:12, GO to D16 .

D13 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Passenger Air Bag Module C256A and C256B (if previously disconnected) .
- Connect: **RCM** C310A and C310B .

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to D18 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0010:13 or B0010:1A, GO to D14 . For DTC B0010:11, GO to D15 . For DTC B0010:12, GO to D16 .

D14 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_05_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_05_R.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to D17 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to D18 .

D15 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL CIRCUITS FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0010:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to D18 .
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No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to D17 .
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D16 CHECK THE PASSENGER FRONTAL STAGE 1 DEPLOYMENT CONTROL CIRCUITS FOR AN INTERMITTENT SHORT TO BATTERY FAULT

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was DTC B0010:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to D18 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to D17 .

D17 CHECK THE HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module [C256A](#) (if not previously disconnected) .
- Disconnect: Passenger Air Bag Module [C256B](#) (if not previously disconnected) .
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to D18 .
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No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to D18 .
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D18 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all SRS components (if previously disconnected).
- If previously directed to depower the SRS , repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and QCSM .*

Are any RCM and/or QCSM DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test E: DTCs B0011:11, B0011:12, B0011:13 and B0011:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) continuously monitors the passenger air bag module stage 2 circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted passenger air bag module

If a fault is detected, the RCM stores DTC B0011:11, B0011:12, B0011:13 or B0011:1A in memory and sends a message to the Instrument Panel Cluster (IPC) module to illuminate the air bag warning indicator.

The RCM analyzes the deployment loop resistance and uses this measurement in combination with an age correction factor to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the RCM . If the value displayed is lower or higher than the desired range (refer to diagram below), the RCM can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The RCM fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an

on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Passenger air bag module
- RCM



DTC Description	Fault Trigger Condition
• B0011:11 — Passenger Frontal Stage 2 Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either passenger air bag stage 2 circuit.
• B0011:12 — Passenger Frontal Stage 2 Deployment Control: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either passenger air bag stage 2 circuit.
• B0011:13 — Passenger Frontal Stage 2 Deployment Control: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between passenger air bag stage 2 circuits.
• B0011:1A — Passenger Frontal Stage 2 Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between passenger air bag stage 2 circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger air bag module
- RCM

PINPOINT TEST E : DTCS B0011:11, B0011:12, B0011:13 AND B0011:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

E1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0011:11, B0011:12, B0011:13 or B0011:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0011:13 or B0011:1A, GO to E2 . For DTC B0011:11, GO to E8 . For DTC B0011:12, GO to E10 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0011:13 or B0011:1A, GO to E13 . For DTC B0011:11, GO to E14 . For DTC B0011:12, GO to E15 .

E2 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_06_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — RCM .
 - Monitor and record PID DEPLOY_06_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to E12 .
No	GO to E3 .

E3 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_06_R) PID WHILE CARRYING OUT THE HARNESS TEST

- While monitoring the DEPLOY_06_R PID, carry out the harness test of the passenger air bag module circuits and accessible connectors (including any in-line connectors), by wiggling connectors and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to E17 .

No	For PID value less than 1.7 ohms, GO to E4 . For PID value greater than 2.78 ohms, GO to E6 .
----	--

E4 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

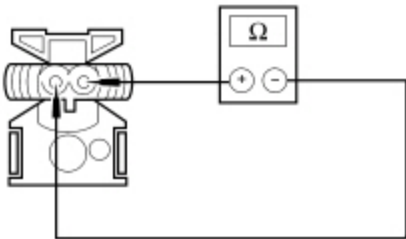
- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger air bag module disconnected, open circuit faults are normally retrieved on passenger air bag stages 1 and 2.

Did the on-demand DTC change from B0011:1A to B0011:13?

Yes	GO to E11 .
No	GO to E5 .

E5 CHECK FOR A SHORT BETWEEN PASSENGER AIR BAG STAGE 2 CIRCUITS

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- Measure the resistance between [C256B](#) Pin 2, circuit CR104 (YE/GY), harness side and [C256B](#) Pin 1, circuit RR104 (WH/BU), harness side.



• A0030495

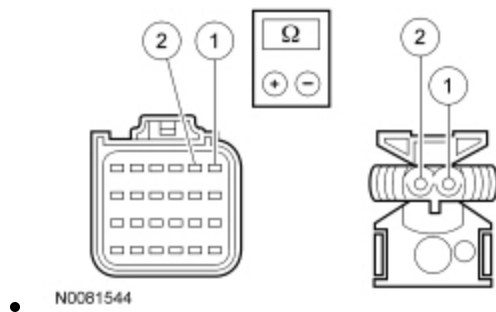
Is the resistance greater than 10,000 ohms?

Yes	GO to E12 .
No	REPAIR circuits CR104 (YE/GY) and RR104 (WH/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to E17 .

E6 CHECK THE PASSENGER AIR BAG STAGE 2 CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310A, harness side and passenger air bag module C256B, harness side using the following chart.

RCM	Circuit	Passenger Air Bag Module Stage 2 Electrical Connector
C310A Pin 10	CR104 (YE/GY)	C256B Pin 2
C310A Pin 9	RR104 (WH/BU)	C256B Pin 1



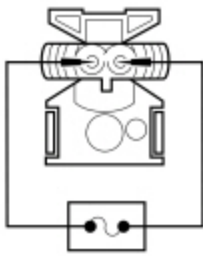
Are the resistances less than 0.5 ohm?

Yes	GO to E7 .
No	REPAIR circuit CR104 (YE/GY) or RR104 (WH/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to E17 .

E7 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Connect: ~~RCM~~ C310A and C310B .
- Connect a fused jumper wire between passenger air bag module stage 2 [C218A](#) Pin 2, circuit CR104 (YE/GY), harness side, and [C218A](#) Pin 1, circuit RR104 (WH/BU), harness side.



• N0113511

- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger air bag module stage 2 circuits shorted together, a low resistance fault is normally retrieved on stage 2. Stage 1 shows an open circuit fault due to the passenger air bag module being disconnected.

Did the on-demand DTC change from B0011:13 to B0011:1A?

Yes	REMOVE the fused jumper wire and GO to E11 .
No	REMOVE the fused jumper wire and GO to E12 .

E8 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

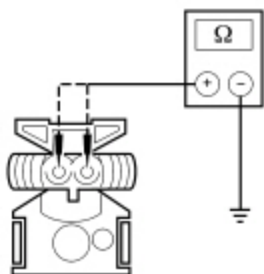
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger air bag module disconnected, open circuit faults are normally retrieved on stages 1 and 2.

Did the on-demand DTC change from B0011:11 to B0011:13?

Yes	GO to E11 .
No	GO to E9 .

E9 CHECK THE PASSENGER AIR BAG STAGE 2 CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between passenger air bag module:
 - [C256B](#) Pin 2, circuit CR104 (YE/GY), harness side and ground.
 - [C256B](#) Pin 1, circuit RR104 (WH/BU), harness side and ground.



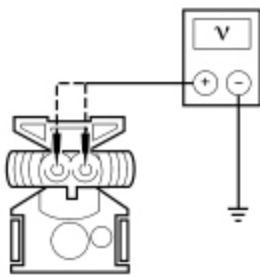
A0094161

Are the resistances greater than 10,000 ohms?

Yes	GO to E12 .
No	REPAIR circuit CR104 (YE/GY) or RR104 (WH/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to E17 .

E10 CHECK THE PASSENGER AIR BAG STAGE 2 CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Disconnect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between passenger air bag module:
 - [C256B](#) Pin 2, circuit CR104 (YE/GY), harness side and ground.
 - [C256B](#) Pin 1, circuit RR104 (WH/BU), harness side and ground.



A0094162

Is voltage present on either circuit?

Yes	REPAIR circuit CR104 (YE/GY) or RR104 (WH/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to E17 .
No	GO to E12 .

E11 CONFIRM THE PASSENGER AIR BAG MODULE FAULT

NOTE: Make sure all restraint system components and the *RCM* electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Passenger Air Bag Module C256A and C256B (if previously disconnected) .
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new passenger air bag module. REFER to Passenger Airbag in this section. GO to E17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new <i>SRS</i> components at this time. Install <i>SRS</i> components only when directed to do so in the pinpoint test. For DTC B0011:13 or B0011:1A, GO to E13 . For DTC B0011:11, GO to E14 . For DTC B0011:12, GO to E15 .

E12 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the *RCM* electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Passenger Air Bag Module C256A and C256B (if previously disconnected) .
- Connect: **RCM** C310A and C310B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to E17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0011:13 or B0011:1A, GO to E13. For DTC B0011:11, GO to E14. For DTC B0011:12, GO to E15.

E13 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL RESISTANCE (DEPLOY_06_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_06_R.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to E16 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to E17.

E14 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0011:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to E17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to E16 .

E15 CHECK THE PASSENGER FRONTAL STAGE 2 DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO BATTERY FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module C256A and C256B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0011:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to E17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to E16 .

E16 CHECK THE HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Air Bag Module [C256A](#) (if not previously disconnected) .
- Disconnect: Passenger Air Bag Module [C256B](#) (if not previously disconnected) .

- inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
- inspect wire harness for any damage, pinched, cut or pierced wires.
- inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to E17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to E17 .

E17 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~OCSM~~ .*

Are any ~~RCM~~ and/or ~~OCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test F: DTCs B0020:11, B0020:12, B0020:13 and B0020:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) continuously monitors the driver seat side air bag module and circuits for the following faults:

- Resistance out of range
- Unexpected voltage

- Short to ground
- Faulted driver seat side air bag module

If a fault is detected, the RCM stores DTC B0020:11, B0020:12, B0020:13 or B0020:1A in memory and sends a message to the Instrument Panel Cluster (IPC) module to illuminate the air bag warning indicator.

The RCM analyzes the deployment loop resistance and uses this measurement in combination with an age correction factor to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the RCM . If the value displayed is lower or higher than the desired range (refer to diagram below), the RCM can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The RCM fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Driver seat side air bag module
- RCM



DTC Description	Fault Trigger Condition
• B0020:11 — Left Side Airbag Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either driver seat side air bag circuit.
• B0020:12 — Left Side Airbag Deployment Control: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either driver seat side air bag circuit.
• B0020:13 — Left Side Airbag Deployment Control: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between the driver seat side air bag circuits.
• B0020:1A — Left Side Airbag Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between the driver seat side air bag circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver seat side air bag module
- RCM

PINPOINT TEST F : DTCS B0020:11, B0020:12, B0020:13 AND B0020:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

F1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0020:11, B0020:12, B0020:13 or B0020:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0020:13 or B0020:1A, GO to F2 . For DTC B0020:11, GO to F8 . For DTC B0020:12, GO to F10 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0020:13 or B0020:1A, GO to F13 . For DTC B0020:11, GO to F14 . For DTC B0020:12, GO to F15 .

F2 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL RESISTANCE (DEPLOY_10_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — RCM .
 - Monitor and record PID DEPLOY_10_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to F12 .
No	GO to F3 .

F3 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL RESISTANCE (DEPLOY_10_R) PID WHILE CARRYING OUT THE HARNESS TEST

- While monitoring the DEPLOY_10_R PID, carry out the harness test of the driver seat side air bag circuits and accessible connectors (including any in-line connectors) by wiggling connectors and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to F17 .
No	For PID value less than 1.7 ohms, GO to F4 . For PID value greater than 2.78 ohms, GO to F6 .

F4 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: *This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.*

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Seat Side Air Bag Module C367 . **NOTE:** *Unzip the trim cover to access the driver seat side air bag module electrical connector.*
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver seat side air bag module disconnected, an open circuit fault is normally retrieved.

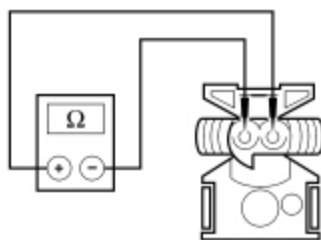
Did the on-demand DTC change from B0020:1A to B0020:13?

Yes	GO to F11 .
No	GO to F5 .

F5 CHECK FOR A SHORT BETWEEN DRIVER SEAT SIDE AIR BAG CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .

- Measure the resistance between driver seat side air bag module [C367](#) Pin 1, circuit CR105 (GN/BU), harness side and [C367](#) Pin 2, circuit RR105 (GY/YE), harness side.



• N0093121

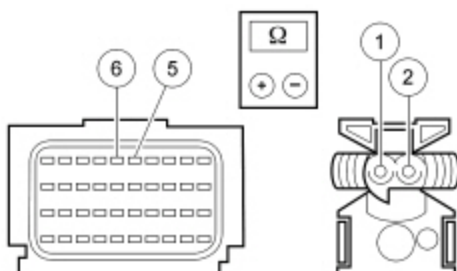
Is the resistance greater than 10,000 ohms?

Yes	GO to F12 .
No	REPAIR circuits CR105 (GN/BU) and RR105 (GY/YE). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to F17 .

F6 CHECK THE DRIVER SEAT SIDE AIR BAG CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Seat Side Air Bag Module C367 . **NOTE:** *Unzip the trim cover to access the driver seat side air bag module electrical connector.*
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310B, harness side and driver seat side air bag module C3051, harness side using the following chart.

RCM	Circuit	Driver Seat Side Air Bag Module Electrical Connector
C310B Pin 6	CR105 (GN/BU)	C367 Pin 1
C310B Pin 5	RR105 (GY/YE)	C367 Pin 2



• N0093122

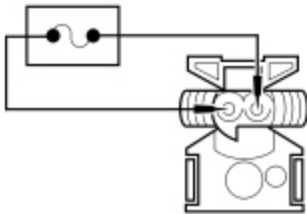
Are the resistances less than 0.5 ohm?

Yes	GO to F7 .
No	REPAIR circuit CR105 (GN/BU) or RR105 (GY/YE). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to F17 .

F7 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Connect: **RCM** C310A and C310B .
- Connect a fused jumper wire between driver seat side air bag [C367](#) Pin 1, circuit CR105 (VT/GY), harness side and [C367](#) Pin 2, circuit RR105 (YE/OG), harness side.



- N0119863
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver seat side air bag module circuits shorted together, a low resistance fault is normally retrieved.

Did the on-demand DTC change from B0020:13 to B0020:1A?

Yes	REMOVE the fused jumper wire and GO to F11 .
No	REMOVE the fused jumper wire and GO to F12 .

F8 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

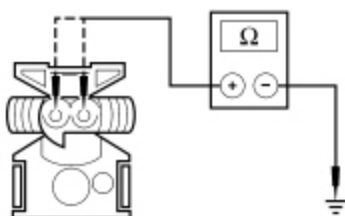
- Disconnect: Driver Seat Side Air Bag Module C367 . **NOTE:** *Unzip the trim cover to access the driver seat side air bag module electrical connector.*
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver seat side air bag module disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B0020:11 to B0020:13?

Yes	GO to F11 .
No	GO to F9 .

F9 CHECK THE DRIVER SEAT SIDE AIR BAG CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between driver seat side air bag module:
 - [C367](#) Pin 1, circuit CR105 (GN/BU), harness side and ground.
 - [C367](#) Pin 2, circuit RR105 (GY/YE), harness side and ground.



• N0093123

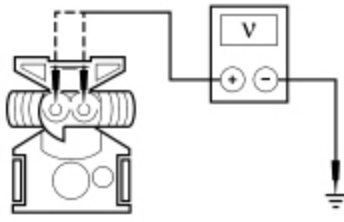
Are the resistances greater than 10,000 ohms?

Yes	GO to F12 .
No	REPAIR circuit CR105 (GN/BU) or RR105 (GY/YE). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to F17 .

F10 CHECK THE DRIVER SEAT SIDE AIR BAG CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Disconnect: Driver Seat Side Air Bag Module C367 . **NOTE:** *Unzip the trim cover to access the driver seat side air bag module electrical connector.*
- Disconnect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between driver seat side air bag module:
 - [C367](#) Pin 1, circuit CR105 (GN/BU), harness side and ground.
 - [C367](#) Pin 2, circuit RR105 (GY/YE), harness side and ground.



• N0093124

Is voltage present on either circuit?

Yes	REPAIR circuit CR105 (GN/BU) or RR105 (GY/YE). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to F17 .
No	GO to F12 .

F11 CONFIRM THE DRIVER SEAT SIDE AIR BAG MODULE FAULT

NOTE: *Make sure all restraint system components and the ~~RCM~~ electrical connectors are connected before carrying out self-test. If not, DTCs will be recorded.*

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Driver Seat Side Air Bag Module C367 .
- Connect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	REMOVE and INSPECT the driver seat side air bag module harness for damage. REFER to Side Airbag in this section. If a concern is found, REPAIR as necessary.
-----	--

Refer to Wiring Diagrams Cell [5](#) for schematic and connector information.

If a concern **was not** found, INSTALL a new driver seat side air bag module. REFER to Side Airbag in this section. GO to [F17](#).

No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0020:13 or B0020:1A, GO to F13. For DTC B0020:11, GO to F14. For DTC B0020:12, GO to F15.
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F12 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the ~~RCM~~ electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Driver Seat Side Air Bag Module C367 .
- Connect: ~~RCM~~ C310A and C310B (if previously disconnected) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to F17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0020:13 or B0020:1A, GO to F13. For DTC B0020:11, GO to F14. For DTC B0020:12, GO to F15.

F13 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL RESISTANCE (DEPLOY_10_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — ~~RCM~~ .
 - Monitor and record PID DEPLOY_10_R.

- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to F16 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to F17 .

F14 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0020:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to F17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to F16 .

F15 CHECK THE LEFT SIDE AIRBAG DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO BATTERY FAULT

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Seat Side Air Bag Module C367 . **NOTE:** *Unzip the trim cover to access the driver seat side air bag module electrical connector.*
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0020:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to F17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to F16 .

F16 CHECK THE HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Seat Side Air Bag Module [C367](#). **NOTE:** *Unzip the trim cover to access the driver seat side air bag module electrical connector.*
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to F17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to F17 .

F17 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test G: DTCs B0021:11, B0021:12, B0021:13 and B0021:1A

Normal Operation

The Restraints Control Module (RCM) continuously monitors the driver Safety Canopy® and circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted driver Safety Canopy®

If a fault is detected, the RCM stores DTC B0021:11, B0021:12, B0021:13 or B0021:1A in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

The RCM analyzes the deployment loop resistance and uses this measurement in combination with an age correction factor to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the RCM . If the value displayed is lower or higher than the desired range (refer to diagram below), the RCM can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The RCM fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Driver Safety Canopy® module
- RCM



DTC Description	Fault Trigger Condition
• B0021:11 — Left Curtain Deployment Control 1: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either driver Safety Canopy® circuit.

DTC Description	Fault Trigger Condition
B0021:12 — Left Curtain Deployment Control 1: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either driver Safety Canopy® circuit.
B0021:13 — Left Curtain Deployment Control 1: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between the driver Safety Canopy® circuits.
B0021:1A — Left Curtain Deployment Control 1: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between the driver Safety Canopy® circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver Safety Canopy® module
- RCM

PINPOINT TEST G : DTCS B0021:11, B0021:12, B0021:13 AND B0021:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

G1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM.

Was DTC B0021:11, B0021:12, B0021:13 or B0021:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0021:13 or B0021:1A, GO to G2. For DTC B0021:11, GO to G8. For DTC B0021:12, GO to G10.
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0021:13 or B0021:1A, GO to G13.

For DTC B0021:11, GO to [G14](#).
For DTC B0021:12, GO to [G15](#).

G2 CHECK THE LEFT CURTAIN DEPLOYMENT CONTROL 1 RESISTANCE (DEPLOY_11_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_11_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to G12 .
No	GO to G3 .

G3 CHECK THE LEFT CURTAIN DEPLOYMENT CONTROL 1 RESISTANCE (DEPLOY_11_R) PID WHILE CARRYING OUT THE HARNESS TEST

- While monitoring the DEPLOY_11_R PID, carry out the harness test of the driver Safety Canopy® circuits and accessible connectors (including any in-line connectors) by wiggling connectors and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to G17 .
No	For PID value less than 1.7 ohms, GO to G4 . For PID value greater than 2.78 ohms, GO to G6 .

G4 CHECK THE LEFT CURTAIN DEPLOYMENT CONTROL 1 DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Canopy® [C3321](#).
 - For regular cab, remove the LH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the LH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

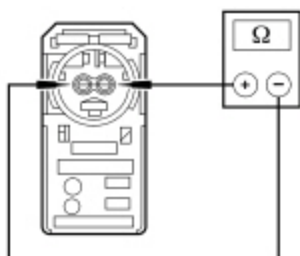
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver Safety Canopy® disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B0021:1A to B0021:13?

Yes	GO to G11 .
No	GO to G5 .

G5 CHECK FOR A SHORT BETWEEN DRIVER SAFETY CANOPY® CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between driver Safety Canopy® [C3321](#) Pin 1, circuit CR109 (BN/BU), harness side and [C3321](#) Pin 2, circuit RR109 (BU/GN), harness side.



• N0102657

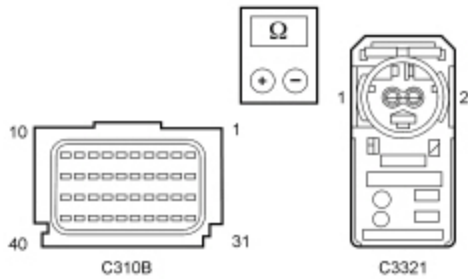
Is the resistance greater than 10,000 ohms?

Yes	GO to G12 .
No	REPAIR circuits CR109 (BN/BU) and RR109 (BU/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to G17 .

G6 CHECK THE DRIVER SAFETY CANOPY® CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Canopy® [C3321](#).
 - For regular cab, remove the LH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the LH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310B, harness side and driver Safety Canopy® C3321, harness side using the following chart.

RCM	Circuit	Driver Safety Canopy® Module Electrical Connector
C310B Pin 7	CR109 (BN/BU)	C3321 Pin 1
C310B Pin 8	RR109 (BU/GN)	C3321 Pin 2



• N0102653

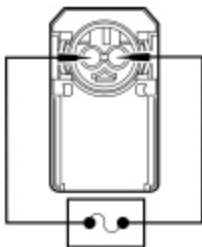
Are the resistances less than 0.5 ohm?

Yes	GO to G7 .
No	REPAIR circuit CR109 (BN/BU) or RR109 (BU/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to G17 .

G7 CHECK THE LEFT CURTAIN DEPLOYMENT CONTROL 1 DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Connect: **RCM** C310A and C310B .
- Connect a fused jumper wire between driver Safety Canopy® [C3321](#) Pin 1, circuit CR109 (BN/BU), harness side and [C3321](#) Pin 2, circuit RR109 (BU/GN), harness side.



• N0114277

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver Safety Canopy® circuits shorted together, a low resistance fault is normally retrieved.

Did the on-demand DTC change from B0021:13 to B0021:1A?

Yes	REMOVE the fused jumper wire and GO to G11 .
No	REMOVE the fused jumper wire and GO to G12 .

G8 CHECK THE LEFT CURTAIN DEPLOYMENT CONTROL 1 DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

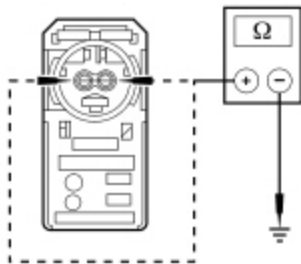
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Canopy® [C3321](#).
 - For regular cab, remove the LH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the LH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver Safety Canopy® disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B0021:11 to B0021:13?

Yes	GO to G11 .
No	GO to G9 .

G9 CHECK THE DRIVER SAFETY CANOPY® CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between driver Safety Canopy®:
 - [C3321](#) Pin 1, circuit CR109 (BN/BU), harness side and ground.
 - [C3321](#) Pin 2, circuit RR109 (BU/GN), harness side and ground.



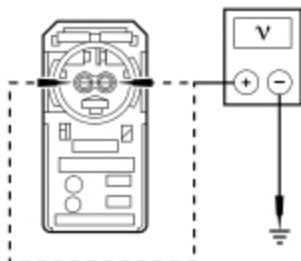
N0102655

Are the resistances greater than 10,000 ohms?

Yes	GO to G12 .
No	REPAIR circuit CR109 (BN/BU) or RR109 (BU/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to G17 .

G10 CHECK THE DRIVER SAFETY CANOPY® CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Canopy® [C3321](#).
 - For regular cab, remove the LH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the LH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Disconnect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between driver Safety Canopy®:
 - [C3321](#) Pin 1, circuit CR109 (BN/BU), harness side and ground.
 - [C3321](#) Pin 2, circuit RR109 (BU/GN), harness side and ground.



N0102655

Is voltage present on either circuit?

Yes	REPAIR circuit CR109 (BN/BU) or RR109 (BU/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to G17 .
-----	---

No	GO to G12 .
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G11 CONFIRM THE DRIVER SAFETY CANOPY® MODULE FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Driver Safety Canopy® C3321 .
- Connect: **RCM** C310A and C310B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new driver Safety Canopy®. REFER to Safety Canopy Module in this section. GO to G17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0021:13 or B0021:1A, GO to G13. For DTC B0021:11, GO to G14. For DTC B0021:12, GO to G15.

G12 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Driver Safety Canopy® C3321 .
- Connect: **RCM** C310A and C310B (if previously disconnected) .

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to G17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0021:13 or B0021:1A, GO to G13 . For DTC B0021:11, GO to G14 . For DTC B0021:12, GO to G15 .

G13 CHECK THE LEFT CURTAIN DEPLOYMENT CONTROL 1 RESISTANCE (DEPLOY_11_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_11_R.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to G16 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to G17 .

G14 CHECK FOR AN INTERMITTENT LEFT CURTAIN DEPLOYMENT CONTROL 1 SHORT TO GROUND FAULT

- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0021:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to G17 .
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No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to G16 .
----	--

G15 CHECK FOR AN INTERMITTENT LEFT CURTAIN DEPLOYMENT CONTROL 1 SHORT TO BATTERY FAULT

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Canopy® [C3321](#).
 - For regular cab, remove the LH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the LH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0021:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to G17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to G16 .

G16 CHECK THE HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Canopy® [C3321](#).
 - For regular cab, remove the LH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the LH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect RCM C310A and C310B CPA lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to G17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to G17 .

G17 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all SRS components (if previously disconnected).
- If previously directed to depower the SRS , repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and OCSM .*

Are any RCM and/or OCSM DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test H: DTCs B0028:11, B0028:12, B0028:13 and B0028:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) continuously monitors the passenger seat side air bag module circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted passenger seat side air bag module

If a fault is detected, the RCM stores DTC B0028:11, B0028:12, B0028:13 or B0028:1A in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

The RCM analyzes the deployment loop resistance and uses this measurement in combination with an age correction factor to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the RCM . If the value displayed is lower or higher than the desired range (refer to diagram below), the RCM can set a DTC. As the

deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The RCM fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Passenger seat side air bag module
- RCM



DTC Description	Fault Trigger Condition
• B0028:11 — Right Side Airbag Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either passenger seat side air bag circuit.
• B0028:12 — Right Side Airbag Deployment Control: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either passenger seat side air bag circuit.
• B0028:13 — Right Side Airbag Deployment Control: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between the passenger seat side air bag circuits.
• B0028:1A — Right Side Airbag Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between the passenger seat side air bag circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger seat side air bag module
- RCM

PINPOINT TEST H : DTCS B0028:11, B0028:12, B0028:13 AND B0028:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases

the risk of serious personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

H1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0028:11, B0028:12, B0028:13 or B0028:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0028:13 or B0028:1A, GO to H2 . For DTC B0028:11, GO to H8 . For DTC B0028:12, GO to H10 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0028:13 or B0028:1A, GO to H13 . For DTC B0028:11, GO to H14 . For DTC B0028:12, GO to H15 .

H2 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL RESISTANCE (DEPLOY_13_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — RCM .
 - Monitor and record PID DEPLOY_13_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to H12 .
No	GO to H3 .

H3 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL RESISTANCE (DEPLOY_13_R) PID WHILE CARRYING OUT THE HARNESS TEST

- While monitoring the DEPLOY_13_R PID, carry out the harness test of the passenger side air bag circuits and accessible connectors (including any in-line connectors) by wiggling connectors and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to H17 .
No	For PID value less than 1.7 ohms, GO to H4 . For PID value greater than 2.78 ohms, GO to H6 .

H4 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

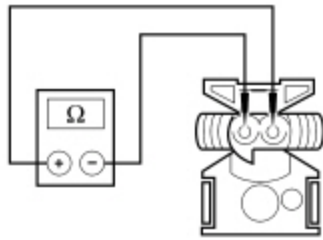
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C337 . **NOTE:** *Unzip the trim cover to access the passenger seat side air bag module electrical connector.*
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger seat side air bag module disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B0028:1A to B0028:13?

Yes	GO to H11 .
No	GO to H5 .

H5 CHECK FOR A SHORT BETWEEN PASSENGER SEAT SIDE AIR BAG CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between passenger seat side air bag [C337](#) Pin 1, circuit CR106 (VT/GY), harness side and [C337](#) Pin 2, circuit RR106 (YE/OG), harness side.



• N0093121

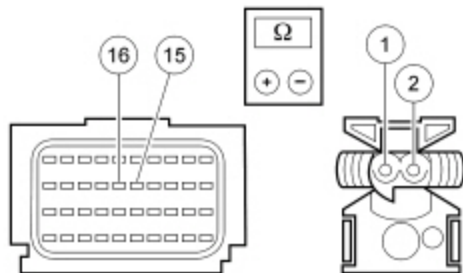
Is the resistance greater than 10,000 ohms?

Yes	GO to H12 .
No	REPAIR circuits CR106 (VT/GY) and RR106 (YE/OG). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to H17 .

H6 CHECK THE PASSENGER SEAT SIDE AIR BAG CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C337 .
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310B, harness side and passenger seat side air bag module C337, harness side using the following chart.

RCM	Circuit	Passenger Seat Side Air Bag Module Electrical Connector
C310B Pin 16	CR106 (VT/GY)	C337 Pin 1
C310B Pin 15	RR106 (YE/OG)	C337 Pin 2



• N0093125

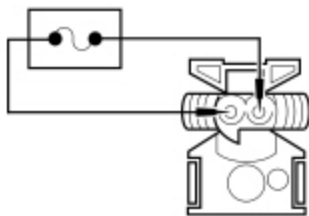
Are the resistances less than 0.5 ohm?

Yes	GO to H7 .
No	REPAIR circuit CR106 (VT/GY) or RR106 (YE/OG). Refer to Wiring Diagrams Cell 5 for schematic and connector information.

H7 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Connect: *RCM* C310A and C310B .
- Connect a fused jumper wire between passenger seat side air bag [C337](#) Pin 1, circuit CR106 (VT/GY), harness side and [C337](#) Pin 2, circuit RR106 (YE/OG), harness side.



- N0119863
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger side air bag circuits shorted together, a low resistance fault is normally retrieved.

Did the on-demand DTC change from B0028:13 to B0028:1A?

Yes	REMOVE the fused jumper wire and GO to H11 .
No	REMOVE the fused jumper wire and GO to H12 .

H8 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C337 . **NOTE:** *Unzip the trim cover to access the passenger seat side air bag module electrical connector.*
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

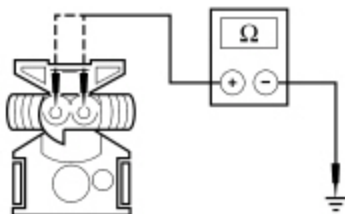
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger seat side air bag module disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B0028:11 to B0028:13?

Yes	GO to H11 .
No	GO to H9 .

H9 CHECK THE PASSENGER SEAT SIDE AIR BAG CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- Measure the resistance between passenger seat side air bag:
 - [C337](#) Pin 1, circuit CR106 (VT/GY), harness side and ground.
 - [C337](#) Pin 2, circuit RR106 (YE/OG), harness side and ground.



• N0093123

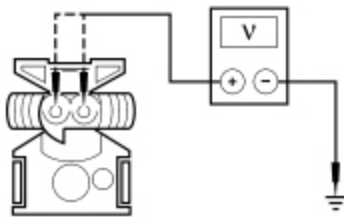
Are the resistances greater than 10,000 ohms?

Yes	GO to H12 .
No	REPAIR circuit CR106 (VT/GY) or RR106 (YE/OG). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to H17 .

H10 CHECK THE PASSENGER SEAT SIDE AIR BAG CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C337 . **NOTE:** *Unzip the trim cover to access the passenger seat side air bag module electrical connector.*
- Disconnect: **RCM** C310A and C310B .

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between passenger seat side air bag:
 - [C337](#) Pin 1, circuit CR106 (VT/GY), harness side and ground.
 - [C337](#) Pin 2, circuit RR106 (YE/OG), harness side and ground.



• N0093124

Is voltage present on either circuit?

Yes	REPAIR circuit CR106 (VT/GY) or RR106 (YE/OG). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to H17 .
No	GO to H12 .

H11 CONFIRM THE PASSENGER SEAT SIDE AIR BAG MODULE FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Passenger Seat Side Air Bag Module [C337](#) .
- Connect: **RCM** [C310A](#) and [C310B](#) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	REMOVE and INSPECT the passenger seat side air bag module wire harness for damage. REFER to Side Airbag in this section. If a concern is found, REPAIR the seat side air bag wire harness. If a concern was not found, INSTALL a new passenger seat side air bag module. REFER to Side Airbag in this section. GO to H17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test.

For DTC B0028:13 or B0028:1A, GO to [H13](#).
For DTC B0028:11, GO to [H14](#).
For DTC B0028:12, GO to [H15](#).

H12 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Passenger Seat Side Air Bag Module C337 .
- Connect: **RCM** C310A and C310B (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to H17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0028:13 or B0028:1A, GO to H13 . For DTC B0028:11, GO to H14 . For DTC B0028:12, GO to H15 .

H13 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL RESISTANCE (DEPLOY_13_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_13_R.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to H16 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to H17 .

H14 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0028:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to H17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to H16 .

H15 CHECK THE RIGHT SIDE AIRBAG DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO BATTERY FAULT

- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C337 . **NOTE:** *Unzip the trim cover to access the passenger seat side air bag module electrical connector.*
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0028:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to H17 .
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No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to H16 .
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H16 CHECK THE HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module [C337](#). **NOTE:** *Unzip the trim cover to access the passenger seat side air bag module electrical connector.*
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to H17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to H17 .

H17 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test I: DTCs B0029:11, B0029:12, B0029:13 and B0029:1A

Normal Operation

The Restraints Control Module (RCM) continuously monitors the passenger Safety Canopy® circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted passenger Safety Canopy®

If a fault is detected, the RCM stores DTC B0029:11, B0029:12, B0029:13 or B0029:1A in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

The RCM analyzes the deployment loop resistance and uses this measurement in combination with an age correction factor to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the RCM . If the value displayed is lower or higher than the desired range (refer to diagram below), the RCM can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The RCM fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Passenger Safety Canopy® module
- RCM



DTC Description	Fault Trigger Condition
• B0029:11 — Right Curtain Deployment Control 1: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either passenger Safety Canopy® circuit.
• B0029:12 — Right Curtain Deployment Control 1: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on either passenger Safety Canopy® circuit.
• B0029:13 — Right Curtain Deployment Control 1: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between the passenger Safety Canopy® circuits.
• B0029:1A — Right Curtain Deployment	A fault is indicated when the RCM measures less than the desired resistance range between the passenger Safety Canopy® circuits.

DTC Description	Fault Trigger Condition
Control 1: Circuit Resistance Below Threshold	

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger Safety Canopy® module
- RCM

PINPOINT TEST I : DTCS B0029:11, B0029:12, B0029:13 AND B0029:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*

NOTE: *Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .*

NOTE: *Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.*

NOTE: *The SRS must be fully operational and free of faults before releasing the vehicle to the customer.*

I1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0029:11, B0029:12, B0029:13 or B0029:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0029:13 or B0029:1A, GO to I2 . For DTC B0029:11, GO to I8 . For DTC B0029:12, GO to I10 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B0029:13 or B0029:1A, GO to I13 . For DTC B0029:11, GO to I14 . For DTC B0029:12, GO to I15 .

I2 CHECK THE RIGHT CURTAIN DEPLOYMENT CONTROL 1 RESISTANCE (DEPLOY_14_R) PID

- Enter the following diagnostic mode on the scan tool: DataLogger — RCM .
 - Monitor and record PID DEPLOY_14_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to I12 .
No	GO to I3 .

I3 CHECK THE RIGHT CURTAIN DEPLOYMENT CONTROL 1 RESISTANCE (DEPLOY_14_R) PID WHILE CARRYING OUT THE HARNESS TEST

- While monitoring the DEPLOY_14_R PID, carry out the harness test of the passenger Safety Canopy® circuits and accessible connectors (including any in-line connectors), by wiggling connectors and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms while carrying out the harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to I17 .
No	For PID value less than 1.7 ohms, GO to I4 . For PID value greater than 2.78 ohms, GO to I6 .

I4 CHECK THE RIGHT CURTAIN DEPLOYMENT CONTROL 1 DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

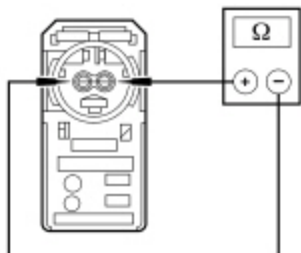
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Canopy® [C3329](#).
 - For regular cab, remove the RH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the RH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- DIAGNOSTIC TIP:** When viewing DTCs with the passenger Safety Canopy® disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B0029:1A to B0029:13?

Yes	GO to I11 .
No	GO to I5 .

I5 CHECK FOR A SHORT BETWEEN PASSENGER SAFETY CANOPY® CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between passenger Safety Canopy® [C3329](#) Pin 1, circuit CR111 (BN/WH), harness side and [C3329](#) Pin 2, circuit RR111 (YE/VT), harness side.



• N0102657

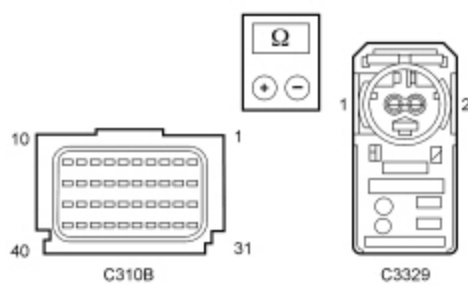
Is the resistance greater than 10,000 ohms?

Yes	GO to I12 .
No	REPAIR circuits CR111 (BN/WH) and RR111 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to I17 .

I6 CHECK THE PASSENGER SAFETY CANOPY® CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Canopy® [C3329](#).
 - For regular cab, remove the RH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the RH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ C310B, harness side and passenger Safety Canopy® [C3329](#), harness side using the following chart.

RCM	Circuit	Passenger Safety Canopy® Electrical Connector
C310B Pin 17	CR111 (BN/WH)	C3329 Pin 1
C310B Pin 18	RR111 (YE/VT)	C3329 Pin 2



• N0102654

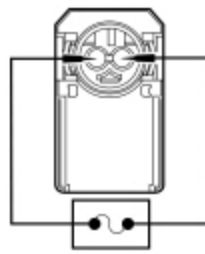
Are the resistances less than 0.5 ohm?

Yes	GO to I7 .
No	REPAIR circuit CR111 (BN/WH) or RR111 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to I17 .

I7 CHECK THE RIGHT CURTAIN DEPLOYMENT CONTROL 1 DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Connect: **RCM** C310A and C310B .
- Connect a fused jumper wire between passenger Safety Canopy® [C3329](#) Pin 1, circuit CR111 (BN/WH), harness side and [C3329](#) Pin 2, circuit RR111 (YE/VT), harness side.



• N0114277

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger Safety Canopy® circuits shorted together, a low resistance fault is normally retrieved.

Did the on-demand DTC change from B0029:13 to B0029:1A?

Yes	REMOVE the fused jumper wire and GO to I11 .
No	REMOVE the fused jumper wire and GO to I12 .

18 CHECK THE RIGHT CURTAIN DEPLOYMENT CONTROL 1 DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

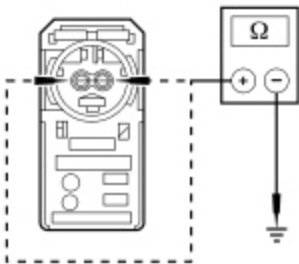
- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Canopy® [C3329](#).
 - For regular cab, remove the RH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the RH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger Safety Canopy® disconnected, an open circuit fault would normally be retrieved.

Did the on-demand DTC change from B0029:11 to B0029:13?

Yes	GO to 111 .
No	GO to 19 .

19 CHECK THE PASSENGER SAFETY CANOPY® CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: *RCM* C310A and C310B .
- Measure the resistance between passenger Safety Canopy®:
 - [C3329](#) Pin 1, circuit CR111 (BN/WH), harness side and ground.
 - [C3329](#) Pin 2, circuit RR111 (YE/VT), harness side and ground.



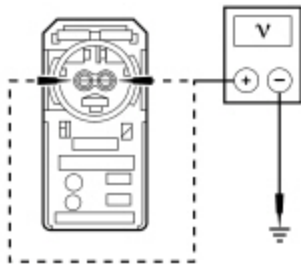
N0102656

Are the resistances greater than 10,000 ohms?

Yes	GO to I12 .
No	REPAIR circuit CR111 (BN/WH) or RR111 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to I17 .

I10 CHECK THE PASSENGER SAFETY CANOPY® CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Canopy® [C3329](#).
 - For regular cab, remove the RH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the RH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Disconnect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between passenger Safety Canopy® C9019:
 - [C3329](#) Pin 1, circuit CR111 (BN/WH), harness side and ground.
 - [C3329](#) Pin 2, circuit RR111 (YE/VT), harness side and ground.



Is voltage present on either circuit?

Yes	REPAIR circuit CR111 (BN/WH) or RR111 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to I17 .
No	GO to I12 .

I11 CONFIRM THE PASSENGER SAFETY CANOPY® MODULE FAULT

NOTE: Make sure all restraint system components and the ~~RCM~~ electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Connect: Passenger Safety Canopy® C3329 .
- Connect: RCM C310A and C310B .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new passenger Safety Canopy®. REFER to Safety Canopy Module in this section. GO to I17 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0029:13 or B0029:1A, GO to I13 . For DTC B0029:11, GO to I14 . For DTC B0029:12, GO to I15 .

I12 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the RCM electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected SRS component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect RCM C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Passenger Safety Canopy® C3329 .
- Connect: RCM C310A and C310B .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to I17 .

No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B0029:13 or B0029:1A, GO to I13. For DTC B0029:11, GO to I14. For DTC B0029:12, GO to I15.
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I13 CHECK THE RIGHT CURTAIN DEPLOYMENT CONTROL 1 RESISTANCE (DEPLOY_14_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — **RCM** .
 - Monitor and record PID DEPLOY_14_R.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to I16.
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to I17.

I14 CHECK FOR AN INTERMITTENT RIGHT CURTAIN DEPLOYMENT CONTROL 1 SHORT TO GROUND FAULT

- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0029:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to I17.
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to I16.

I15 CHECK FOR AN INTERMITTENT RIGHT CURTAIN DEPLOYMENT CONTROL 1 SHORT TO BATTERY FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Canopy® [C3329](#).

- For regular cab, remove the RH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the RH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
 - Ignition ON.
 - Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness frequently.
 - Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was DTC B0029:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to I17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to I16 .

I16 CHECK THE HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Canopy® [C3329](#).
 - For regular cab, remove the RH side B-pillar trim panel.
 - For SuperCab and Crew Cab, remove the RH C-pillar trim panel.
 - Refer to Interior Trim in Section 501-05.
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to I17 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to I17 .

I17 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the **RCM** and **OCSM** .*

Are any **RCM and/or **OCSM** DTCs retrieved on-demand during self-test?**

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test J: DTCs B0050:11, B0050:12, B0050:13 and B0050:1D

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors the driver safety belt buckle switch circuits for the following faults:

- Open circuit
- Short to voltage
- Short to ground
- Current out of range
- Faulted driver safety belt buckle switch

If a fault is detected, the **RCM** stores DTC B0050:11, B0050:12, B0050:13 or B0050:1D in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Condition
• B0050:11 — Driver Seatbelt Sensor: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on the driver safety belt buckle switch circuit.
• B0050:12 — Driver Seatbelt Sensor: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on the driver safety belt buckle switch circuit.
• B0050:13 — Driver Seatbelt Sensor: Circuit Open	A fault is indicated when the RCM senses an open on either driver safety belt buckle switch circuit.

DTC Description	Fault Trigger Condition
B0050:1D — Driver Seatbelt Sensor: Circuit Current Out of Range	A fault is indicated when the RCM senses current out of an acceptable range between the driver safety belt buckle switch circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver safety belt buckle and pretensioner
- RCM

PINPOINT TEST J : DTCS B0050:11, B0050:12, B0050:13 AND B0050:1D

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: <i>Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.</i>
NOTE: <i>Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .</i>
NOTE: <i>Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.</i>
NOTE: <i>The SRS must be fully operational and free of faults before releasing the vehicle to the customer.</i>

J1 RETRIEVE RCM DTCS	
- Ignition ON. - Enter the following diagnostic mode on the scan tool: Self Test — RCM .	
Was DTC B0050:11, B0050:12, B0050:13 or B0050:1D retrieved on-demand during self-test?	
Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0050:11 or B0050:1D, GO to J2 . For DTC B0050:12, GO to J4 . For DTC B0050:13, GO to J5 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to J10 .

J2 CHECK THE DRIVER SEATBELT SENSOR DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND OR CURRENT OUT OF RANGE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

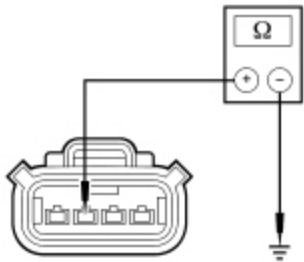
- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.
- Disconnect: Driver Safety Belt Buckle C3065 .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver safety belt buckle disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B0050:11 or B0050:1D to B0050:13?

Yes	GO to J8 .
No	For DTC B0050:11, GO to J3 . For DTC B0050:1D, GO to J9 .

J3 CHECK THE DRIVER SAFETY BELT BUCKLE SWITCH CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- Measure the resistance between driver safety belt buckle [C3065](#) Pin 3, circuit CR201 (GN/BU), harness side and ground.



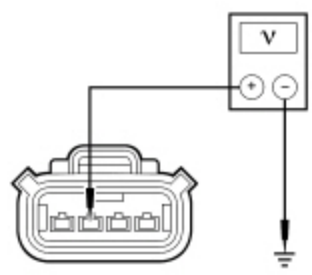
• N0065413

Is the resistance greater than 10,000 ohms?

Yes	GO to J9 .
No	REPAIR circuit CR201 (GN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to J11 .

J4 CHECK THE DRIVER SAFETY BELT BUCKLE SWITCH CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Seat Side Air Bag Module C3051 .
- Disconnect: Driver Safety Belt Buckle C3065 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between driver safety belt buckle [C3065](#) Pin 3, circuit CR201 (GN/BU), harness side and ground.



• N0065410

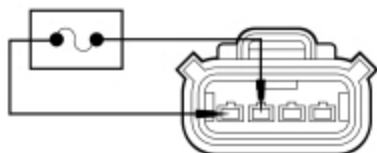
Is any voltage present?

Yes	REPAIR circuit CR201 (GN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to J11 .
No	GO to J9 .

J5 CHECK THE DRIVER SEATBELT SENSOR DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Drive Seat Side Air Bag Module C3051 .
- Disconnect: Driver Safety Belt Buckle C3065 .
- Connect a fused jumper wire between driver safety belt buckle [C3065](#) Pin 3, circuit CR201 (GN/BU), harness side and [C3065](#) Pin 4, circuit RR137 (BU/GY), harness side.



• N0119864

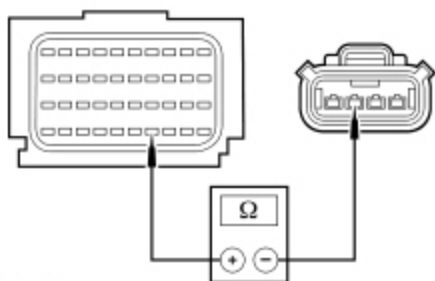
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver safety belt buckle switch circuits shorted together, a short to ground fault is normally retrieved.

Did the on-demand DTC change from B0050:13 to B0050:11?

Yes	REMOVE the fused jumper wire and GO to J8 .
No	REMOVE the fused jumper wire and GO to J6 .

J6 CHECK THE DRIVER SAFETY BELT BUCKLE SWITCH CIRCUIT FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ [C310B](#) Pin 34, circuit CR201 (GN/BU), harness side and driver safety belt buckle [C3065](#) Pin 3, circuit CR201 (GN/BU), harness side.



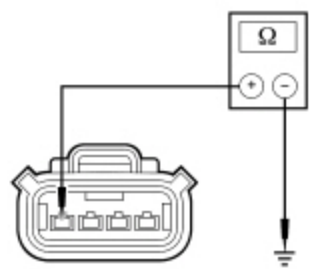
• N0091864

Is the resistance less than 0.5 ohm?

Yes	GO to J7 .
No	REPAIR circuit CR201 (GN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to J11 .

J7 CHECK THE DRIVER SAFETY BELT BUCKLE SWITCH GROUND CIRCUIT FOR AN OPEN

- Measure the resistance between driver safety belt buckle [C3065](#) Pin 4, circuit RR137 (BU/GY)/GD174 (BK/WH), harness side and ground.



• N0065411

Is the resistance less than 3 ohms?

Yes	GO to J9 .
No	REPAIR circuit RR137 (BU/GY)/GD174 (BK/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to J11 .

J8 CONFIRM THE DRIVER SAFETY BELT BUCKLE SWITCH FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: **RCM** C310A and C310B (if previously disconnected) .
- Connect: Driver Safety Belt Buckle C3065 .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new driver safety belt buckle and pretensioner. REFER to Section 501-20A. GO to J11 .
-----	---

No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to J10 .
----	--

J9 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B **CPA** lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Driver Safety Belt Buckle C3065 .
- Connect: **RCM** C310A and C310B (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to J11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to J10 .

J10 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Belt Buckle [C3065](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .

- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was DTC B0050:11, B0050:12, B0050:13 or B0050:1D retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0050:11 or B0050:1D, GO to J2 . For DTC B0050:12, GO to J4 . For DTC B0050:13, GO to J5 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to J11 .

J11 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test K: DTCs B0052:11, B0052:12, B0052:13 and B0052:1D

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors the passenger safety belt buckle switch circuits for the following faults:

- Open circuit

- Short to voltage
- Short to ground
- Current out of range
- Faulted passenger safety belt buckle switch

If a fault is detected, the RCM stores DTC B0052:11, B0052:12, B0052:13 or B0052:1D in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Condition
• B0052:11 — Passenger Seatbelt Sensor: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on the passenger safety belt buckle switch circuit.
• B0052:12 — Passenger Seatbelt Sensor: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on the passenger safety belt buckle switch circuit.
• B0052:13 — Passenger Seatbelt Sensor: Circuit Open	A fault is indicated when the RCM senses an open on either passenger safety belt buckle switch circuit.
• B0052:1D — Passenger Seatbelt Sensor: Circuit Current Out of Range	A fault is indicated when the RCM senses current out of an acceptable range between the passenger safety belt buckle switch circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger safety belt buckle and pretensioner
- RCM

PINPOINT TEST K : DTCS B0052:11, B0052:12, B0052:13 AND B0052:1D

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

K1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0052:11, B0052:12, B0052:13 or B0052:1D retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0052:11 or B0052:1D, GO to K2 . For DTC B0052:12, GO to K4 . For DTC B0052:13, GO to K5 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to K10 .

K2 CHECK THE PASSENGER SEATBELT SENSOR DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND OR CURRENT OUT OF RANGE INDICATED)

NOTE: *This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.*

- Ignition OFF.
- Disconnect: Passenger Safety Belt Buckle C3066 . **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger safety belt buckle disconnected, an open circuit fault is normally retrieved.

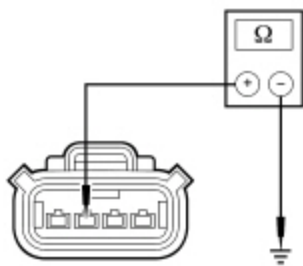
Did the on-demand DTC change from B0052:11 or B0052:1D to B0052:13?

Yes	GO to K8 .
No	For DTC B0052:11, GO to K3 . For DTC B0052:1D, GO to K9 .

K3 CHECK THE PASSENGER SAFETY BELT BUCKLE SWITCH CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .

- Measure the resistance between passenger safety belt buckle [C3066](#) Pin 3, circuit CR203 (GY/VT), harness side and ground.



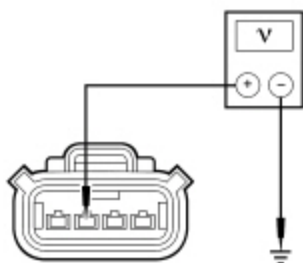
N0065413

Is the resistance greater than 10,000 ohms?

Yes	GO to K9 .
No	REPAIR circuit CR203 (GY/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to K11 .

K4 CHECK THE PASSENGER SAFETY BELT BUCKLE SWITCH CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: Passenger Safety Belt Buckle C3066 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between passenger safety belt buckle [C3066](#) Pin 3, circuit CR203 (GY/VT), harness side and ground.



N0065410

Is any voltage present?

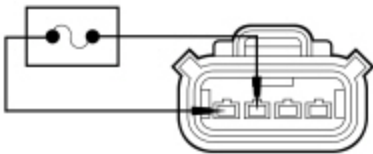
Yes	REPAIR circuit CR203 (GY/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to K11 .
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No	GO to K9 .
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K5 CHECK THE PASSENGER SEATBELT SENSOR DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: Passenger Safety Belt Buckle C3066 .
- Connect a fused jumper wire between passenger safety belt buckle [C3066](#) Pin 3, circuit CR203 (GY/VT), harness side and [C3066](#) Pin 4, circuit GD901 (BK), harness side.



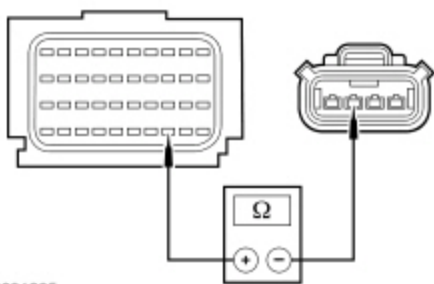
- N0119864
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger safety belt buckle circuits shorted together, a short to ground fault is normally retrieved.

Did the on-demand DTC change from B0052:13 to B0052:11?

Yes	REMOVE the fused jumper wire and GO to K8 .
No	REMOVE the fused jumper wire and GO to K6 .

K6 CHECK THE PASSENGER SAFETY BELT BUCKLE SWITCH CIRCUIT FOR AN OPEN

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: *RCM* C310A and C310B .
- Measure the resistance between *RCM* [C310B](#) Pin 33, circuit CR203 (GY/VT), harness side and the passenger safety belt buckle [C3066](#) Pin 3, circuit CR203 (GY/VT), harness side.



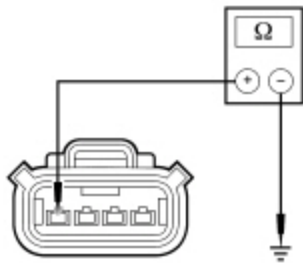
• N0091865

Is the resistance less than 0.5 ohm?

Yes	GO to K7 .
No	REPAIR circuit CR203 (GY/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to K11 .

K7 CHECK THE PASSENGER SAFETY BELT BUCKLE SWITCH GROUND CIRCUIT FOR AN OPEN

- Measure the resistance between passenger safety belt buckle [C3066](#) Pin 4, circuit GD901 (BK)/GD174 (BK/WH), harness side and ground.



• N0065411

Is the resistance less than 3 ohms?

Yes	GO to K9 .
No	REPAIR circuit GD901 (BK)/GD174 (BK/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to K11 .

K8 CONFIRM THE PASSENGER SAFETY BELT BUCKLE SWITCH FAULT

NOTE: Make sure all restraint system components and the ~~RCM~~ electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.

- inspect ~~RCM~~ C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
- repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.

- Connect: ~~RCM~~ C310A and C310B (if previously disconnected) .
- Connect: Passenger Safety Belt Buckle C3066 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new passenger safety belt buckle and pretensioner. REFER to Section 501-20A. GO to K11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to K10 .

K9 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the ~~RCM~~ electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Passenger Safety Belt Buckle C3066 .
- Connect: ~~RCM~~ C310A and C310B (if previously disconnected) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to K11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to K10 .

K10 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Belt Buckle [C3066](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0052:11, B0052:12, B0052:13 or B0052:1D retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0052:11 or B0052:1D, GO to K2 . For DTC B0052:12, GO to K4 . For DTC B0052:13, GO to K5 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to K11 .

K11 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE: When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and OCSM .**

Are any RCM and/or OCSM DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test L: DTCs B0090:11, B0090:81, B0090:93, B0090:96

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors the LH front impact severity sensor and circuits for the following faults:

- Open circuit
- Short to voltage
- Short to ground
- Faulted sensor

If a fault is detected, the RCM stores DTC B0090:11, B0090:81, B0090:93 or B0090:96 in memory and sends a message to theInstrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Condition
• B0090:11 — Left Frontal Restraints Sensor: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on the LH front severity impact sensor feed circuit.
• B0090:81 — Left Frontal Restraints Sensor: Invalid Serial Data Received	A fault is indicated when the RCM receives data that is invalid from the LH front impact severity sensor or the wrong sensor is installed to the vehicle. A short to voltage on the feed circuit also sets this DTC, depending on the voltage of the short.
• B0090:93 — Left Frontal Restraints Sensor: No Operation	A fault is indicated when the RCM senses a faulted sensor, a short to voltage on the feed circuit or an open feed or return circuit.
• B0090:96 — Left Frontal Restraints Sensor: Component Internal Failure	A fault is indicated when the RCM senses an incorrect sensor is installed or an internal failure with the LH front severity impact sensor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- LH front impact severity sensor
- RCM

PINPOINT TEST L : DTCS B0090:11, B0090:81, B0090:93, B0090:96

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the Ford Catalog Advantage™ or equivalent to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: Do not probe any impact sensor. The impact sensor cannot be tested using a multi-meter.

L1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0090:11, B0090:81, B0090:93 or B0090:96 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0090:11, GO to L2 . For DTC B0090:81 or B0090:93, GO to L5 . For B0090:96, INSTALL a new LH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to L11 .
No	The fault is intermittent when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to L10 .

L2 CHECK THE LEFT FRONTAL RESTRAINTS SENSOR DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

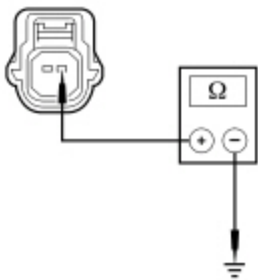
- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: LH Front Impact Severity Sensor [C1465](#).
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Did the on-demand DTC change from B0090:11 to B0090:93?

Yes	INSTALL a new LH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to L11 .
No	GO to L3 .

L3 CHECK THE LH FRONT IMPACT SEVERITY SENSOR FEED CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between LH front impact severity sensor [C1465](#) Pin 1, circuit VR213 (VT/GN), harness side and ground.



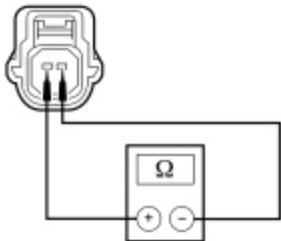
• N0092553

Is the resistance greater than 10,000 ohms?

Yes	GO to L4 .
No	REPAIR circuit VR213 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to L11 .

L4 CHECK FOR A SHORT BETWEEN LH FRONT IMPACT SEVERITY SENSOR CIRCUITS

- Measure the resistance between LH front impact severity sensor [C1465](#) Pin 1, circuit VR213 (VT/GN), harness side and [C1465](#) Pin 2, circuit RR129 (YE/GY), harness side.



• N0036987

Is the resistance greater than 10,000 ohms?

Yes	GO to L9 .
No	REPAIR circuits VR213 (VT/GN) and RR129 (YE/GY). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to L11 .

L5 CHECK THE LH FRONT IMPACT SEVERITY SENSOR AND RCM CONNECTIONS (NO OPERATION INDICATED)

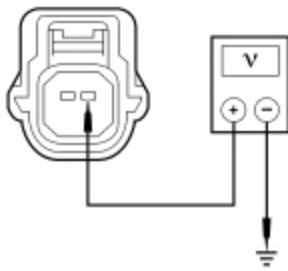
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Inspect LH front impact severity sensor C1465 to be fully seated and locked. Seat and lock connector as necessary.
- Gain access to the ~~RCM~~ and inspect C310A and C310B to be fully seated and locked. Seat and lock the connector(s) as necessary. Refer to Restraints Control Module (RCM) in this section.
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was DTC B0090:81 or B0090:93 retrieved on-demand during self-test?

Yes	GO to L6 .
No	Fault corrected. GO to L11 .

L6 CHECK THE LH FRONT IMPACT SEVERITY SENSOR FEED CIRCUIT FOR A SHORT TO VOLTAGE (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: LH Front Impact Severity Sensor [C1465](#).
- Disconnect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between LH front impact severity sensor:
 - [C1465](#) Pin 1, circuit VR213 (VT/GN), harness side and ground.



• N0106003

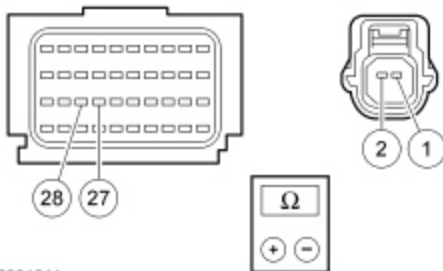
Is any voltage present?

Yes	REPAIR circuit VR213 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to L11 .
No	For DTC B0090:81, INSTALL a new LH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to L11 . For DTC B0090:93, GO to L7 .

L7 CHECK THE LH FRONT IMPACT SEVERITY SENSOR CIRCUITS FOR AN OPEN (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Measure the resistance between ~~RCM~~ C310B, harness side and LH front impact severity sensor C1465, harness side using the following chart.

RCM	Circuit	LH Front Impact Severity Sensor
C310B Pin 27	VR213 (VT/GN)	C1465 Pin 1
C310B Pin 28	RR129 (YE/GY)	C1465 Pin 2



• N0091811

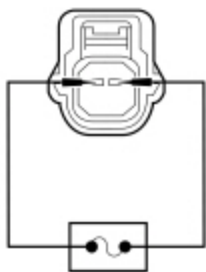
Are the resistances less than 0.5 ohm?

Yes	GO to L8 .
No	REPAIR circuit VR213 (VT/GN) or RR129 (YE/GY). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to L11 .

L8 CHECK THE LEFT FRONTAL RESTRAINTS SENSOR DTC FOR A FAULT STATUS CHANGE (NO OPERATION INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Connect: *RCM* C310A and C310B .
- Connect a fused jumper wire between LH front impact severity sensor [C1465](#) Pin 1, circuit VR213 (WH/BU), harness side and [C1465](#) Pin 2, circuit RR129 (GN/BU), harness side.



- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .

Did the on-demand DTC change from B0090:93 to B0090:11?

Yes	REMOVE the fused jumper wire and INSTALL a new LH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to L11 .
No	REMOVE the fused jumper wire and GO to L9 .

L9 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the *RCM* electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected *SRS* component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect *RCM* C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: *RCM* C310A and C310B .

- Connect: LH Front Impact Severity Sensor C1465 .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to L11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to L10 .

L10 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: LH Front Impact Severity Sensor [C1465](#).
- Disconnect: **RCM** C310A and C310B .
- Inspect for the following:
 - inspect harness and component connectors for loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B **CPA** lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.

Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0090:11, B0090:81, B0090:93 or B0090:96 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0090:11, GO to L2 . For DTC B0090:81 or B0090:93, GO to L5 . For B0090:96, INSTALL a new LH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to L11 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to L11 .

L11 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Reconnect all SRS components (if previously disconnected).
- If previously directed to depower the SRS , repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and OCSM .*

Are any RCM and/or OCSM DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all <u>RCM</u> and <u>OCSM</u> CMDTCs . PROVE OUT the <u>SRS</u> . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test M: DTCs B0091:11, B0091:81, B0091:93, B0091:96, B0097:11, B0097:81, B0097:93, B0097:96

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors the driver front door impact sensor and passenger second row side impact sensor (SuperCab and SuperCrew vehicles only) circuits for the following faults:

- Open circuit
- Short to voltage
- Short to ground
- Faulted driver front door side impact sensor
- Faulted passenger second row side impact sensor (SuperCab and SuperCrew vehicles only)

If a fault is detected, the RCM stores DTC B0091:11, B0091:81, B0091:93, B0091:96, B0097:11, B0097:81, B0097:93 or B0097:96 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

The RCM communicates with each impact sensor, if the RCM is unable to communicate with the sensor, a DTC is stored in memory. A Continuous Memory Diagnostic Trouble Code (CMDTC) **invalid serial data received** (failure type :81) or **no operation** (failure type :93) can be set by a short to voltage on the return circuit of the impact sensor.

NOTE: *A short to ground on either impact sensor feed circuit sets DTCs B0091:11 and B0097:11 on SuperCab and SuperCrew vehicles.*

A short to voltage on either impact sensor feed or return circuit can set DTCs B0091:81 and B0097:81 or B0091:93 and B0097:93 on SuperCab and SuperCrew vehicles, depending on the voltage of the short.

DTC Description	Fault Trigger Condition
• B0091:11 — Left Side Restraints Sensor 1: Circuit Short to Ground	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a shorted driver front door side impact sensor, a shorted passenger second row side impact sensor, a short to ground on the driver front door side impact sensor feed circuit or a short to ground on the passenger second row side impact sensor feed circuit. For regular cab vehicles, a fault is indicated when the RCM senses a shorted driver front door side impact sensor or a short to ground on the driver front door side impact sensor feed circuit.
• B0091:81 — Left Side Restraints Sensor 1: Invalid Serial Data Received	When the RCM receives data that is invalid from the driver front door side impact sensor or the wrong sensor is installed to the vehicle, a fault is indicated. A short to voltage on the feed circuit can also set this DTC, depending on the voltage of the short.
• B0091:93 — Left Side Restraints Sensor 1: No Operation	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a faulted driver front door side impact sensor, a short to voltage on the feed or return circuit of the driver front door side impact sensor, a short to voltage on the feed or return circuit of passenger second row side impact sensor or an open feed or return circuit of the driver front door side impact sensor. For regular cab vehicles, a fault is indicated when the RCM senses a faulted driver front door side impact sensor, a short to voltage on the feed or return circuit of the driver front door side impact sensor or an open feed or return circuit of the driver front door side impact sensor.
• B0091:96 — Left Side Restraints Sensor 1: Component Internal Failure	When the RCM senses a internal failure with the driver front door side impact sensor or if the wrong sensor is installed to the vehicle, a fault is indicated.
• B0097:11 — Right Side Restraints Sensor 2: Circuit Short to Ground	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a shorted driver front door side impact sensor, a shorted passenger second row side impact sensor, a short to ground on the driver front door side impact sensor feed circuit or a short to ground on the passenger second row side impact sensor feed circuit.
• B0097:81 — Right Side Restraints Sensor 2: Invalid Serial Data Received	When the RCM receives data that is invalid from the passenger second row side impact sensor or if a wrong sensor is installed to the vehicle, a fault is indicated. A short to voltage on the feed circuit can also set this DTC, depending on the voltage of the short.
• B0097:93 — Right Side Restraints Sensor 2: No Operation	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a faulted passenger second row side impact sensor, a short to voltage on the feed or return circuit of passenger second row side impact sensor, a short to voltage on the feed or return circuit of the driver front door side impact sensor or an open feed or return circuit of the passenger second row side impact sensor.
• B0097:96 — Right Side Restraints Sensor 2: Component Internal Failure	When the RCM senses a internal failure with the passenger second row side impact sensor or a wrong sensor is installed to the vehicle, a fault is indicated.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver front door side impact sensor
- Passenger second row side impact sensor
- RCM

PINPOINT TEST M : DTCS B0091:11, B0091:81, B0091:93, B0091:96, B0097:11, B0097:81, B0097:93, B0097:96

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the Ford Catalog Advantage™ or equivalent to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: Do not probe any impact sensor. The impact sensor cannot be tested using a multi-meter.

M1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Were any impact sensor DTCs retrieved on-demand during self-test?

Yes	Fault(s) cannot be cleared until corrected and the DTC(s) are no longer retrieved on-demand during self-test. All vehicles For DTC B0097:96, INSTALL a new passenger second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to M16 . For DTC B0091:96, INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to M16 . SuperCab and SuperCrew Vehicles For DTC B0097:11 AND B0091:11, GO to M2 . For DTC B0097:81 AND B0091:81, GO to M6 . For DTC B0097:93 AND B0091:93, GO to M6 . For DTC B0091:93 ONLY , GO to M8 . For DTC B0097:93 ONLY , GO to M11 . For DTC B0097:81 ONLY , INSTALL a new passenger second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to M16 . For DTC B0091:81 ONLY , INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to M16 . Regular Cab Vehicles For DTC B0091:11, GO to M2 . For DTC B0091:81 or B0091:93, GO to M8 .
No	The fault is intermittent when present as a CMDTC only. GO to M15 .

M2 CHECK THE LEFT SIDE RESTRAINTS SENSOR 1 DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Front Door Side Impact Sensor C567 .
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver front door side impact sensor disconnected, an open circuit (no operation) fault is normally retrieved.

For SuperCab and SuperCrew vehicles, did the on-demand DTCs change from B0091:11 and B0097:11 to B0091:93?

For regular cab vehicles, did the on-demand DTC change from B0091:11 to B0091:93?

Yes	INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to M16 .
No	For SuperCab and SuperCrew vehicles, GO to M3 . For regular cab vehicles, GO to M4 .

M3 CHECK THE RIGHT SIDE RESTRAINTS SENSOR 2 DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

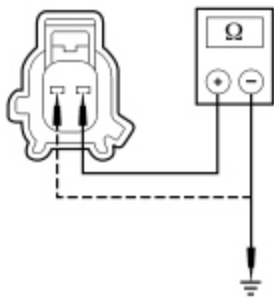
- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Driver Front Door Side Impact Sensor C567 .
- Disconnect: Passenger Second Row Side Impact Sensor C3249 .
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger second row side impact sensor disconnected, an open circuit (no operation) fault is normally retrieved.

Did the on-demand DTC change from B0091:11 and B0097:11 to B0097:93?

Yes	INSTALL a new passenger second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to M16 .
No	GO to M4 .

M4 CHECK THE DRIVER FRONT DOOR SIDE IMPACT SENSOR FEED CIRCUIT FOR A SHORT TO GROUND AND SHORT BETWEEN SENSOR CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Front Door Side Impact Sensor C567 (if not previously disconnected) .
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between driver front door side impact sensor:
 - [C567](#) Pin 1, circuit VR217 (GY/YE), harness side and ground.
 - [C567](#) Pin 1, circuit VR217 (GY/YE), harness side and [C567](#) Pin 2, circuit RR131 (VT/GY), harness side.



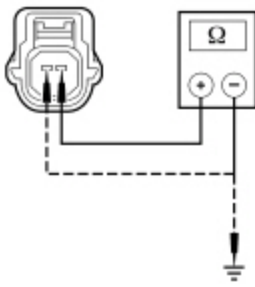
N0105402

Are the resistances greater than 10,000 ohms?

Yes	For SuperCab and SuperCrew vehicles, GO to M5 . For regular cab vehicles, GO to M14 .
No	REPAIR circuit VR217 (GY/YE) and/or RR131 (VT/GY). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to M16 .

M5 CHECK THE PASSENGER SECOND ROW SIDE IMPACT SENSOR FEED CIRCUIT FOR A SHORT TO GROUND AND SHORT BETWEEN SENSOR CIRCUITS

- Measure the resistance between passenger second row side impact sensor:
 - [C3249](#) Pin 1, circuit VR220 (VT/OG), harness side and ground.
 - [C3249](#) Pin 1, circuit VR220 (VT/OG), harness side and [C3249](#) Pin 2, circuit RR134 (BN/BU), harness side.



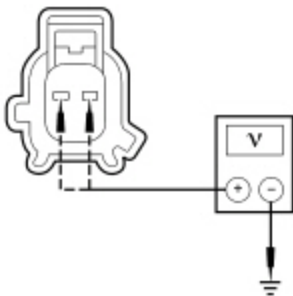
• N0105403

Are the resistances greater than 10,000 ohms?

Yes	GO to M14 .
No	REPAIR circuit VR220 (VT/OG) and/or RR134 (BN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to M16 .

M6 CHECK THE DRIVER FRONT DOOR SIDE IMPACT SENSOR CIRCUITS FOR A SHORT TO VOLTAGE (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B (if not previously disconnected) .
- Disconnect: Driver Front Door Side Impact Sensor C567 (if not previously disconnected) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between driver front door side impact sensor:
 - [C567](#) Pin 1, circuit VR217 (GY/YE), harness side and ground.
 - [C567](#) Pin 2, circuit RR131 (VT/GY), harness side and ground.



• N0052817

Is voltage present on either circuit?

Yes	REPAIR circuit VR217 (GY/YE) or RR131 (VT/GY). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to M16 .
No	SuperCab and SuperCrew vehicles GO to M7 .

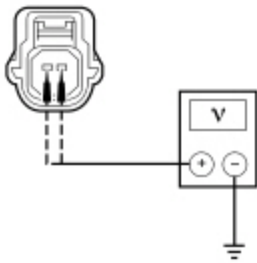
Regular Cab vehicles

For DTC B0091:81, INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to [M16](#).

For DTC B0091:93, GO to [M9](#).

M7 CHECK THE PASSENGER SECOND ROW SIDE IMPACT SENSOR CIRCUITS FOR A SHORT TO VOLTAGE (NO OPERATION INDICATED)

- Ignition OFF.
- Disconnect: Passenger Second Row Side Impact Sensor C3249 .
- Ignition ON.
- Measure the voltage between passenger second row side impact sensor:
 - [C3249](#) Pin 1, circuit VR220 (VT/OG), harness side and ground.
 - [C3249](#) Pin 2, circuit RR134 (BN/BU), harness side and ground.



• N0036990

Is voltage present on either circuit?

Yes	REPAIR circuit VR220 (VT/OG) or RR134 (BN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to M16 .
No	GO to M14 .

M8 CHECK THE DRIVER FRONT DOOR SIDE IMPACT SENSOR AND RCM CONNECTIONS (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Gain access to the driver front door side impact sensor and inspect C567 to be fully seated and locked. Seat and lock the connector as necessary. Refer to Side Impact Sensor — Front Door in this section.
- Gain access to the **RCM** and inspect C310A and C310B to be fully seated and locked. Seat and lock the connector(s) as necessary. Refer to Restraints Control Module (RCM) in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

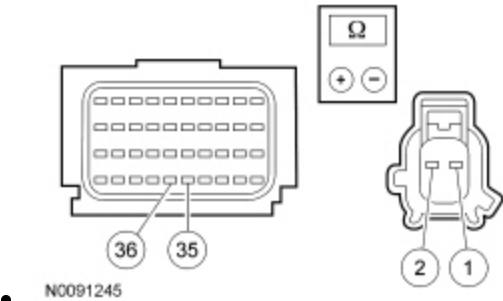
Was DTC B0091:93 retrieved on-demand during self-test?

Yes	For SuperCab and SuperCrew vehicles, GO to M9 . For regular cab vehicles, GO to M6 .
No	Fault corrected. GO to M16 .

M9 CHECK THE DRIVER FRONT DOOR SIDE IMPACT SENSOR CIRCUITS FOR AN OPEN (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B (if not previously disconnected) .
- Disconnect: Driver Front Door Side Impact Sensor C567 (if not previously disconnected) .
- Measure the resistance between ~~RCM~~ C310B, harness side and driver front door side impact sensor C567, harness side using the following chart.

RCM	Circuit	Driver Front Door Side Impact Sensor
C310B Pin 36	VR217 (GY/YE)	C567 Pin 1
C310B Pin 35	RR131 (VT/GY)	C567 Pin 2



Are the resistances less than 0.5 ohm?

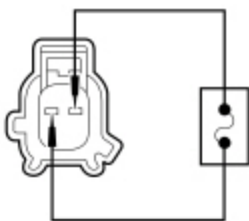
Yes	GO to M10 .
No	REPAIR circuit VR217 (GY/YE) or RR131 (VT/GY). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to M16 .

M10 CHECK THE LEFT SIDE RESTRAINTS SENSOR 1 DTC FOR A FAULT STATUS CHANGE (NO OPERATION INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Connect: ~~RCM~~ C310A and C310B .

- Connect a fused jumper wire between driver front door side impact sensor [C567](#) Pin 1, circuit VR217 (GY/YE), harness side and [C567](#) Pin 2, circuit RR131 (VT/GY), harness side.



• N0109792

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver front door side impact sensor circuits shorted together, a short to ground fault is normally retrieved.

For SuperCab and SuperCrew vehicles, did the DTC change from B0091:93 to B0091:11 and B0097:11?

For regular cab vehicles, did the DTC change from B0091:93 to B0091:11?

Yes	REMOVE the fused jumper wire and INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to M16 .
No	REMOVE the fused jumper wire and GO to M14 .

M11 CHECK THE PASSENGER SECOND ROW SIDE IMPACT SENSOR AND RCM CONNECTIONS (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Gain access to the passenger second row side impact sensor and inspect C3249 to be fully seated and locked. Seat and lock the connector as necessary. Refer to Side Impact Sensor — Second Row in this section.
- Gain access to the **RCM** and inspect C310A and C310B to be fully seated and locked. Seat and lock the connector(s) as necessary. Refer to Restraints Control Module (RCM) in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0097:93 retrieved on-demand during self-test?

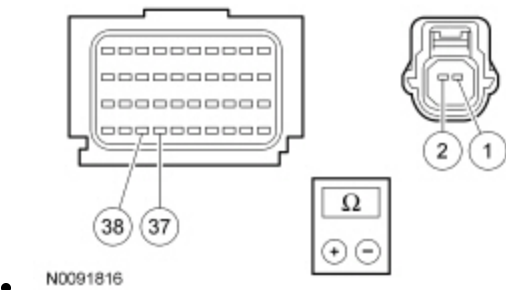
Yes	GO to M12 .
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No	Fault corrected. GO to M16 .
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M12 CHECK THE PASSENGER SECOND ROW SIDE IMPACT SENSOR CIRCUITS FOR AN OPEN (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Disconnect: Passenger Second Row Side Impact Sensor C3249 .
- Measure the resistance between ~~RCM~~ C310B, harness side and passenger second row side impact sensor C3249, harness side using the following chart.

RCM	Circuit	Passenger Second Row Side Impact Sensor
C310B Pin 37	VR220 (VT/OG)	C3249 Pin 1
C310B Pin 38	RR134 (BN/BU)	C3249 Pin 2



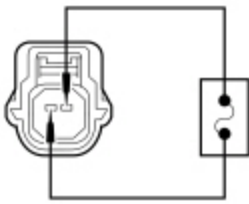
Are the resistances less than 0.5 ohm?

Yes	GO to M13 .
No	REPAIR circuit VR220 (VT/OG) or RR134 (BN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to M16 .

M13 CHECK THE RIGHT SIDE RESTRAINTS SENSOR 2 DTC FOR A FAULT STATUS CHANGE (NO OPERATION INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Connect: ~~RCM~~ C310A and C310B .
- Connect a fused jumper wire between passenger second row side impact sensor [C3249](#) Pin 1, circuit VR220 (VT/OG), harness side and [C3249](#) Pin 2, circuit RR134 (BN/BU), harness side.



• N0109791

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger second row side impact sensor circuits shorted together, a short to ground fault is normally retrieved.

Did the DTC change from B0097:93 to B0097:11 and B0091:11?

Yes	REMOVE the fused jumper wire and INSTALL a new passenger second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to M16 .
No	REMOVE the fused jumper wire and GO to M14 .

M14 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.

Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: **RCM** C310A and C310B .
- Connect: Driver Front Door Side Impact Sensor C567 (if previously disconnected) .
- Connect: Passenger Second Row Side Impact Sensor C3249 (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to M16 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to M15 .

M15 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Affected Impact Sensor Driver Front Door Side Impact Sensor C567 and/or Passenger Second Row Side Impact Sensor C3249 (if equipped) .
- Disconnect: RCM C310A and C310B .
- Inspect for the following:
 - inspect harness and component connectors for loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect RCM C310A and C310B CPA lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Were any impact sensor DTCs retrieved on-demand during self-test?

Yes	Fault(s) cannot be cleared until corrected and the DTC(s) are no longer retrieved on-demand during self-test. All vehicles For DTC B0097:96, INSTALL a new passenger second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to M16. For DTC B0091:96, INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to M16. SuperCab and SuperCrew Vehicles For DTC B0097:11 AND B0091:11, GO to M2. For DTC B0097:81 AND B0091:81, GO to M6. For DTC B0097:93 AND B0091:93, GO to M6. For DTC B0091:93 ONLY, GO to M8. For DTC B0097:93 ONLY, GO to M11. For DTC B0097:81 ONLY, INSTALL a new passenger second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to M16. For DTC B0091:81 ONLY, INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to M16.
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Regular Cab VehiclesFor DTC B0091:11, GO to [M2](#).For DTC B0091:81 or B0091:93, GO to [M8](#).

No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to M16 .
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M16 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF.
- If not previously directed to do so, depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Reconnect all ~~SRS~~ components (if previously disconnected).
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test N: DTCs B0092:11, B0092:81, B0092:93, B0092:96, B0096:11, B0096:81, B0096:93, B0096:96

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors the passenger front door impact sensor, driver second row side impact sensor (SuperCab and SuperCrew vehicles only) circuits for the following faults:

- Open circuit
- Short to voltage
- Short to ground
- Faulted passenger front door side impact sensor
- Faulted driver second row side impact sensor (SuperCab and SuperCrew vehicles only)

If a fault is detected, the ~~RCM~~ stores DTC B0092:11, B0092:81, B0092:93, B0092:96, B0096:11, B0096:81, B0096:93 or B0096:96 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

The ~~RCM~~ communicates with each impact sensor, if the ~~RCM~~ is unable to communicate with the sensor, a DTC is stored in memory. A Continuous Memory Diagnostic Trouble Code (CMDTC) **invalid serial data received** (failure type :81) or **no operation** (failure type :93) can be set by a short to voltage on the return circuit of the impact sensor.

NOTE: A short to ground on either impact sensor feed circuit sets DTCs B0092:11 and B0096:11 on SuperCab and SuperCrew vehicles.

A short to voltage on either impact sensor feed or return circuit can set DTCs B0092:81 and B0096:81 or B0092:93 and B0096:93 on SuperCab and SuperCrew vehicles, depending on the voltage of the short.

DTC Description	Fault Trigger Condition
B0092:11 — Left Side Restraints Sensor 2: Circuit Short to Ground	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a shorted passenger front door side impact sensor, a shorted driver second row side impact sensor, a short to ground on the passenger front door side impact sensor feed circuit or a short to ground on the driver second row side impact sensor feed circuit.
B0092:81 — Left Side Restraints Sensor 2: Invalid Serial Data Received	When the RCM receives data that is invalid from the driver second row side impact sensor or if the wrong sensor is installed in the vehicle, a fault is indicated. A short to voltage on the feed circuit can also set this DTC, depending on the voltage of the short.
B0092:93 — Left Side Restraints Sensor 2: No Operation	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a faulted driver second row side impact sensor, a short to voltage on the feed or return circuit of driver second row side impact sensor, a short to voltage on the feed or return circuit of the passenger front door side impact sensor or an open feed or return circuit of the driver second row side impact sensor.
B0092:96 — Left Side Restraints Sensor 2: Component Internal Failure	When the RCM senses a internal failure with the driver second row side impact sensor or if the wrong sensor is installed to the vehicle, a fault is indicated.
B0096:11 — Right Side Restraints Sensor 1: Circuit Short to Ground	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a shorted passenger front door side impact sensor, a shorted driver second row side impact sensor, a short to ground on the passenger front door side impact sensor feed circuit or a short to ground on the driver second row side impact sensor feed circuit. For regular cab vehicles, a fault is indicated when the RCM senses a shorted passenger front door side impact sensor or a short to ground on the passenger front door side impact sensor feed circuit.
B0096:81 — Right Side Restraints Sensor 1: Invalid Serial Data Received	When the RCM receives data that is invalid from the passenger front door side impact sensor or if the wrong sensor is installed in the vehicle, a fault is indicated. A short to voltage on the feed circuit can also set this DTC, depending on the voltage of the short.
B0096:93 — Right Side Restraints Sensor 1: No Operation	For SuperCab and SuperCrew vehicles, a fault is indicated when the RCM senses a faulted passenger front door side impact sensor, a short to voltage on the feed or return circuit of the passenger front door side impact sensor, a short to voltage on the feed or return circuit of driver second row side impact sensor or an open feed or return circuit of the passenger front door side impact sensor. For regular cab vehicles, a fault is indicated when the RCM senses a faulted passenger front door side impact sensor, a short to voltage on the feed or return circuit of the passenger front door side impact sensor or an open feed or return circuit of the passenger front door side impact sensor.
B0096:96 — Right Side Restraints Sensor 1:	When the RCM senses a internal failure with the passenger front door side impact sensor or if the wrong sensor is installed to the vehicle, a fault is indicated.

DTC Description	Fault Trigger Condition
Component Internal Failure	

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger front door side impact sensor
- Driver second row side impact sensor
- RCM

PINPOINT TEST N : DTCS B0092:11, B0092:81, B0092:93, B0092:96, B0096:11, B0096:81, B0096:93, B0096:96

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: <i>Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .</i>
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the Ford Catalog Advantage™ or equivalent to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: Do not probe any impact sensor. The impact sensor cannot be tested using a multi-meter.
N1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Were any impact sensor DTCs retrieved on-demand during self-test?

Yes	Fault(s) cannot be cleared until corrected and the DTC(s) are no longer retrieved on-demand during self-test. All Vehicles For DTC B0092:96, INSTALL a new driver second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. For DTC B0096:96, INSTALL a new passenger front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. SuperCab and SuperCrew Vehicles For DTC B0092:11 AND B0096:11, GO to N2. For DTC B0092:81 AND B0096:81, GO to N6. For DTC B0092:93 AND B0096:93, GO to N6. For DTC B0092:93 ONLY, GO to N11. For DTC B0096:93 ONLY, GO to N8. For DTC B0092:81 ONLY, INSTALL a new driver second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. For DTC B0096:81 ONLY, INSTALL a new passenger front door side impact sensor. REFER to
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Side Impact Sensor — Front Door in this section.

Regular Cab Vehicles

For DTC B0096:11, GO to [N2](#).

For DTC B0096:81 or B0096:93, GO to [N8](#).

No	The fault is intermittent when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to N15 .
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N2 CHECK THE RIGHT SIDE RESTRAINTS SENSOR 1 DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Front Door Side Impact Sensor C644 .
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger front door side impact sensor disconnected, an open circuit (no operation) fault is normally retrieved.

For SuperCab and SuperCrew vehicles, did the on-demand DTCs change from B0092:11 and B0096:11 to B0096:93?

For regular cab vehicles, did the on-demand DTC change from B0096:11 to B0096:93?

Yes	INSTALL a new passenger front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to N16 .
No	For SuperCab and SuperCrew vehicles, GO to N3 . For regular cab vehicles, GO to N4 .

N3 CHECK THE LEFT SIDE RESTRAINTS SENSOR 2 DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: Passenger Front Door Side Impact Sensor C644 .
- Disconnect: Driver Second Row Side Impact Sensor C3248 .

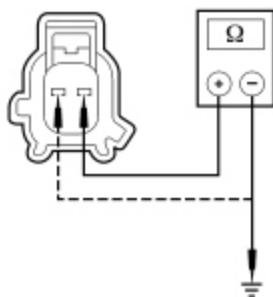
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver second row side impact sensor disconnected, an open circuit (no operation) fault is normally retrieved.

Did the on-demand DTC change from B0092:11 and B0096:11 to B0092:93?

Yes	INSTALL a new driver second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to N16 .
No	GO to N4 .

N4 CHECK THE PASSENGER FRONT DOOR SIDE IMPACT SENSOR FEED CIRCUIT FOR A SHORT TO GROUND AND SHORT BETWEEN SENSOR CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Front Door Side Impact Sensor C644 (if not previously disconnected) .
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between passenger front door side impact sensor:
 - [C644](#) Pin 1, circuit VR218 (YE/OG), harness side and ground.
 - [C644](#) Pin 1, circuit VR218 (YE/OG), harness side and [C644](#) Pin 2, circuit RR132 (BU/WH), harness side.



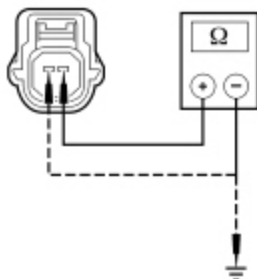
N0106402

Are the resistances greater than 10,000 ohms?

Yes	For SuperCab and SuperCrew vehicles, GO to N5 . For regular cab vehicles, GO to N14 .
No	REPAIR circuit VR218 (YE/OG) and/or RR132 (BU/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to N16 .

N5 CHECK THE DRIVER SECOND ROW SIDE IMPACT SENSOR FEED CIRCUIT FOR A SHORT TO GROUND AND SHORT BETWEEN SENSOR CIRCUITS

- Measure the resistance between driver second row side impact sensor:
 - [C3248](#) Pin 1, circuit VR219 (GN/WH), harness side and ground.
 - [C3248](#) Pin 1, circuit VR219 (GN/WH), harness side and [C3248](#) Pin 2, circuit RR133 (GY/BN), harness side.



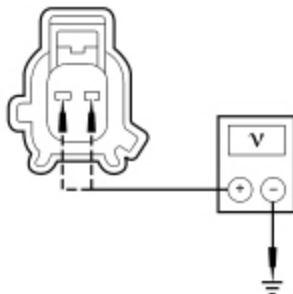
N0105403

Are the resistances greater than 10,000 ohms?

Yes	GO to N14 .
No	REPAIR circuit VR219 (GN/WH) and/or RR133 (GY/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to N16 .

N6 CHECK THE PASSENGER FRONT DOOR SIDE IMPACT SENSOR CIRCUITS FOR A SHORT TO VOLTAGE (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B (if not previously disconnected) .
- Disconnect: Passenger Front Door Side Impact Sensor C644 (if not previously disconnected) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between passenger front door side impact sensor:
 - [C644](#) Pin 1, circuit VR218 (YE/OG), harness side and ground.
 - [C644](#) Pin 2, circuit RR132 (BU/WH), harness side and ground.



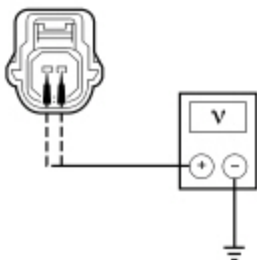
N0052817

Is voltage present on either circuit?

Yes	REPAIR circuit VR218 (YE/OG) or RR132 (BU/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to N16 .
No	SuperCab and SuperCrew vehicles GO to N7 . Regular Cab vehicles For DTC B0096:81, INSTALL a new driver front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to N16 . For DTC B0096:93, GO to N9 .

N7 CHECK THE DRIVER SECOND ROW SIDE IMPACT SENSOR CIRCUITS FOR A SHORT TO VOLTAGE (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Second Row Side Impact Sensor C3248 .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between driver second row side impact sensor:
 - [C3248](#) Pin 1, circuit VR219 (GN/WH), harness side and ground.
 - [C3248](#) Pin 2, circuit RR133 (GY/BN), harness side and ground.



N0036990

Is voltage present on either circuit?

Yes	REPAIR circuit VR219 (GN/WH) or RR133 (GY/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to N16 .
No	GO to N14 .

N8 CHECK THE PASSENGER FRONT DOOR SIDE IMPACT SENSOR AND RCM CONNECTIONS (NO OPERATION INDICATED)

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Gain access to the passenger front door side impact sensor and inspect C644 to be fully seated and locked. Seat and lock the connector as necessary. Refer to Side Impact Sensor — Front Door in this section.
- Gain access to the **RCM** and inspect C310A and C310B to be fully seated and locked. Seat and lock the connector(s) as necessary. Refer to Restraints Control Module (RCM) in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

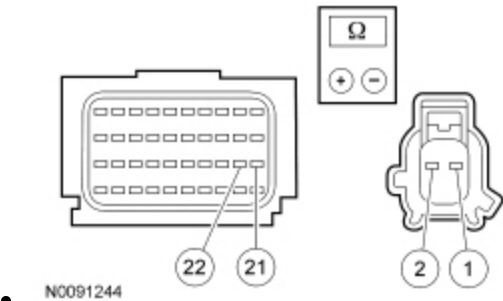
Was DTC B0096:93 retrieved on-demand during self-test?

Yes	For SuperCab and SuperCrew vehicles, GO to N9 . For regular cab vehicles, GO to N6 .
No	Fault corrected. GO to N16 .

N9 CHECK THE PASSENGER FRONT DOOR SIDE IMPACT SENSOR CIRCUITS FOR AN OPEN (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B (if not previously disconnected) .
- Disconnect: Passenger Front Door Side Impact Sensor C644 (if not previously disconnected) .
- Measure the resistance between **RCM** C310B, harness side and passenger front door side impact sensor C644, harness side using the following chart.

RCM	Circuit	Passenger Front Door Side Impact Sensor
C310B Pin 22	VR218 (YE/OG)	C644 Pin 1
C310B Pin 21	RR132 (BU/WH)	C644 Pin 2



Are the resistances less than 0.5 ohm?

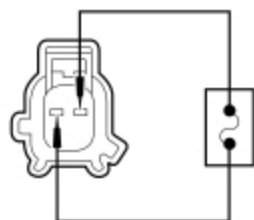
Yes	GO to N10 .
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No	REPAIR circuit VR218 (YE/OG) or RR132 (BU/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to N16 .
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N10 CHECK THE RIGHT SIDE RESTRAINTS SENSOR 1 DTC FOR A FAULT STATUS CHANGE (NO OPERATION INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Connect: *RCM* C310A and C310B .
- Connect a fused jumper wire between passenger front door side impact sensor [C644](#) Pin 1, circuit VR218 (YE/OG), harness side and [C644](#) Pin 2, circuit RR132 (BU/WH), harness side.



- N0109792
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger front door side impact sensor circuits shorted together, a short to ground fault is normally retrieved.

For SuperCab and SuperCrew vehicles, did the DTC change from B0096:93 to B0092:11 and B0096:11?

For regular cab vehicles, did the DTC change from B0096:93 to B0096:11?

Yes	REMOVE the fused jumper wire and INSTALL a new passenger front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section. GO to N16 .
No	REMOVE the fused jumper wire and GO to N14 .

N11 CHECK THE DRIVER SECOND ROW SIDE IMPACT SENSOR AND RCM CONNECTIONS (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Gain access to the driver second row side impact sensor and inspect C3248 to be fully seated and locked. Seat and lock the connector as necessary. Refer to Side Impact Sensor — Second Row in this section.

- Gain access to the **RCM** and inspect C310A and C310B to be fully seated and locked. Seat and lock the connector(s) as necessary. Refer to Restraints Control Module (RCM) in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

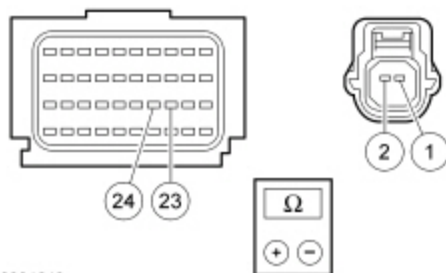
Was DTC B0092:93 retrieved on-demand during self-test?

Yes	GO to N12 .
No	Fault corrected. GO to N16 .

N12 CHECK THE DRIVER SECOND ROW SIDE IMPACT SENSOR CIRCUITS FOR AN OPEN (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- Disconnect: Driver Second Row Side Impact Sensor C3248 .
- Measure the resistance between **RCM** C310B, harness side and driver second row side impact sensor C3248, harness side using the following chart.

RCM	Circuit	Driver Second Row Side Impact Sensor
C310B Pin 23	VR219 (GN/WH)	C3248 Pin 1
C310B Pin 24	RR133 (GY/BN)	C3248 Pin 2



N0091813

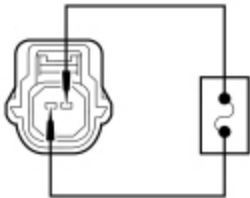
Are the resistances less than 0.5 ohm?

Yes	GO to N13 .
No	REPAIR circuit VR219 (GN/WH) or RR133 (GY/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to N16 .

N13 CHECK THE LEFT SIDE RESTRAINTS SENSOR 2 DTC FOR A FAULT STATUS CHANGE (NO OPERATION INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Connect: *RCM* C310A and C310B .
- Connect a fused jumper wire between driver second row side impact sensor [C3248](#) Pin 1, circuit VR219 (GN/WH), harness side and [C3248](#) Pin 2, circuit RR133 (GY/BN), harness side.



- N0109791
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver second row side impact sensor circuits shorted together, a short to ground fault would normally be retrieved.

Did the DTC change from B0092:93 to B0092:11 and B0096:11?

Yes	REMOVE the fused jumper wire and INSTALL a new driver second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section. GO to N16 .
No	REMOVE the fused jumper wire and GO to N14 .

N14 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the *RCM* electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected *SRS* component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect *RCM* C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.

- repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.

- Connect: ~~RCM~~ C310A and C310B .
- Connect: Passenger Front Door Side Impact Sensor C644 (if previously disconnected) .
- Connect: Driver Second Row Side Impact Sensor C3248 (if previously disconnected) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to N16 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to N15 .

N15 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Affected Impact Sensor Passenger Front Door Side Impact Sensor C644 and/or Driver Second Row Side Impact Sensor C3248 (if equipped) .
- Disconnect: ~~RCM~~ C310A and C310B .
- Inspect for the following:
 - inspect harness and component connectors for loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Were any impact sensor DTCs retrieved on-demand during self-test?

Yes	Fault(s) cannot be cleared until corrected and the DTC(s) are no longer retrieved on-demand during self-test.
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All Vehicles

For DTC B0092:96, INSTALL a new driver second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section.

For DTC B0096:96, INSTALL a new passenger front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section.

SuperCab and SuperCrew Vehicles

For DTC B0092:11 **AND** B0096:11, GO to [N2](#).

For DTC B0092:81 **AND** B0096:81, GO to [N6](#).

For DTC B0092:93 **AND** B0096:93, GO to [N6](#).

For DTC B0092:93 **ONLY**, GO to [N11](#).

For DTC B0096:93 **ONLY**, GO to [N8](#).

For DTC B0092:81 **ONLY**, INSTALL a new driver second row side impact sensor. REFER to Side Impact Sensor — Second Row in this section.

For DTC B0096:81 **ONLY**, INSTALL a new passenger front door side impact sensor. REFER to Side Impact Sensor — Front Door in this section.

Regular Cab Vehicles

For DTC B0096:11, GO to [N2](#).

For DTC B0096:81 or B0096:93, GO to [N8](#).

No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to N16 .
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N16 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF.
- If not previously directed to do so, depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Reconnect all ~~SRS~~ components (if previously disconnected).
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test O: DTCs B0095:11, B0095:81, B0095:93, B0095:96

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors the RH front impact severity sensor circuits for the following faults:

- Open circuit
- Short to voltage
- Short to ground
- Faulted sensor

If a fault is detected, the RCM stores DTC B0095:11, B0095:81, B0095:93 or B0095:96 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Condition
• B0095:11 — Right Frontal Restraints Sensor: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on the RH front severity impact sensor feed circuit.
• B0095:81 — Right Frontal Restraints Sensor: Invalid Serial Data Received	A fault is indicated when the RCM receives data that is invalid from the RH front impact severity sensor or the wrong sensor is installed to the vehicle. A short to voltage on the feed circuit can also set this DTC, depending on the voltage of the short.
• B0095:93 — Right Frontal Restraints Sensor: No Operation	A fault is indicated when the RCM senses a faulted sensor, a short to voltage on the feed circuit or an open feed or return circuit.
• B0095:96 — Right Frontal Restraints Sensor: Component Internal Failure	A fault is indicated when the RCM senses an incorrect sensor is installed or an internal failure with the RH front severity impact sensor.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RH front impact severity sensor
- RCM

PINPOINT TEST O : DTCS B0095:11, B0095:81, B0095:93, B0095:96

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the Ford Catalog Advantage™ or equivalent to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: Do not probe any impact sensor. The impact sensor cannot be tested using a multi-meter.

O1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B0095:11, B0095:81, B0095:93 or B0095:96 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0095:11, GO to O2 . For DTC B0095:81 or B0095:93, GO to O5 . For DTC B0095:96, INSTALL a new RH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to O11 .
No	The fault is intermittent when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to O10 .

O2 CHECK THE RIGHT FRONTAL RESTRAINTS SENSOR DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RH Front Impact Severity Sensor [C1466](#).
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

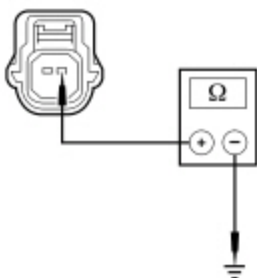
Did the on-demand DTC change from B0095:11 to B0095:93?

Yes	INSTALL a new RH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to O11 .
No	GO to O3 .

O3 CHECK THE RH FRONT IMPACT SEVERITY SENSOR FEED CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RCM C310A and C310B .

- Measure the resistance between RH front impact severity sensor [C1466](#) Pin 1, circuit VR214 (WH/BU), harness side and ground.



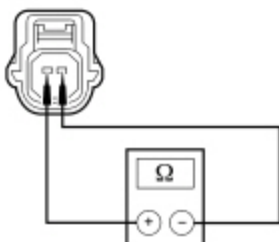
• N0092553

Is the resistance greater than 10,000 ohms?

Yes	GO to O4 .
No	REPAIR circuit VR214 (WH/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to O11 .

O4 CHECK FOR A SHORT BETWEEN RH FRONT IMPACT SEVERITY SENSOR CIRCUITS

- Measure the resistance between RH front impact severity sensor [C1466](#) Pin 1, circuit VR214 (WH/BU), harness side and [C1466](#) Pin 2, circuit RR130 (GN/BU), harness side.



• N0036987

Is the resistance greater than 10,000 ohms?

Yes	GO to O9 .
No	REPAIR circuits VR214 (WH/BU) and RR130 (GN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to O11 .

O5 CHECK THE RH FRONT IMPACT SEVERITY SENSOR AND RCM CONNECTIONS (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Inspect RH front impact severity sensor C1466 to be fully seated and locked. Seat and lock the connector as necessary.

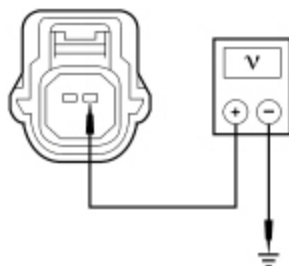
- Gain access to the **RCM** and inspect C310A and C310B to be fully seated and locked. Refer to Restraints Control Module (RCM) in this section.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B0095:81 or B0095:93 retrieved on-demand during self-test?

Yes	GO to O6 .
No	Fault corrected. GO to O11 .

O6 CHECK THE RH FRONT IMPACT SEVERITY SENSOR FEED CIRCUIT FOR A SHORT TO VOLTAGE (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RH Front Impact Severity Sensor [C1466](#).
- Disconnect: **RCM** C310A and C310B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between RH front impact severity sensor:
 - [C1466](#) Pin 1, circuit VR214 (WH/BU), harness side and ground.



• N0106003

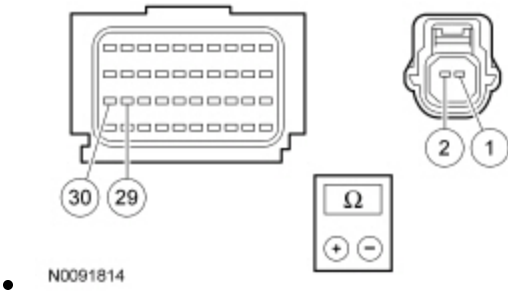
Is any voltage present?

Yes	REPAIR circuit VR214 (WH/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to O11 .
No	For DTC B0095:81, INSTALL a new RH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to O11 . For DTC B0095:93, GO to O7 .

O7 CHECK THE RH FRONT IMPACT SEVERITY SENSOR CIRCUITS FOR AN OPEN (NO OPERATION INDICATED)

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Measure the resistance between RCM C310B, harness side and RH front impact severity sensor C1466, harness side using the following chart.

RCM	Circuit	RH Front Impact Severity Sensor
C310B Pin 30	VR214 (WH/BU)	C1466 Pin 1
C310B Pin 29	RR130 (GN/BU)	C1466 Pin 2



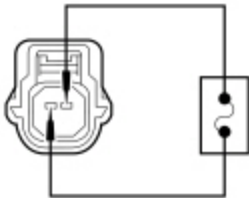
Are the resistances less than 0.5 ohm?

Yes	GO to O8 .
No	REPAIR circuit VR214 (WH/BU) or RR130 (GN/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to O11 .

O8 CHECK THE RIGHT FRONTAL RESTRAINTS SENSOR DTC FOR A FAULT STATUS CHANGE (NO OPERATION INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Connect: RCM C310A and C310B .
- Connect a fused jumper wire between RH front impact severity sensor [C1466](#) Pin 1, circuit VR214 (WH/BU), harness side and [C1466](#) Pin 2, circuit RR130 (GN/BU), harness side.



- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Did the DTC change from B0095:93 to B0095:11?

Yes	REMOVE the fused jumper wire and INSTALL a new RH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to O11 .
No	REMOVE the fused jumper wire and GO to O9 .

O9 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: RH Front Impact Severity Sensor C1466 .
- Connect: **RCM** C310A and C310B (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to O11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to O10 .

O10 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.

- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RH Front Impact Severity Sensor [C1466](#).
- Disconnect: ~~RCM~~ C310A and C310B .
- Inspect for the following:
 - inspect harness and component connectors for loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was DTC B0095:11, B0095:81, B0095:93 or B0095:96 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B0095:11, GO to Q2 . For DTC B0095:81 or B0095:93, GO to Q5 . For DTC B0095:96, INSTALL a new RH front impact severity sensor. REFER to Front Impact Severity Sensor in this section. GO to Q11 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to Q11 .

O11 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF.
- If not previously directed to do so, depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Reconnect all ~~SRS~~ components (if previously disconnected).
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test R: DTCs B00A0:09, B00A0:4A, B00A0:63, B00A0:64 and B00A0:68

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) is in constant communication with various control modules on the High Speed Controller Area Network (HS-CAN) , one of them is the Occupant Classification System Module (OCSM) . The RCM continuously monitors the HS-CAN for fault messages reported by the OCSM . The RCM also checks for the correct identification of the OCSM . If the RCM receives fault message(s) from the OCSM , it stores DTC B00A0:09, B00A0:4A, B00A0:63, B00A0:64 or B00A0:68 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

This pinpoint test is intended to diagnose the following:

- OCSM
- RCM

DTC Description	Fault Trigger Condition
• B00A0:09 — Occupant Classification System: Component Failures	A fault is indicated when the RCM receives a message from the OCSM that a fault exists within the OCS .
• B00A0:4A — Occupant Classification System: Incorrect Component Installed	A fault is indicated when the RCM senses the OCSM is present but not configured.
• B00A0:63 — Occupant Classification System: Circuit / Component Protection Time-Out	A fault is indicated when the RCM receives 8 or more invalid states from the OCSM .
• B00A0:64 — Occupant Classification System: Signal Plausibility Failure	A fault is indicated when the RCM receives messages containing upper or lower data limits other than limits stored in memory.
• B00A0:68 — Occupant Classification System: Event Information	A fault is indicated when the RCM receives a message from the OCSM containing incorrect data.

PINPOINT TEST R : DTCS B00A0:09, B00A0:4A, B00A0:63, B00A0:64 AND B00A0:68

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

R1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B00A0:09, B00A0:4A, B00A0:63, B00A0:64 or B00A0:68 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B00A0:09, B00A0:63 or B00A0:68, RETRIEVE DTCs from the OCSM and REFER to the Occupant Classification System Module (OCSM) DTC Chart for diagnostic direction. For DTC B00A0:4A or B00A0:64, GO to R2 .
No	CHECK for causes of the intermittent fault. Refer to Wiring Diagrams Cell 46 , Supplemental Restraint System for schematic and connector information. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to R5.

R2 VERIFY THE OCSM PART NUMBER

- Enter the following diagnostic mode on the scan tool: Log Viewer .
- Retrieve and record the OCSM part number from Log Viewer.
- Check the part number recorded against the part number listed in the parts catalog.

Is the correct OCSM installed in the vehicle?

Yes	GO to R3 .
No	INSTALL the correct OCSM . REFER to Occupant Classification System Module (OCSM) in this section. GO to R5 .

R3 VERIFY THE RCM PART NUMBER

- Retrieve and record the RCM part number from Log Viewer.

- Check the part number recorded against the part number listed in the parts catalog.

Is the correct **RCM** installed in the vehicle?

Yes	GO to R4 .
No	INSTALL the correct RCM . REFER to Restraints Control Module (RCM) in this section. GO to R5 .

R4 CHECK THE RCM

- Ignition ON.
- Carry out Programmable Module Installation (PMI) for the **RCM** . Manually enter As-Built data. Refer to Section 418-01.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B00A0:4A retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to R5 .
No	Fault corrected. GO to R5 .

R5 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the **RCM** and **QCSM** .*

Are any **RCM** and/or **QCSM** DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test S: DTCs B00B5:11, B00B5:12, B00B5:13 and B00B5:1D

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors the driver seat track position sensor circuits for the following faults:

- Short to ground
- Short to voltage
- Open circuit
- Current out of range
- Faulted driver seat track position sensor

If a fault is detected, the RCM stores DTC B00B5:11, B00B5:12, B00B5:13 or B00B5:1D in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Condition
• B00B5:11 — Driver Seat Track Position Restraints Sensor: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on the driver seat track position sensor circuit.
• B00B5:12 — Driver Seat Track Position Restraints Sensor: Circuit Short to Battery	A fault is indicated when the RCM senses a short to voltage on the driver seat track position sensor circuit.
• B00B5:13 — Driver Seat Track Position Restraints Sensor: Circuit Open	A fault is indicated when the RCM senses an open on the driver seat track position sensor circuit. An open driver seat track position sensor ground circuit sets this fault.
• B00B5:1D — Driver Seat Track Position Restraints Sensor: Circuit Current Out of Range	A fault is indicated when the RCM senses current out of an acceptable range between the driver seat track position sensor circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver seat track position sensor
- RCM

PINPOINT TEST S : DTCS B00B5:11, B00B5:12, B00B5:13 AND B00B5:1D

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

S1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B00B5:11, B00B5:12, B00B5:13 or B00B5:1D retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B00B5:11 or B00B5:1D, GO to S2 . For DTC B00B5:12, GO to S4 . For DTC B00B5:13, GO to S5 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to S10 .

S2 CHECK THE DRIVER SEAT TRACK POSITION RESTRAINTS SENSOR DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND OR CURRENT OUT OF RANGE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the RCM by inducing a different fault condition. If the fault reported changes, this indicates the RCM is functioning correctly and is not the source of the fault.

- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.
- Disconnect: Driver Seat Track Position Sensor C356 .
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver seat track position sensor disconnected, an open circuit fault is normally retrieved.

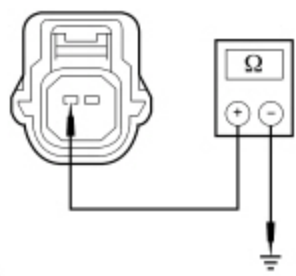
Did the on-demand DTC change from B00B5:11 or B00B5:1D to B00B5:13?

Yes	GO to S8 .
No	For DTC B00B5:1D, GO to S9 . For DTC B00B5:11, GO to S3 .

S3 CHECK THE DRIVER SEAT TRACK POSITION SENSOR CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF.

- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between driver seat track position sensor [C356](#) Pin 2, circuit VR215 (YE/VT), harness side and ground.



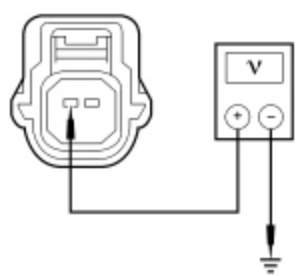
• N0061722

Is the resistance greater than 10,000 ohms?

Yes	GO to S9 .
No	REPAIR circuit VR215 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to S11 .

S4 CHECK THE DRIVER SEAT TRACK POSITION SENSOR CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Seat Side Air Bag C3051 .
- Disconnect: ~~RCM~~ C310A and C310B .
- Disconnect: Driver Seat Track Position Sensor C356 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between driver seat track position sensor [C356](#) Pin 2, circuit VR215 (YE/VT), harness side and ground.



• N0061721

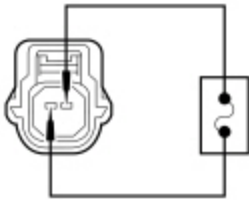
Is any voltage present?

Yes	REPAIR circuit VR215 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to S11 .
No	GO to S9 .

S5 CHECK THE DRIVER SEAT TRACK POSITION RESTRAINTS SENSOR DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Seat Side Air Bag Module C3051 .
- Disconnect: Driver Seat Track Position Sensor C356 .
- Connect a fused jumper wire between driver seat track position sensor [C356](#) Pin 2, circuit VR215 (YE/VT), harness side and [C356](#) Pin 1, circuit RR137 (BU/GY), harness side.



- N0109791
- Repower the *SRS* . **Do not** prove out the system at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *RCM* .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver seat track position sensor circuits shorted together, a short to ground fault is normally retrieved.

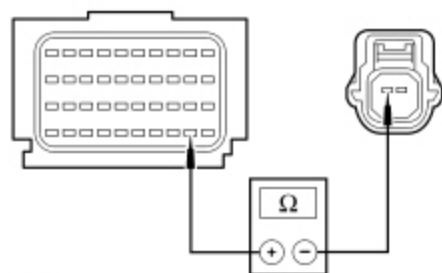
Did the on-demand DTC change from B00B5:13 to B00B5:11?

Yes	REMOVE the fused jumper wire and GO to S8 .
No	REMOVE the fused jumper wire and GO to S6 .

S6 CHECK THE DRIVER SEAT TRACK POSITION SENSOR CIRCUIT FOR AN OPEN

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Disconnect: ~~RCM~~ C310A and C310B .
- Measure the resistance between ~~RCM~~ [C310B](#) Pin 32, circuit VR215 (YE/VT), harness side and driver seat track position sensor [C356](#) Pin 2, circuit VR215 (YE/VT), harness side.



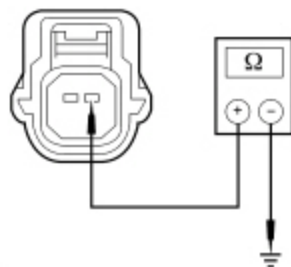
• N0091866

Is the resistance less than 0.5 ohm?

Yes	GO to S7 .
No	REPAIR circuit VR215 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to S11 .

S7 CHECK THE DRIVER SEAT TRACK POSITION SENSOR GROUND CIRCUIT FOR AN OPEN

- Measure the resistance between driver seat track position sensor [C356](#) Pin 1, circuit RR137 (BU/GY) and ground.



• N0061723

Is the resistance less than 5 ohm?

Yes	GO to S9 .
No	REPAIR circuit RR137 (BU/GY)/GD174 (BK/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to S11 .

S8 CONFIRM THE DRIVER SEAT TRACK POSITION SENSOR FAULT

NOTE: Make sure all restraint system components and the ~~RCM~~ electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.

- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: ~~RCM~~ C310A and C310B (if previously disconnected) .
- Connect: Driver Seat Track Position Sensor C356 .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new driver seat track position sensor. REFER to Seat Position Sensor in this section. GO to S11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to S10 .

S9 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the ~~RCM~~ electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect ~~RCM~~ C310A and C310B ~~CPA~~ lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Driver Seat Track Position Sensor C356 .
- Connect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to S11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to S10 .

S10 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: Driver Seat Track Position Sensor [C356](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was **DTC B00B5:11, B00B5:12, B00B5:13 or B00B5:1D** retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B00B5:11 or B00B5:1D, GO to S2 . For DTC B00B5:12, GO to S4 . For DTC B00B5:13, GO to S5 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to S11 .

S11 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE: When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and QCSCM .**

Are any RCM and/or OCSM DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test T: DTCs B00D5:11, B00D5:12 and B00D5:13

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The RCM briefly activates the Passenger Air Bag Deactivation (PAD) indicator to prove-out and verify to the occupants correct functional operation of the PAD indicator.

The RCM monitors the PAD indicator circuits for the following faults:

- Open circuit
- Short to voltage
- Short to ground
- Faulted PAD indicator

If a fault is detected, the RCM stores DTC B00D5:11, B00D5:12 or B00D5:13 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

- DTC B00D5:11 (Restraint System Passenger Disable Indicator: Circuit Short to Ground) — A fault is indicated when the RCM senses a short to ground on the PAD indicator circuit.
- DTC B00D5:12 (Restraint System Passenger Disable Indicator: Circuit Short to Battery) — A fault is indicated when the RCM senses a short to voltage on the PAD indicator circuit.
- DTC B00D5:13 (Restraint System Passenger Disable Indicator: Circuit Open) — A fault is indicated when the RCM senses an open circuit on the PAD indicator. An incorrect RCM configuration or an open ignition circuit to the PAD indicator also sets this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- PAD indicator
- RCM configuration
- RCM

PINPOINT TEST T : DTCS B00D5:11, B00D5:12 AND B00D5:13

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification

before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

T1 RETRIEVE RCM DTCS

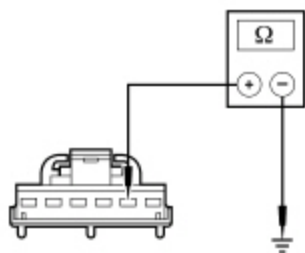
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B00D5:11, B00D5:12 or B00D5:13 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B00D5:11, GO to T2 . For DTC B00D5:13, GO to T4 For DTC B00D5:12, GO to T8 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to T11 .

T2 CHECK THE PAD INDICATOR CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.
- Disconnect: PAD Indicator C2355 .
- Measure the resistance between PAD indicator [C2355](#) Pin 2, circuit RR116 (BN/GN), harness side and ground.



• N0091249

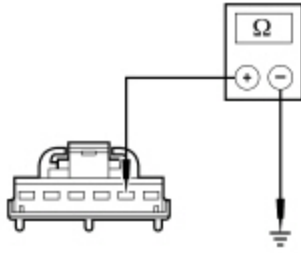
Is the resistance greater than 10,000 ohms?

Yes	GO to T10 .
No	GO to T3 .

T3 CHECK THE RCM FOR LOW RESISTANCE

- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Disconnect: **RCM** C310A and C310B .
- Measure the resistance between **PAD** indicator [C2355](#) Pin 2, circuit RR116 (BN/GN), harness side and ground.



• N0091249

Is the resistance greater than 10,000 ohms?

Yes	GO to T10 .
No	REPAIR circuit RR116 (BN/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to T12 .

T4 CHECK THE RCM CONFIGURATION

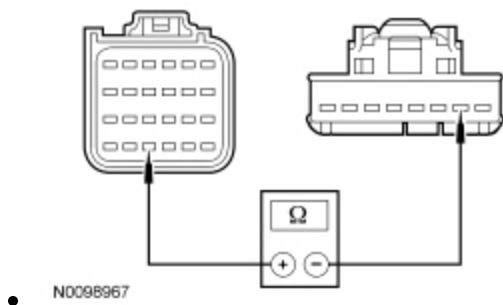
- Carry out **PMI** for the **RCM** . Use As-Built data when carrying out **PMI** .
- Enter the following diagnostic mode on the scan tool: Clear DTCs — **RCM** .
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Is DTC B00D5:13 retrieved on-demand during self-test?

Yes	GO to T5 .
No	The fault is no longer present and was caused by an incorrect RCM configuration. GO to T12 .

T5 CHECK THE PAD INDICATOR CIRCUIT FOR AN OPEN

- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- Disconnect: **PAD** Indicator C2355 .
- Measure the resistance between **PAD** indicator [C2355](#) Pin 2, circuit RR116 (BN/GN) and **RCM** [C310A](#) Pin 22, circuit RR116 (BN/GN), harness side.

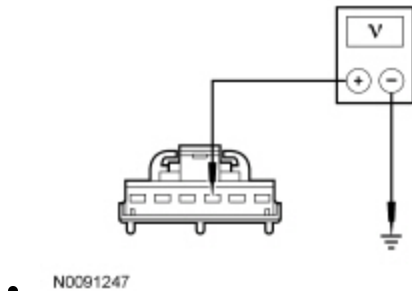


Is the resistance less than 0.5 ohm?

Yes	GO to T6 .
No	REPAIR circuit RR116 (BN/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to T12 .

T6 CHECK THE IGNITION CIRCUIT AT THE PAD INDICATOR FOR VOLTAGE

- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between ~~PAD~~ indicator [C2355](#) Pin 3, circuit CBP41 (BU), harness side and ground.

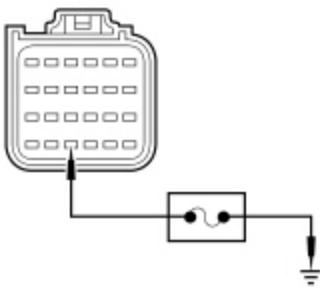


Is the voltage greater than 10 volts?

Yes	GO to T7 .
No	VERIFY Body Control Module (BCM) fuse 41 (7.5A) is OK. If OK, REPAIR circuit CBP41 (BU). If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. GO to T12 .

T7 CHECK THE PAD INDICATOR

- Ignition OFF.
- Connect: ~~PAD~~ Indicator [C2355](#) .
- Connect a fused jumper wire between ~~RCM~~ [C310A](#) Pin 22, circuit RR116 (BN/GN), harness side and ground.



• N0091819

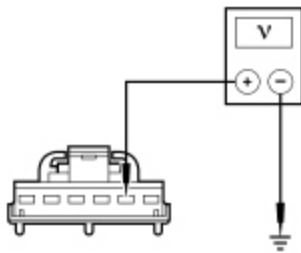
- Monitor the PAD indicator.

Does the PAD indicator illuminate?

Yes	REMOVE the fused jumper wire and GO to T10 .
No	REMOVE the fused jumper wire and INSTALL a new PAD indicator. REFER to Passenger Airbag Deactivation (PAD) Indicator in this section. GO to T12 .

T8 CHECK THE PAD INDICATOR CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RCM C310A and C310B .
- Disconnect: PAD Indicator C2355 .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between PAD indicator [C2355](#) Pin 2, circuit RR116 (BN/GN), harness side and ground.



• N0091252

Is any voltage present?

Yes	REPAIR circuit RR116 (BN/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to T12 .
No	GO to T9 .

T9 CHECK THE RCM

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Connect: **RCM** C310A and C310B .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the **PAD** indicator disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B00D5:12 to B00D5:13?

Yes	INSTALL a new PAD indicator. REFER to Passenger Airbag Deactivation (PAD) Indicator in this section.
No	GO to T10 .

T10 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: **PAD** Indicator C2355 .
- Connect: **RCM** C310A and C310B .
- Make sure the connectors seat and latch correctly.
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.

- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to T12 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to T11 .

T11 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: **PAID** Indicator [C2355](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B00D5:11, B00D5:12 or B00D5:13 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B00D5:11, GO to T2 . For DTC B00D5:13, GO to T4 . For DTC B00D5:12, GO to T8 .
No	CHECK for causes of intermittent open, short to ground or short to voltage on circuit RR116 (BN/GN). ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to T12 .

T12 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.

- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** When selecting Restraints from the Self Test menu, DTCs are retrieved from the *RCM* and *QCSM* .

Are any on-demand *RCM* and/or *QCSM* DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all <i>RCM</i> and <i>QCSM</i> CMDTCs . PROVE OUT the <i>SRS</i> . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test U: DTCs B1211:11, B1211:12, B1211:13 and B1211:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) continuously monitors driver safety belt retractor pretensioner circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted driver safety belt retractor pretensioner

If a fault is detected, the *RCM* sets and stores DTC B1211:11, B1211:12, B1211:13 or B1211:1A in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

The *RCM* analyzes the deployment loop resistance and uses this measurement in combination with an age correction factor to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the *RCM* . If the value displayed is lower or higher than the desired range (refer to diagram below), the *RCM* can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The *RCM* fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses a process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Driver safety belt retractor pretensioner
- *RCM*



DTC Description	Fault Trigger Condition
B1211:11 — Driver Seatbelt Retractor Pretensioner Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either driver safety belt retractor pretensioner circuit.
B1211:12 — Driver Seatbelt Retractor Pretensioner Deployment Control: Short to Battery	A fault is indicated when the RCM senses a short to battery on either driver safety belt retractor pretensioner circuit.
B1211:13 — Driver Seatbelt Retractor Pretensioner Deployment Control: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between the driver safety belt retractor pretensioner circuits.
B1211:1A — Driver Seatbelt Retractor Pretensioner Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between the driver safety belt retractor pretensioner circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Driver safety belt retractor pretensioner
- RCM

PINPOINT TEST U : DTCS B1211:11, B1211:12, B1211:13 AND B1211:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

U1 RETRIEVE RCM DTCS

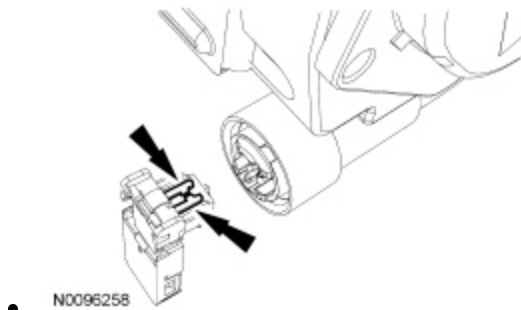
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B1211:11, B1211:12, B1211:13 or B1211:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test. For regular cab and Crew Cab vehicles with DTC B1211:13 or B1211:1A, GO to U2 . For SuperCab vehicles with DTC B1211:13 or B1211:1A, GO to U3 . For DTC B1211:11, GO to U9 . For DTC B1211:12, GO to U11 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B1211:13 or B1211:1A, GO to U14 . For DTC B1211:11, GO to U15 . For DTC B1211:12, GO to U16 .

U2 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER ELECTRICAL CONNECTOR

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Gain access to the safety belt retractor pretensioner and inspect for the following:
 - the Connector Position Assurance (CPA) tabs are not broken or the clip is not damaged.
 - the **CPA** tab is fully and correctly seated.



Is the **CPA clip installed correctly, not damaged or broken and fully seated?**

Yes	GO to U3 .
No	If the CPA clip is damaged or broken, INSTALL a pigtail connector for the pretensioner using pigtail kit 8U2Z-14S411-YA. If the CPA clip is not correctly installed or seated, REPAIR as necessary. GO to U18 .

U3 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL RESISTANCE (DEPLOY_23_R) PID

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — RCM .
 - Monitor and record PID DEPLOY_23_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to U13 .
No	GO to U4 .

U4 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL RESISTANCE (DEPLOY_23_R) PID WHILE CARRYING OUT A HARNESS TEST

- While monitoring the DEPLOY_23_R PID, carry out a harness test of the driver safety belt retractor pretensioner circuits and accessible connectors (including any in-line connectors), by wiggling connectors and flexing the wire harness.

Does the PID value read between 1.7 and 2.78 ohms while carrying out a harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to U18 .
No	For PID value less than 1.7 ohms, GO to U5 . For PID value greater than 2.78 ohms, GO to U7 .

U5 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Belt Retractor Pretensioner C3201 (regular cab and Crew Cab) or C727 (SuperCab) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver safety belt retractor pretensioner disconnected, an open circuit fault is normally retrieved.

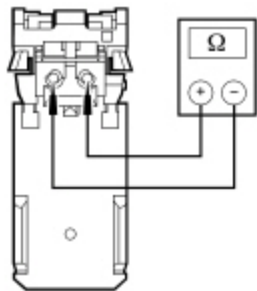
Did the on-demand DTC change from B1211:1A to a B1211:13?

Yes	GO to U12 .
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No	GO to U6 .
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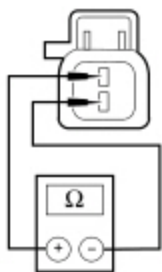
U6 CHECK FOR A SHORT BETWEEN DRIVER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- For **regular cab and Crew Cab**, measure the resistance between the driver safety belt retractor pretensioner [C3201](#) Pin 1, circuit CR146 (GY/VT), harness side and [C3201](#) Pin 2, circuit RR146 (BU/BN), harness side.



• N0061724

- For **SuperCab**, measure the resistance between the driver safety belt retractor pretensioner [C727](#) Pin 1, circuit CR146 (GY/VT), harness side and [C727](#) Pin 2, circuit RR146 (BU/BN), harness side.



• A0029887

Is the resistance greater than 10,000 ohms?

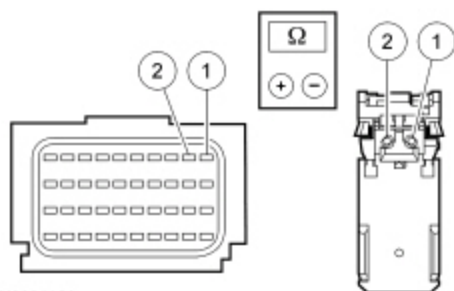
Yes	GO to U13 .
No	REPAIR circuits CR146 (GY/VT) and RR146 (BU/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to U18 .

U7 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Belt Retractor Pretensioner C3201 (regular cab and Crew Cab) or C727 (SuperCab) .
- Disconnect: ~~RCM~~ C310A and C310B .

- For **regular cab and Crew Cab**, measure the resistance between **RCM** , harness side and driver safety belt retractor pretensioner C3201, harness side using the following chart.

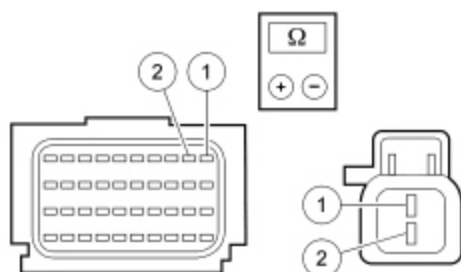
RCM	Circuit	Driver Safety Belt Retractor Pretensioner
C310B Pin 2	CR146 (GY/VT)	C3201 Pin 1
C310B Pin 1	RR146 (BU/BN)	C3201 Pin 2



N0091182

- For **SuperCab**, measure the resistance between **RCM** , harness side and driver safety belt retractor pretensioner C727, harness side using the following chart.

RCM	Circuit	Driver Safety Belt Retractor Pretensioner
C310B Pin 2	CR146 (GY/VT)	C727 Pin 1
C310B Pin 1	RR146 (BU/BN)	C727 Pin 2



N0092067

Are the resistances less than 0.5 ohm?

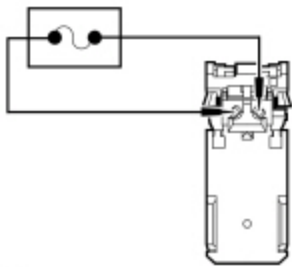
Yes	GO to U8 .
No	REPAIR circuit CR146 (GY/VT) or RR146 (BU/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to U18 .

U8 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the

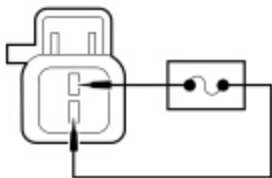
fault.

- Connect: ~~RCM~~ C310A and C310B .
- For **regular cab and Crew Cab**, connect a fused jumper wire between the driver safety belt retractor pretensioner [C3201](#) Pin 1, circuit CR146 (GY/VT), harness side and [C3201](#) Pin 2, circuit RR146 (BU/BN), harness side.



• N0119865

- For **SuperCab**, connect a fused jumper wire between the driver safety belt retractor pretensioner [C727](#) Pin 1, circuit CR146 (GY/VT), harness side and [C727](#) Pin 2, circuit RR146 (BU/BN), harness side.



• N0109770

- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver safety belt retractor pretensioner circuits shorted together, a low resistance fault is normally retrieved.

Did the on-demand DTC change from B1211:13 to B1211:1A?

Yes	REMOVE the fused jumper wire and GO to U12 .
No	REMOVE the fused jumper wire and GO to U13 .

U9 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

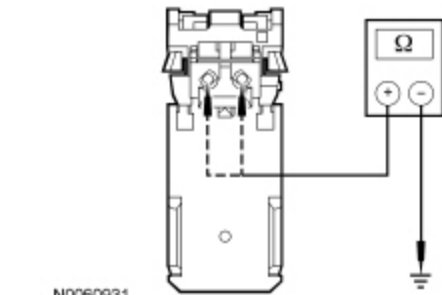
- Disconnect: Driver Safety Belt Retractor Pretensioner C3201 (regular cab and Crew Cab) or C727 (SuperCab) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the driver safety belt retractor pretensioner disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B1211:11 to B1211:13?

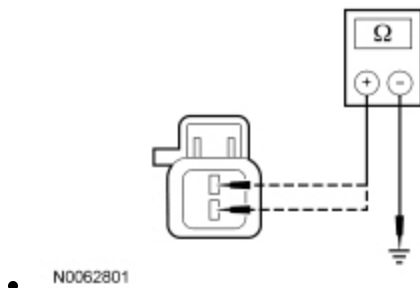
Yes	GO to U12 .
No	GO to U10 .

U10 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- For **regular cab and Crew Cab**, measure the resistance between the driver safety belt retractor pretensioner:
 - [C3201](#) Pin 1, circuit CR146 (GY/VT), harness side and ground.
 - [C3201](#) Pin 2, circuit RR146 (BU/BN), harness side and ground.



- For **SuperCab**, measure the resistance between the driver safety belt retractor pretensioner:
 - [C727](#) Pin 1, circuit CR146 (GY/VT), harness side and ground.
 - [C727](#) Pin 2, circuit RR146 (BU/BN), harness side and ground.

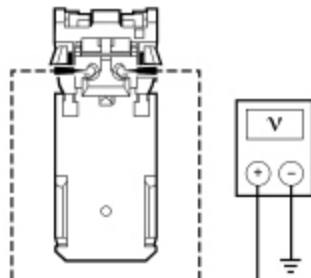


Are the resistances greater than 10,000 ohms?

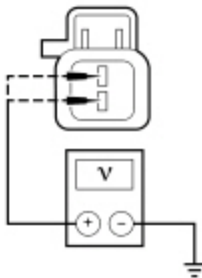
Yes	GO to U13 .
No	REPAIR circuit CR146 (GY/VT) or RR146 (BU/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to U18 .

U11 CHECK THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Belt Retractor Pretensioner C3201 (regular cab and Crew Cab) or C727 (SuperCab) .
- Disconnect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- For **regular cab and Crew Cab**, measure the voltage between the driver safety belt retractor pretensioner:
 - [C3201](#) Pin 1, circuit CR146 (GY/VT), harness side and ground.
 - [C3201](#) Pin 2, circuit RR146 (BU/BN), harness side and ground.



- For **SuperCab**, measure the voltage between the driver safety belt retractor pretensioner:
 - [C727](#) Pin 1, circuit CR146 (GY/VT), harness side and ground.
 - [C727](#) Pin 2, circuit RR146 (BU/BN), harness side and ground.



Is voltage present on either circuit?

Yes	REPAIR circuit CR146 (GY/VT) or RR146 (BU/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to U18 .
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U12 CONFIRM THE DRIVER SAFETY BELT RETRACTOR PRETENSIONER FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Driver Safety Belt Retractor Pretensioner C3201 (regular cab and Crew Cab) or C727 (SuperCab) .
- Connect: **RCM** C310A and C310B (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new driver safety belt retractor pretensioner. REFER to Section 501-20A. GO to U18 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B1211:13 or B1211:1A, GO to U14. For DTC B1211:11, GO to U15. For DTC B1211:12, GO to U16.

U13 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.

- inspect C310A and C310B CPA lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
- Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.

- Connect: Driver Safety Belt Retractor Pretensioner C3201 (regular cab and Crew Cab) or C727 (SuperCab) .
- Connect: RCM C310A and C310B (if previously disconnected) .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to U18 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B1211:13 or B1211:1A, GO to U14. For DTC B1211:11, GO to U15. For DTC B1211:12, GO to U16.

U14 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL RESISTANCE (DEPLOY_23_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — RCM .
 - Monitor and record PID DEPLOY_23_R.
- Attempt to recreate the fault by wiggling the connectors (including any in-line connectors) and flexing the wire harness.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to U17 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to U18.

U15 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness.

- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B1211:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to U18 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to U17 .

U16 CHECK THE DRIVER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO VOLTAGE FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Belt Retractor Pretensioner C3201 (regular cab and Crew Cab) or C727 (SuperCab) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B1211:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to U18 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to U17 .

U17 CHECK HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Driver Safety Belt Retractor Pretensioner [C3201](#) (regular cab and Crew Cab) or [C727](#) (SuperCab) .
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to U18 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to U18 .

U18 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all SRS components (if previously disconnected).
- If previously directed to depower the SRS , repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE: When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and QCSM .**

Are any RCM and/or QCSM DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test V: DTCs B1214:11, B1214:12, B1214:13 and B1214:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) continuously monitors the passenger safety belt retractor pretensioner circuits for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted passenger safety belt retractor

If a fault is detected, the RCM stores DTC B1214:11, B1214:12, B1214:13 or B1214:1A in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

The RCM analyzes the deployment loop resistance and uses this measurement in combination with an age correction factor to determine if a fault exists. The value displayed in the PID is the deployment loop resistance as measured by the RCM . If

the value displayed is lower or higher than the desired range (refer to diagram below), the RCM can set a DTC. As the deployment loop resistance drifts farther outside the desired range, the chance for a DTC increases. The RCM fault trigger threshold may vary as the vehicle ages. Small variations in resistance can occur due to the effect of road vibrations on terminal fit and/or temperature changes. These variables can result in an intermittent fault. For this reason, the test requires the PID value to be within the desired range before the fault is considered repaired, regardless if the module is reporting an on-demand DTC at the time of diagnosis. Following this direction helps make sure that minor changes in resistance do not create a repeat concern. This test uses a process of elimination to diagnose each part of the deployment loop circuit including:

- Wiring
- Connections
- Passenger safety belt retractor pretensioner
- RCM



DTC Description	Fault Trigger Condition
• B1214:11 — Passenger Seatbelt Retractor Pretensioner Deployment Control: Circuit Short to Ground	A fault is indicated when the RCM senses a short to ground on either passenger safety belt retractor pretensioner circuit.
• B1214:12 — Passenger Seatbelt Retractor Pretensioner Deployment Control: Short to Battery	A fault is indicated when the RCM senses a short to battery on either passenger safety belt retractor pretensioner circuit.
• B1214:13 — Passenger Seatbelt Retractor Pretensioner Deployment Control: Circuit Open	A fault is indicated when the RCM measures greater than the desired resistance range between the passenger safety belt retractor pretensioner circuits.
• B1214:1A — Passenger Seatbelt Retractor Pretensioner Deployment Control: Circuit Resistance Below Threshold	A fault is indicated when the RCM measures less than the desired resistance range between the passenger safety belt retractor pretensioner circuits.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger safety belt retractor pretensioner
- RCM

PINPOINT TEST V : DTCS B1214:11, B1214:12, B1214:13 AND B1214:1A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.

WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

V1 RETRIEVE RCM DTCS

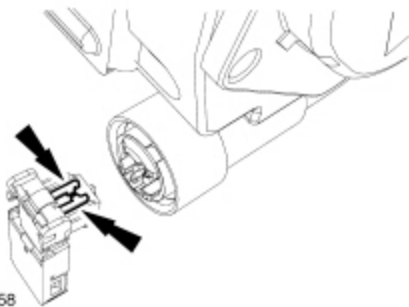
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B1214:11, B1214:12, B1214:13 or B1214:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test. For regular cab and Crew Cab vehicles with DTC B1214:13 or B1214:1A, GO to V2 . For SuperCab vehicles with DTC B1214:13 or B1214:1A, GO to V3 . For DTC B1214:11, GO to V9 . For DTC B1214:12, GO to V11 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. For DTC B1214:13 or B1214:1A, GO to V14 . For DTC B1214:11, GO to V15 . For DTC B1214:12, GO to V16 .

V2 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER ELECTRICAL CONNECTOR

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Gain access to the safety belt retractor pretensioner and inspect for the following:
 - the Connector Position Assurance (CPA) tabs are not broken or the clip is not damaged.
 - the CPA tab is fully and correctly seated.



Is the ~~CPA~~ clip installed correctly, not damaged or broken and fully seated?

Yes	GO to V3 .
No	If the CPA clip is damaged or broken, INSTALL a pigtail connector for the pretensioner using pigtail kit 8U2Z-14S411-YA. If the CPA clip is not correctly installed or seated, REPAIR as necessary. GO to V18 .

V3 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL RESISTANCE (DEPLOY_24_R) PID

- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — ~~RCM~~ .
 - Monitor and record PID DEPLOY_24_R.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	GO to V13 .
No	GO to V4 .

V4 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL RESISTANCE (DEPLOY_24_R) PID WHILE CARRYING OUT A HARNESS TEST

- While monitoring the DEPLOY_24_R PID, carry out a harness test of the passenger safety belt retractor pretensioner circuits and accessible connectors (including any in-line connectors), by wiggling connectors and flexing the wire harness.

Does the PID value read between 1.7 and 2.78 ohms while carrying out a harness test?

Yes	DEPOWER the SRS and REPAIR the connector, terminals or wire harness as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to V18 .
No	For PID value less than 1.7 ohms, GO to V5 . For PID value greater than 2.78 ohms, GO to V7 .

V5 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (LOW RESISTANCE INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the ~~RCM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~RCM~~ is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

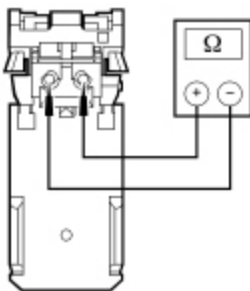
- Disconnect: Passenger Safety Belt Retractor Pretensioner C3202 (regular cab and Crew Cab) or C830 (SuperCab) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger safety belt retractor pretensioner disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B1214:1A to B1214:13?

Yes	GO to V12 .
No	GO to V6 .

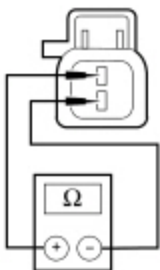
V6 CHECK FOR A SHORT BETWEEN PASSENGER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- For **regular cab and Crew Cab**, measure the resistance between the passenger safety belt retractor pretensioner [C3202](#) Pin 1, circuit CR147 (GN/OG), harness side and [C3202](#) Pin 2, circuit RR147 (YE/VT), harness side.



N0061724

- For **SuperCab**, measure the resistance between the passenger safety belt retractor pretensioner [C830](#) Pin 1, circuit CR147 (GN/OG), harness side and [C830](#) Pin 2, circuit RR147 (YE/VT), harness side.



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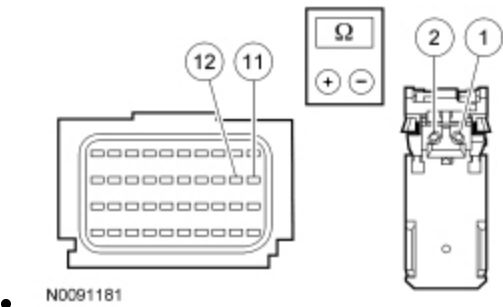
Is the resistance greater than 10,000 ohms?

Yes	GO to V13 .
No	REPAIR circuits CR147 (GN/OG) and RR147 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to V18 .

V7 CHECK THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS FOR AN OPEN

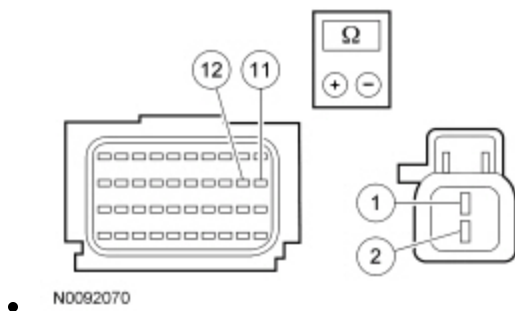
- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C3202 (regular cab and Crew Cab) or C830 (SuperCab) .
- Disconnect: ~~RCM~~ C310A and C310B .
- For **regular cab and Crew Cab**, measure the resistance between ~~RCM~~ C310B, and passenger safety belt retractor pretensioner C3202, harness side using the following chart.

RCM	Circuit	Passenger Safety Belt Retractor Pretensioner
C310B Pin 11	RR147 (YE/VT)	C3202 Pin 2
C310B Pin 12	CR147 (GN/OG)	C3202 Pin 1



- For **SuperCab**, measure the resistance between ~~RCM~~ C310B, and passenger safety belt retractor pretensioner C830, harness side using the following chart.

RCM	Circuit	Passenger Safety Belt Retractor Pretensioner
C310B Pin 11	RR147 (YE/VT)	C830 Pin 2
C310B Pin 12	CR147 (GN/OG)	C830 Pin 1



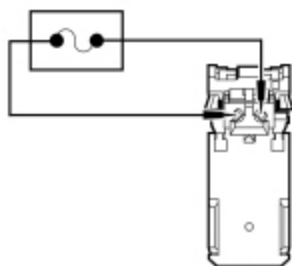
Are the resistances less than 0.5 ohm?

Yes	GO to V8 .
No	REPAIR circuit CR147 (GN/OG) or RR147 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to V18 .

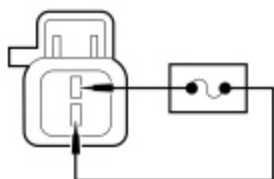
V8 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *RCM* by inducing a different fault condition. If the fault reported changes, this indicates the *RCM* is functioning correctly and is not the source of the fault.

- Connect: *RCM* C310A and C310B .
- For **regular cab or Crew Cab**, connect a fused jumper wire between the passenger safety belt retractor pretensioner [C3202](#) Pin 1, circuit CR147 (GN/OG), harness side and [C3202](#) Pin 2, circuit RR147 (YE/VT), harness side.



- For **SuperCab**, connect a fused jumper wire between the passenger safety belt retractor pretensioner [C830](#) Pin 1, circuit CR147 (GN/OG), harness side and [C830](#) Pin 2, circuit RR147 (YE/VT), harness side.



- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger safety belt retractor pretensioner circuits shorted together, a low resistance fault would normally be retrieved.

Did the on-demand DTC change from B1214:13 to B1214:1A?

Yes	REMOVE the fused jumper wire and GO to V12 .
No	REMOVE the fused jumper wire and GO to V13 .

V9 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL DTC FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: *This pinpoint test step attempts to change the fault reported by the **RCM** by inducing a different fault condition. If the fault reported changes, this indicates the **RCM** is functioning correctly and is not the source of the fault.*

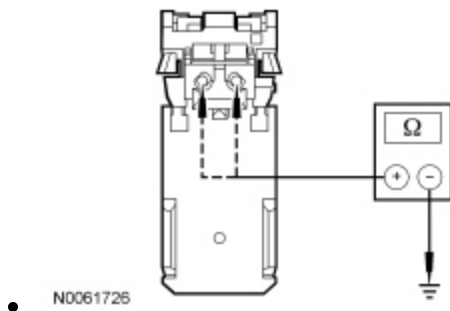
- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C3202 (regular cab and Crew Cab) or C830 (SuperCab) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the passenger safety belt retractor pretensioner disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B1214:11 to B1214:13?

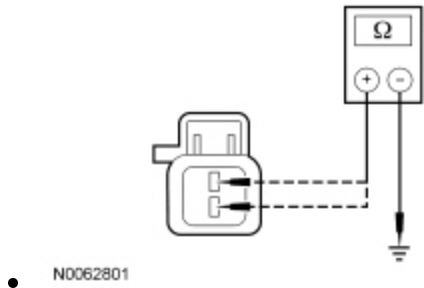
Yes	GO to V12 .
No	GO to V10 .

V10 CHECK THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** C310A and C310B .
- For **regular cab and Crew Cab**, measure the resistance between the passenger safety belt retractor pretensioner:
 - [C3202](#) Pin 1, circuit CR147 (GN/OG), harness side and ground.
 - [C3202](#) Pin 2, circuit RR147 (YE/VT), harness side and ground.



- For **SuperCab**, measure the resistance between the passenger safety belt retractor pretensioner:
 - [C830](#) Pin 1, circuit CR147 (GN/OG), harness side and ground.
 - [C830](#) Pin 2, circuit RR147 (YE/VT), harness side and ground.

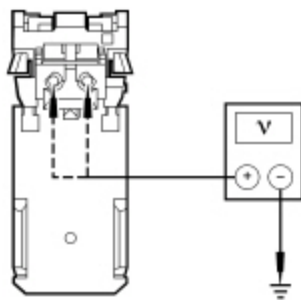


Are the resistances greater than 10,000 ohms?

Yes	GO to V13 .
No	REPAIR circuit CR147 (GN/OG) or RR147 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to V18 .

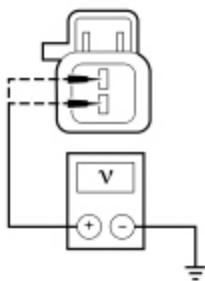
V11 CHECK THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C3202 (regular cab and Crew Cab) or C830 (SuperCab) .
- Disconnect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- For **regular cab and Crew Cab**, measure the voltage between the passenger safety belt retractor pretensioner:
 - [C3202](#) Pin 1, circuit CR147 (GN/OG), harness side and ground.
 - [C3202](#) Pin 2, circuit RR147 (YE/VT), harness side and ground.



• N0061727

- For **SuperCab**, measure the voltage between the passenger safety belt retractor pretensioner:
 - [C830](#) Pin 1, circuit CR147 (GN/OG), harness side and ground.
 - [C830](#) Pin 2, circuit RR147 (YE/VT), harness side and ground.



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Is voltage present on either circuit?

Yes	REPAIR circuit CR147 (GN/OG) or RR147 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to V18 .
No	GO to V13 .

V12 CONFIRM THE PASSENGER SAFETY BELT RETRACTOR PRETENSIONER FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Passenger Safety Belt Retractor Pretensioner C3202 (regular cab and Crew Cab) or C830 (SuperCab) .
- Connect: **RCM** C310A and C310B (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new passenger safety belt retractor pretensioner. REFER to Section 501-20A. GO to V18 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B1214:13 or B1214:1A, GO to V14 . For DTC B1214:11, GO to V15 . For DTC B1214:12, GO to V16 .

V13 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect C310A and C310B **CPA** lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: Passenger Safety Belt Retractor Pretensioner C3202 (regular cab and Crew Cab) or C830 (SuperCab) .
- Connect: **RCM** C310A and C310B (if previously disconnected) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to V18 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. For DTC B1214:13 or B1214:1A, GO to V14 . For DTC B1214:11, GO to V15 . For DTC B1214:12, GO to V16 .

V14 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL RESISTANCE (DEPLOY_24_R) PID FOR AN INTERMITTENT LOW RESISTANCE OR OPEN CIRCUIT FAULT

- Enter the following diagnostic mode on the scan tool: DataLogger — RCM .
 - Monitor and record PID DEPLOY_24_R.
- Attempt to recreate the fault by wiggling the connectors (including any in-line connectors) and flexing the wire harness.

Does the PID value read between 1.7 and 2.78 ohms?

Yes	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to V17 .
No	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to V18 .

V15 CHECK THE PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL FOR AN INTERMITTENT SHORT TO GROUND FAULT

- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B1214:11 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to V18 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to V17 .

V16 CHECK FOR AN INTERMITTENT PASSENGER SEATBELT RETRACTOR PRETENSIONER DEPLOYMENT CONTROL SHORT TO VOLTAGE FAULT

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Belt Retractor Pretensioner C3202 (regular cab and Crew Cab) or C830 (SuperCab) .

- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Attempt to recreate the fault by wiggling connectors (including any in-line connectors) and flexing the wire harness.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B1214:12 retrieved on-demand during self-test?

Yes	DEPOWER the SRS and REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to V18 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to V17 .

V17 CHECK HARNESS AND CONNECTORS

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Safety Belt Retractor Pretensioner [C3202](#) (regular cab and Crew Cab) or [C830](#) (SuperCab) .
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.

Were any concerns found?

Yes	REPAIR as necessary. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to V18 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to V18 .

V18 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.

- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** When selecting *Restraints from the Self Test menu*, DTCs are retrieved from the *RCM* and *QCSM* .

Are any **RCM** and/or **QCSM** DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test W: DTCs B0001:2B, B0002:2B, B0010:2B, B0011:2B, B0020:2B, B0021:2B, B0028:2B, B0029:2B, B0050:2B, B0052:2B, B00B5:2B, B1211:2B and B1214:2B

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors all deployable device and sensor circuits for a signal cross coupled (short) between circuits. If the **RCM** detects a short between the circuits of one device and another device, it stores a DTC for each of the signal cross coupled faults in memory. The **RCM** sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator. When a normal loop fault is present (one loop is either shorted to battery/ground, open circuit or low resistance), then signal cross coupled diagnostics are not active. Once the normal loop fault is repaired, then the signal cross coupled diagnostics resume.

DTC Description	Fault Trigger Condition
• B0001:2B — Driver Frontal Stage 1 Deployment Control: Signal Cross Coupled	If the RCM detects a short between the driver air bag stage 1 circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
• B0002:2B — Driver Frontal Stage 2 Deployment Control: Signal Cross Coupled	If the RCM detects a short between the driver air bag stage 2 circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
• B0010:2B — Passenger Frontal Stage 1 Deployment Control: Signal Cross Coupled	If the RCM detects a short between the passenger air bag stage 1 circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
• B0011:2B — Passenger Frontal Stage 2 Deployment Control: Signal Cross Coupled	If the RCM detects a short between the passenger air bag stage 2 circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
• B0020:2B — Left Side Airbag Deployment Control: Signal Cross Coupled	If the RCM detects a short between the driver seat side air bag circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.

DTC Description	Fault Trigger Condition
B0021:2B — Left Curtain Deployment Control: Signal Cross Coupled	If the RCM detects a short between the driver Safety Canopy® circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
B0028:2B — Right Side Airbag Deployment Control: Signal Cross Coupled	If the RCM detects a short between the passenger seat side air bag circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
B0029:2B — Right Curtain Deployment Control: Signal Cross Coupled	If the RCM detects a short between the passenger Safety Canopy® circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
B0050:2B — Driver Seatbelt Sensor: Signal Cross Coupled	If the RCM detects a short between the driver safety belt buckle switch circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
B0052:2B — Passenger Seatbelt Sensor: Signal Cross Coupled	If the RCM detects a short between the passenger safety belt buckle switch circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
B00B5:2B — Driver Seat Track Position Restraints Sensor: Signal Cross Coupled	If the RCM detects a short between the driver seat track position sensor circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
B1211:2B — Driver Seatbelt Retractor Pretensioner Deployment Control: Signal Cross Coupled	If the RCM detects a short between the driver safety belt buckle pretensioner circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.
B1214:2B — Passenger Seatbelt Retractor Pretensioner Deployment Control: Signal Cross Coupled	If the RCM detects a short between the passenger safety belt buckle pretensioner circuits and the circuits of another SRS component, it sets this DTC and the DTC of the corresponding SRS component.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RCM

PINPOINT TEST W : DTCS B0001:2B, B0002:2B, B0010:2B, B0011:2B, B0020:2B, B0021:2B, B0028:2B, B0029:2B, B0050:2B, B0052:2B, B00B5:2B, B1211:2B AND B1214:2B

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: The Supplemental Restraint System (SRS) must be fully operational and free of faults before releasing the vehicle to the customer.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

W1 RETRIEVE RCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM . **NOTE:** Signal cross coupled DTCs are retrieved only in pairs.

Were any signal cross coupled DTCs retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. GO to W2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to W4 .

W2 CHECK THE AFFECTED DEPLOYABLE CIRCUITS FOR A SIGNAL CROSS COUPLED FAULT

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RCM C310A and C310B .
- Disconnect ALL of the affected SRS devices.
- **NOTE:** Most signal cross coupled faults are due to connector and/or wiring harness damage. Carry out a thorough inspection of connector(s) and/or wiring harness for damage.

Using the following table, measure resistance between the circuits of the affected SRS devices.

Signal Cross Coupled Fault Table

Device/Loop (DTC)	Connector	Circuit
Driver air bag module stage 1 (B0001:2B)	• Driver air bag module stage 1 electrical connector	• RR101 (YE/GN) • CR101 (VT/BN)
Driver air bag module stage 2 (B0002:2B)	• Driver air bag module stage 2 electrical connector	• RR102 (WH) • CR102 (BU)
Passenger air bag module stage 1 (B0010:2B)	• C256A Pin 2 • C256A Pin 1	• CR103 (GY/BU) • RR103 (VT/GN)
Passenger air bag module stage 2 (B0011:2B)	• C256B Pin 2 • C256B Pin 1	• CR104 (YE/GY) • RR104 (WH/BU)

Device/Loop (DTC)	Connector	Circuit
Driver seat side air bag module (B0020:2B)	• C3051 Pin 1 • C3051 Pin 2	• CR105 (GN/BN) • RR105 (GY/YE)
Driver Safety Canopy® module (B0021:2B)	• C3321 Pin 1 • C3321 Pin 2	• CR109 (BN/BU) • RR109 (BU/GN)
Passenger seat side air bag module (B0028:2B)	• C339 Pin 1 • C339 Pin 2	• CR106 (VT/GY) • RR106 (YE/OG)
Passenger Safety Canopy® module (B0029:2B)	• C3329 Pin 1 • C3329 Pin 2	• CR111 (BN/WH) • RR111 (YE/VT)
Driver safety belt buckle switch (B0050:2B)	• C3065 Pin 3	• CR201 (GN/BU)
Passenger safety belt buckle switch (B0052:2B)	• C3066 Pin 3	• CR203 (GY/VT)
Driver seat track position sensor (B00B5:2B)	• C356 Pin 2	• VR215 (YE/VT)
Driver safety belt retractor pretensioner (B1211:2B)	• Regular cab and Crew Cab C3201 Pin 1/ SuperCab C727 Pin 1 • Regular cab and Crew Cab C3201 Pin 2/ SuperCab C727 Pin 2	• CR146 (GY/VT) • RR146 (BU/BN)
Passenger safety belt retractor pretensioner (B1214:2B)	• Regular cab and Crew Cab C3202 Pin 1/ SuperCab C830 Pin 1 • Regular cab and Crew Cab C3202 Pin 2/ SuperCab C830 Pin 2	• CR147 (GN/OG) • RR147 (YE/VT)

Are the resistances greater than 10,000 ohms between the affected circuits?

Yes	GO to W3 .
No	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to W5 .

W3 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the *RCM* electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Prior to reconnecting any previously disconnected *SR* component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.

- inspect C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
- repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.

- Connect: All Previously Disconnected Restraint System Component(s) .
- Connect: ~~RCM~~ C310A and C310B .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Were any signal cross coupled DTCs retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to W5 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to W4 .

W4 CHECK FOR AN INTERMITTENT FAULT

NOTE: *Most cross link faults are due to connector and/or wiring harness damage. Carry out a thorough inspection of connector(s) and/or wiring harness for damage.*

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect **ALL of the affected**~~SRS~~ devices.
- Inspect the wiring harness(es) for any type of damage. Repair as necessary.
Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~RCM~~ .

Were any signal cross coupled DTCs retrieved on-demand during self-test?

Yes	The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test. GO to W2 .
No	The fault is not present and cannot be recreated at this time. CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to W5 .

W5 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**

- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the **RCM** and **OCSM** .*

Are any **RCM** and/or**OCSM** DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM DTCs. PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test X: DTCs B0001:4A, B0002:4A, B0005:4A, B0010:4A, B0011:4A, B0020:4A, B0021:4A, B0028:4A, B0029:4A, B0050:4A, B0052:4A, B0080:4A, B0082:4A, B0090:4A, B0091:4A, B0092:4A, B0095:4A, B0096:4A, B0097:4A, B00B5:4A, B00C5:4A, B00D5:4A, B1212:4A, B1213:4A and U3000:4A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) monitors all Supplemental Restraint System (SRS) device connections. If the **RCM** detects a connection or device connected that is not stored in memory or is not configured for the component, the **RCM** sets a DTC and sends a message to the Instrument Panel Cluster (IPC) to illuminate air bag warning indicator.

DTC Description	Fault Trigger Condition
• B0001:4A — Driver Frontal Stage 1 Deployment Control: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
• B0002:4A — Driver Frontal Stage 2 Deployment Control: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
• B0005:4A — Collapsible Steering Column Deployment Control: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
• B0010:4A — Passenger Frontal Stage 1 Deployment Control: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.

DTC Description	Fault Trigger Condition
B0011:4A — Passenger Frontal Stage 2 Deployment Control: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0020:4A — Left Side Airbag Deployment Control: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0021:4A — Left Curtain Deployment Control: Signal Cross Coupled	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0028:4A — Right Side Airbag Deployment Control: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0029:4A — Right Curtain Deployment Control: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0050:4A — Driver Seatbelt Sensor: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0052:4A — Passenger Seatbelt Sensor: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0080:4A — Driver Seatbelt Load Limiter Deployment Control: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0082:4A — Passenger Seatbelt Load Limiter Deployment Control: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0090:4A — Left Frontal Restraints Sensor: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0091:4A — Left Side Restraints Sensor 1: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0092:4A — Left Side Restraints Sensor 2: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0095:4A — Right Frontal Restraints Sensor: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0096:4A — Right Side Restraints Sensor 1: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.
B0097:4A — Right Side Restraints Sensor 2: Incorrect Component Installed	This DTC sets if the <u>RCM</u> detects the device but it is not configured for this component.

DTC Description	Fault Trigger Condition
B00B5:4A — Driver Seat Track Position Restraints Sensor: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
B00C5:4A — Passenger Seat Track Position Restraints Sensor: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
B00D5:4A — Restraint System Passenger Disable Indicator: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
B1212:4A — Driver Seatbelt Buckle Pretensioner Deployment Control: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
B1213:4A — Passenger Seatbelt Buckle Pretensioner Deployment Control: Incorrect Component Installed	This DTC sets if the RCM detects the device but it is not configured for this component.
U3000:4A — Control Module: Incorrect Component Installed	If the RCM detects the incorrect vehicle identification or model year identification from the Occupant Classification System Module (OCSM) , PCM or ABS module, this DTC sets.

This pinpoint test is intended to diagnose the following:

- Incorrect ABS module
- Incorrect ~~OCSM~~
- Incorrect ~~RCM~~
- ~~RCM~~ not configured

PINPOINT TEST X : DTCS B0001:4A, B0002:4A, B0005:4A, B0010:4A, B0011:4A, B0020:4A, B0021:4A, B0028:4A, B0029:4A, B0050:4A, B0052:4A, B0080:4A, B0082:4A, B0090:4A, B0091:4A, B0092:4A, B0095:4A, B0096:4A, B0097:4A, B00B5:4A, B00C5:4A, B00D5:4A, B1212:4A, B1213:4A AND U3000:4A

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: <i>Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.</i>
NOTE: <i>Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .</i>

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

X1 RETRIEVE RCM DTCs

- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Were any incorrect component installed DTCs retrieved?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during self-test. GO to X2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to X7 .

X2 CHECK THE VEHICLE HISTORY

- Ignition OFF.
- Check the vehicle repair history.

Has a new RCM or OCSM been installed during this or a prior repair?

Yes	GO to X3 .
No	GO to X4 .

X3 CHECK THE RCM AND OCSM VEHICLE IDENTIFICATION PIDS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — RCM . View and record the following PID values:
 - V_ID_14229_RCM
 - OCS_MY
- Enter the following diagnostic mode on the scan tool: DataLogger — OCSM . View and record the following PID values:
 - V_ID_14229_OCS
 - MY_14229
- Verify the correct vehicle application.

Do the PID values match the vehicle application?

Yes	GO to X4 .
No	INSTALL the correct <u>RCM</u> or <u>OCSM</u> . REFER to Restraints Control Module (RCM) or Occupant Classification System Module (OCSM) in this section. GO to X8 .

X4 VERIFY THE ABS MODULE PART NUMBER

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Log Viewer .
- Retrieve and record the ABS module part number from Log Viewer.
- Check the part number recorded against the part number listed In the Ford Catalog Advantage™ or equivalent.

Is the correct ABS module installed in the vehicle?

Yes	GO to X5 .
No	INSTALL the correct ABS module. REFER to the ABS Module Removal and Installation in Section 206-09. GO to X8 .

X5 CHECK THE RCM

- Ignition ON.
- Carry out Programmable Module Installation (PMI) for the RCM . Use As-Built data when carrying out PMI . Refer to Section 418-01.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Were any incorrect component installed DTCs retrieved on-demand during self-test?

Yes	GO to X6 .
No	Fault corrected. GO to X8 .

X6 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components and the RCM electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Prior to reconnecting any previously disconnected SRS component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect RCM C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - repair any concerns found.
Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Were any incorrect component installed DTCs retrieved?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to X8 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to X7 .

X7 CHECK FOR AN INTERMITTENT FAULT

- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Were any DTCs retrieved?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to X8 .
No	CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to X8 .

X8 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the **RCM** and **QCSM** .*

Are any **RCM** and/or **QCSM** DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test Y: DTCs U0028:08 and U0028:88

Refer to Wiring Diagrams Cell [42](#) , Vehicle Dynamic Systems for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) supplies the stability/traction control system with the yaw rate, roll rate, lateral and longitudinal accelerometer over a dedicated Controller Area Network (CAN) . If the RCM detects a fault with communications with the ABS module or dedicated CAN circuits, it sets DTC U0028:08 or U0028:88.

The RCM does not request the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator if either DTC U0028:08 or U0028:88 is set. The air bag warning indicator remains off if either of these DTCs are set.

- DTC U0028:08 (Vehicle Communication Bus A: Bus Signal / Message Failures) — When the RCM does not receive dedicated CAN bus messages from the ABS module for 5 seconds, a fault is indicated. An open dedicated CAN circuit, incorrect installed ABS module or loss of power and/or ground to the ABS module can set this DTC.
- DTC U0028:88 (Vehicle Communication Bus A: Bus Off) — When the RCM detects the dedicated CAN circuits are shorted to ground, shorted to voltage or shorted to each other, a fault is indicated.

The ABS module may set similar DTCs, U0151 (Lost Communication With Restraints Control Module) and U0452 (Invalid Data Received From Restraints Control Module).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RCM
- ABS module

PINPOINT TEST Y : DTCS U0028:08 AND U0028:88

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: <i>Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.</i>
NOTE: <i>Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .</i>
NOTE: <i>Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.</i>
NOTE: <i>The SRS must be fully operational and free of faults before releasing the vehicle to the customer.</i>

Y1 RETRIEVE RCM DTCS
• Ignition ON. • Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC U0028:08 or U0028:88 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the self-test. For DTC U0028:08, GO to Y2 . For DTC U0028:88, GO to Y6 .
------------	---

No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to Y13 .
-----------	---

Y2 VERIFY THE RCM PART NUMBER

- Enter the following diagnostic mode on the scan tool: Log Viewer .
- Retrieve and record the RCM part number from Log Viewer.
- Check the part number recorded against the part number listed in the Ford Catalog AdvantageTM or equivalent.

Is the correct RCM installed in the vehicle?

Yes	GO to Y3 .
No	INSTALL the correct <u>RCM</u> . REFER to Restraints Control Module (RCM) in this section. GO to Y14 .

Y3 VERIFY THE ABS MODULE PART NUMBER

- Retrieve and record the ABS module part number from Log Viewer.
- Check the part number recorded against the part number listed in the Ford Catalog AdvantageTM or equivalent.

Is the correct ABS module installed in the vehicle?

Yes	GO to Y4 .
No	INSTALL the correct ABS module. REFER to the Removal and Installation procedure in Section 206-09. GO to Y14 .

Y4 CHECK THE HS-CAN NETWORK COMMUNICATIONS (ABS MODULE COMMUNICATIONS)

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Network Test . **NOTE:** *The Network Test does not test the dedicated CAN communications between the ABS module and RCM .*

Did the Network Test pass (ABS module communicates with the scan tool)?

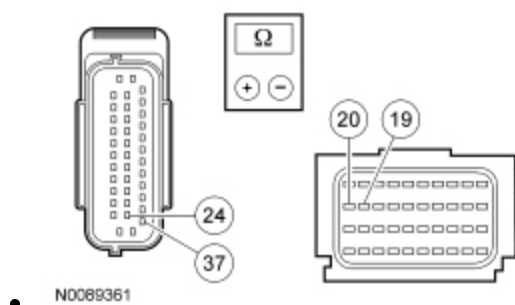
Yes	GO to Y5 .
No	REFER to Section 418-00 to diagnose the High Speed Controller Area Network (HS-CAN) / ABS module communication concern. GO to Y14 .

Y5 CHECK THE DEDICATED CAN CIRCUITS FOR AN OPEN

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RCM C310A and C310B .

- Disconnect: ABS Module C135 .
- Measure the resistance between the ABS module C135, harness side and the RCM C310B, harness side as indicated in the following chart:

ABS Module	Circuit	RCM
C135 Pin 37	VCA23 (BU/WH)	C310B Pin 20
C135 Pin 24	VCA24 (GN/OG)	C310B Pin 19

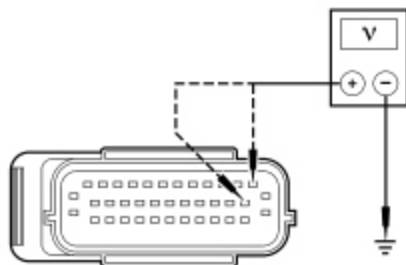


Are the resistances less than 3 ohms?

Yes	GO to Y9 .
No	REPAIR circuits VCA24 (GN/OG) and/or VCA23 (BU/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to Y14 .

Y6 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~RCM~~ C310A and C310B .
- Disconnect: ABS Module C135 .
- Repower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between ABS module:
 - [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side and ground.
 - [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side and ground.



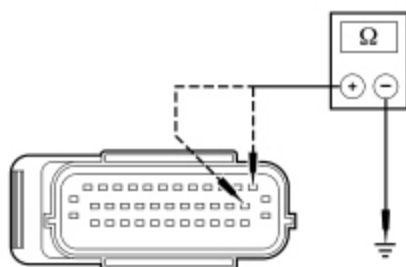
N0089358

Is voltage present on either circuit?

Yes	REPAIR circuit VCA24 (GN/OG) or VCA23 (BU/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to Y14 .
No	GO to Y7 .

Y7 CHECK THE DEDICATED CAN CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between ABS module:
 - [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side and ground.
 - [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side and ground.



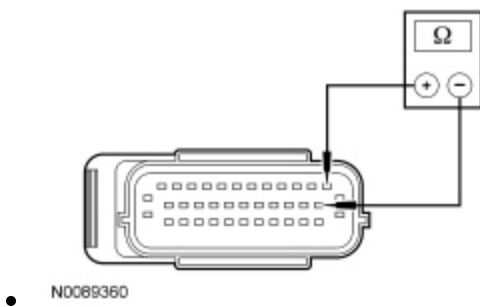
N0089359

Are the resistances greater than 10,000 ohms?

Yes	GO to Y8 .
No	REPAIR circuit VCA24 (GN/OG) or VCA23 (BU/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to Y14 .

Y8 CHECK FOR SHORT BETWEEN THE DEDICATED CAN CIRCUITS

- Measure the resistance between ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side and [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.

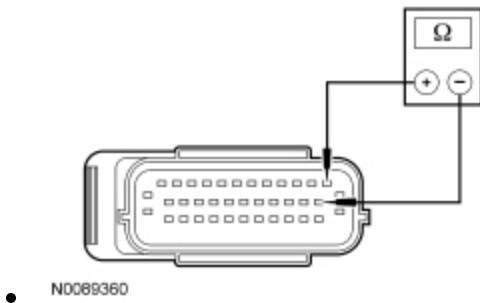


Is the resistance greater than 10,000 ohms?

Yes	GO to Y9 .
No	REPAIR circuits VCA24 (GN/OG) and VCA23 (BU/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to Y14 .

Y9 CHECK THE RCM TERMINATING RESISTOR

- Connect: ~~RCM~~ C310B .
- Measure the resistance between ABS module [C135](#) Pin 37, circuit VCA23 (BU/WH), harness side and [C135](#) Pin 24, circuit VCA24 (GN/OG), harness side.

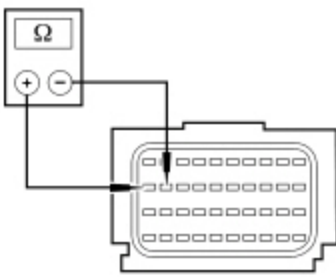


Is the resistance between 108 and 132 ohms?

Yes	GO to Y10 .
No	GO to Y11 .

Y10 CHECK ABS MODULE TERMINATING RESISTOR

- Disconnect: ~~RCM~~ C310B .
- Connect: ABS Module C135 .
- Measure the resistance between ~~RCM~~ [C310B](#) Pin 20, circuit VCA23 (BU/WH), harness side and [C310B](#) Pin 19, circuit VCA24 (GN/OG), harness side.



N0093009

Is the resistance between 108 and 132 ohms?

Yes	For DTC U0028:08, GO to Y11 . For DTC U0028:88, GO to Y12 .
No	GO to Y12 .

Y11 CONFIRM THE RCM FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- If previously directed to repower the **SRS**, depower the **SRS**. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B Connector Position Assurance (CPA) lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - inspect ABS module C135 for dirt, corrosion, water intrusion, pushed out, loose or spread terminals and loose or frayed wire connections at terminals.
 - repair any concerns found.

Refer to Wiring Diagrams Cell [5](#), Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s).
- Repower the **SRS**. **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM**.

Was DTC U0028:08 retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to Y14 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to Y13 .

Y12 CONFIRM THE ABS MODULE FAULT

NOTE: Make sure all restraint system components, sensor electrical connectors and the **RCM** electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- If previously directed to repower the **SRS** , depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - inspect **RCM** C310A and C310B **CPA** lever/lock for correct operation. Refer to Restraints Control Module (RCM) in this section.
 - inspect ABS module C135 for dirt, corrosion, water intrusion, pushed out, loose or spread terminals and loose or frayed wire connections at terminals.
 - repair any concerns found.Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC U0028:88 retrieved on-demand during self-test?

Yes	INSTALL a new ABS module. REFER to the Removal and Installation procedure in Section 206-09. GO to Y14 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to Y13 .

Y13 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** [C310B](#) .
- Disconnect: ABS module [C135](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC U0028:08 or U0028:88 retrieved on-demand during self-test?

Yes	The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test. CHECK for causes of the intermittent fault in the areas previously worked in, particularly the pins and terminals of any electrical connectors that were disconnected. REPAIR any intermittent wiring, terminal or connector concerns found. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to Y14.
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to Y14.

Y14 CHECK FOR ADDITIONAL RCM AND ABS MODULE DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **SRS** components (if previously disconnected).
- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .
- Enter the following diagnostic mode on the scan tool: Self Test — **ABS Module** .

Were any DTCs retrieved on-demand from the **RCM or ABS module during self-test?**

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the appropriate DTC Chart in this section and/or Section 206-09 for pinpoint test direction.
No	CLEAR all CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test Z: DTCs U3003:16 and U3003:17

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

NOTE: DTCs B1317, B1318, U3003:16 and U3003:17 can be set if the vehicle has been recently jump started, the battery has been recently charged or the battery has been discharged. The battery may become discharged due to excessive load(s) on the charging system from aftermarket accessories or if the battery has been left unattended with the accessories on.

Normal Operation — Restraints Control Module (RCM)

The Restraints Control Module (RCM) continuously monitors the input voltage for correct operation. If the RCM detects input voltage below 9.5 volts, it stores DTC U3003:16 in memory. If the RCM detects input voltage above 16 volts, it stores DTC U3003:17 in memory. If the RCM sets DTC U3003:16 or U3003:17, it sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

Normal Operation — Occupant Classification System Module (OCSM)

The Occupant Classification System Module (OCSM) continuously monitors the input voltage for correct operation. If the OCSM detects input voltage below 8 volts, it stores DTC U3003:16 in memory. If the OCSM detects input voltage above 17.2 volts, it stores DTC U3003:17 in memory. If the OCSM sets DTC U3003:16 or U3003:17, it sends a fault message to the RCM . The RCM then sends a message to the IPC to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Conditions
U3003:16 — Battery Voltage: Circuit Voltage Below Threshold	If the RCM detects voltage below 9.5 volts with a Supplemental Restraint System (SRS) fault, it sets this DTC.
U3003:17 — Battery Voltage: Circuit Voltage Above Threshold	If the RCM detects voltage above 16 volts with a SRS fault or above 20 volts without a SRS fault, it sets this DTC.
U3003:16 — Battery Voltage: Circuit Voltage Below Threshold	If the OCSM detects voltage below 9.6 volts when another Occupant Classification System (OCS) fault is present, or if the OCSM detects voltage below 7.6 volts, regardless of fault status, it sets this DTC.
U3003:17 — Battery Voltage: Circuit Voltage Above Threshold	If the OCSM detects voltage above 16.5 volts when another Occupant Classification System (OCS) fault is present, or if the OCSM detects voltage above 18.5 volts, regardless of fault status, it sets this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- RCM
- OCSM

PINPOINT TEST Z : DTCS U3003:16 AND U3003:17

WARNING: Always tighten the fasteners of the restraints control module (RCM) and impact sensor (if equipped) to the specified torque. Failure to do so may result in incorrect restraint system operation, which increases the risk of personal injury or death in a crash.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: SRS components should only be disconnected or reconnected when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the

correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

Z1 RETRIEVE ALL CMDTCS IN ALL MODULES

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ALL CMDTCS (Continuous Memory) .

Is DTC U3003:16 or U3003:17 present in one or more modules AND P0563, P0620, P0625, P0626 or P065B retrieved from the PCM?

Yes	REFER to Section 414-00 to diagnose the charging system.
No	GO to Z2 .

Z2 TEST BATTERY CONDITION

- Ignition OFF.
- Carry out the battery condition test. Refer to Section 414-01.

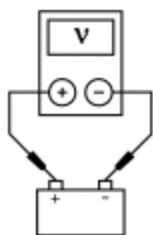
Did the battery pass the condition test?

Yes	If the battery passed the condition test but required a recharge, REFER to Section 414-00 to diagnose the charging system. CLEAR all Continuous Memory Diagnostic Trouble Codes (CMDTCS) . TEST the system for normal operation. If the battery passed the condition test and did not require a recharge, GO to Z3 .
No	INSTALL a new battery. CLEAR all CMDTCS . TEST the SRS for normal operation.

Z3 CHECK THE CHARGING SYSTEM VOLTAGE

NOTE: Do not allow the engine speed to increase above 2,000 rpm while performing this step or the generator may self excite and result in default charging system output voltage. If engine speed goes above 2,000 rpm, shut the vehicle OFF and restart the engine before performing this step.

- Measure the voltage of the battery:
 - For DTC U3003:17, turn off all accessories and run the engine at 1,500 rpm for a minimum of 2 minutes while measuring the battery voltage.
 - For DTC U3003:16, turn on headlights and HVAC fan on high and run engine at 1,500 rpm for a minimum of 2 minutes while measuring the battery voltage.



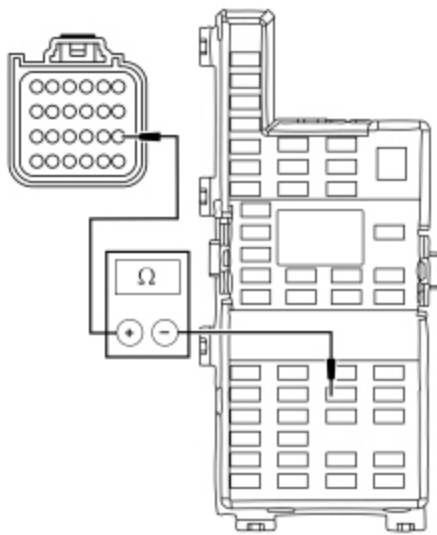
AJ0210-A

Is the voltage between 13 and 15.2 volts?

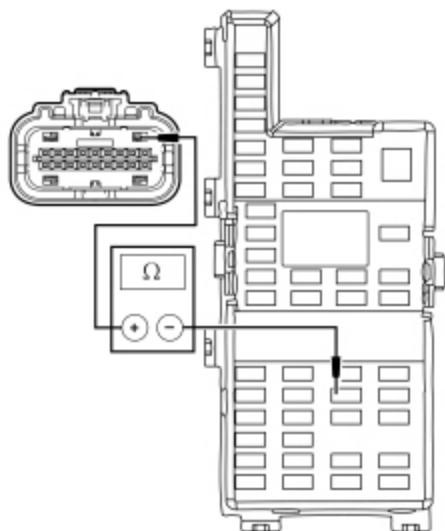
Yes	For DTC U3003:16, GO to Z4 . For DTC U3003:17, GO to Z6 .
No	REFER to Section 414-00 to diagnose the charging system. CLEAR all <u>CMDTCs</u> . TEST the <u>SRS</u> for normal operation.

Z4 CHECK FOR OPEN IGNITION CIRCUIT TO THE MODULE

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RCM [C310A](#) and [C310B](#).
- Disconnect: OCSM [C3159](#) .
- For the RCM , measure the resistance between RCM [C310A](#) Pin 13, circuit CBP36 (BU/BN), harness side and Body Control Module (BCM) fuse 36 (10A).



- N0119861
- For the OCSM , measure the resistance between OCSM [C3159](#) Pin 1, circuit CBX02 (BN/RD), harness side and BCM fuse 36 (10A).



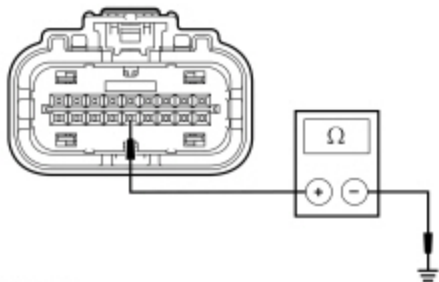
N0119862

Is the resistance less than 1 ohm?

Yes	GO to Z5 .
No	REPAIR the circuit as necessary. Refer to Wiring Diagrams Cell 5 for schematic and connector information. CLEAR all CMDTCs . TEST the SRS for normal operation.

Z5 CHECK FOR OPEN GROUND TO THE MODULE

- Measure the resistance of the ground circuit:
 - For the RCM , measure the resistance between the RCM case and a good ground.
 - For the QCSM , measure the resistance between QCSM [C3159](#) Pin 14, circuit GD901 (BK), harness side and ground.



N0118518

Is the resistance less than 1 ohm?

Yes	GO to Z6 .
No	For the RCM , MAKE SURE there is no corrosion between the RCM case and mounting. MAKE SURE the fasteners are clean and tightened to specification with no corrosion. REPAIR as necessary. For the QCSM , REPAIR circuit GD901 (BK)/GD174 (BK/WH) as necessary. Refer to Wiring Diagrams Cell 5 for schematic and connector information.

CLEAR all CMDTCs . TEST the SRS for normal operation.

Z6 CHECK FOR MODULE OPERATION

- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect the module with the concern:
 - For the RCM , disconnect C310A and C310B.
 - For the OCSM , disconnect C3159.
- Check for the following:
 - corrosion.
 - damaged pins.
 - pushed-out pins.
- Connect the RCM C310A and C310B or OCSM C3159 and make sure the connector seats correctly and engage Connector Position Assurance (CPA) lever/lock.
- Repower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new module. REFER to Restraints Control Module (RCM) or Occupant Classification System Module (OCSM) in this section.
No	The system is operating correctly at this time. The concern may have been caused by a loose or corroded connector. CLEAR all <u>CMDTCs</u> . TEST the <u>SRS</u> for normal operation.

Pinpoint Test AA: DTCs B0061:11, B0061:12, B0061:13 and B0061:1C

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The safety Belt Tension Sensor (BTS) operates in conjunction with the Occupant Classification System (OCS) . The OCS interprets a variable voltage signal provided by the BTS to identify the possible presence of a child safety seat in the front passenger seat. The voltage output of the BTS is proportional to the amount of tension applied to the sensor by the belt. No tension outputs a low voltage (approximately 0.95 volt), and high tension outputs a high voltage (approximately 3.8 volts), in relation to the reference voltage.

The Occupant Classification System Module (OCSM) monitors the BTS and all circuitry for the following faults:

- Resistance out of range
- Unexpected voltage
- Short to ground
- Faulted BTS (part of RH front safety belt retractor)

If a fault is detected, the OCSM stores DTC B0061:11, B0061:12, B0061:13 or B0061:1C in memory and sends a message to the Restraints Control Module (RCM) , which stores DTC B00A0:09 or B00A0:68 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Condition
B0061:11 — Passenger Seatbelt Tension Sensor: Circuit Short to Ground	A fault is indicated if the BTS VREF circuit is open or if the BTS signal circuit is shorted to ground. An internally-shortened BTS can also set this DTC. This DTC is detected by calculating the ratio of the measured BTS signal divided by the BTS VREF value. If the ratio equals less than .1, this DTC sets.
B0061:12 — Passenger Seatbelt Tension Sensor: Circuit Short to Battery	A fault is indicated when the BTS signal circuit is shorted to battery voltage or BTS VREF. An open BTS return circuit also sets this DTC. This DTC is detected by calculating the ratio of the measured BTS signal divided by the BTS VREF value. If the ratio equals greater than .9, this DTC sets.
B0061:13 — Passenger Seatbelt Tension Sensor: Circuit Open	A fault is indicated when the QCSM measures resistance above a calibrated threshold on the BTS signal circuit.
B0061:1C — Passenger Seatbelt Tension Sensor: Circuit Voltage Out of Range	A fault is indicated when the QCSM measures greater than 2.0 volts on the BTS signal circuit while the passenger safety belt is not buckled.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- **BTS** (part of RH front passenger safety belt retractor)
- **QCSM**

PINPOINT TEST AA : DTCS B0061:11, B0061:12, B0061:13 AND B0061:1C

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

AA1 RETRIEVE OCSM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .

Was DTC B0061:11, B0061:12, B0061:13 or B0061:1C retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test. For DTC B0061:11 or B0061:12, GO to AA2 . For DTC B0061:13, GO to AA8 . For DTC B0061:1C, GO to AA10 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AA12 .

AA2 CHECK THE BTS

NOTE: *This pinpoint test step attempts to change the fault reported by the OCSM by inducing a different fault condition. If the fault reported changes, this indicates the OCSM is functioning correctly and is not the source of the fault.*

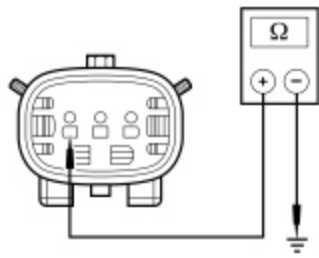
- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: BTS C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the BTS disconnected, an open circuit fault is normally retrieved.

Was DTC B0061:11 or B0061:12 retrieved on-demand during self-test?

Yes	For DTC B0061:11, GO to AA3 . For DTC B0061:12, GO to AA5 .
No	INSTALL a new passenger safety belt retractor pretensioner assembly (includes <u>BTS</u>). REFER to Section 501-20A. GO to AA13 .

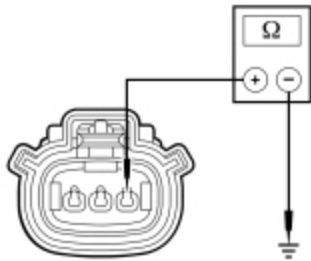
AA3 CHECK THE BTS CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: OCSM C3159 .
- For regular cab and Crew Cab vehicles, measure the resistance between BTS [C3238](#) Pin 1, circuit VR210 (BU/BN), harness side and ground.



• N0132341

- For SuperCab vehicles, measure the resistance between **BTS C829** Pin 1, circuit VR210 (BU/BN), harness side and ground.



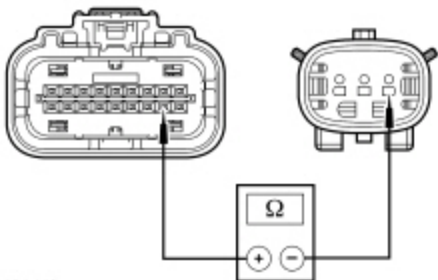
• N0132340

Are the resistances greater than 10,000 ohms?

Yes	GO to AA4 .
No	REPAIR circuit VR210 (BU/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AA13 .

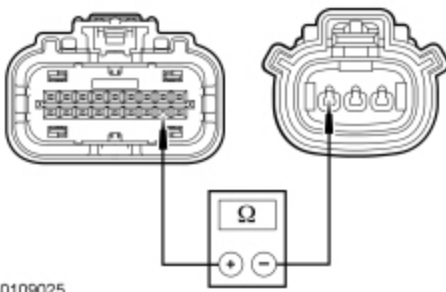
AA4 CHECK THE BTS VREF CIRCUIT FOR AN OPEN

- For regular cab and Crew Cab vehicles, measure the resistance between **BTS C3238** Pin 3, circuit LR142 (YE/BU), harness side and **QCSM C3159** Pin 11, circuit LR142 (YE/BU), harness side.



• N0101448

- For SuperCab vehicles, measure the resistance between **BTS C829** Pin 3, circuit LR142 (YE/BU), harness side and **QCSM C3159** Pin 11, circuit LR142 (YE/BU), harness side.



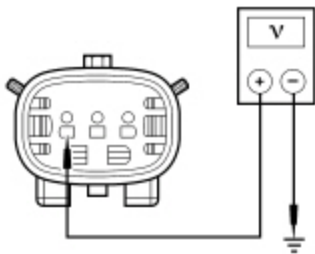
• N0109025

Is the resistance less than 0.5 ohm?

Yes	GO to AA5 .
No	REPAIR circuit LR142 (YE/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AA13 .

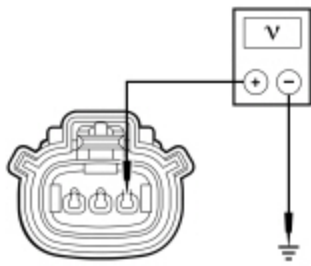
AA5 CHECK THE BTS SIGNAL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: ~~BTS~~ C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Disconnect: ~~OCSM~~ C3159 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- For regular cab and Crew Cab vehicles, measure the voltage between ~~BTS~~ [C3238](#) Pin 1, circuit VR210 (BU/BN), harness side and ground.



• N0101447

- For SuperCab vehicles, measure the voltage between ~~BTS~~ [C829](#) Pin 1, circuit VR210 (BU/BN), harness side and ground.



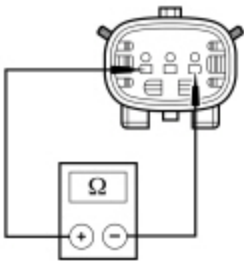
• N0109026

Is any voltage present?

Yes	REPAIR circuit VR210 (BU/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AA13 .
No	GO to AA6 .

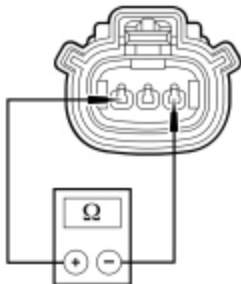
AA6 CHECK THE BTS SIGNAL CIRCUIT FOR A SHORT TO THE BTS VREF CIRCUIT

- Ignition OFF.
- For regular cab and Crew Cab vehicles, measure the resistance between **BTS C3238** Pin 1, circuit VR210 (BU/BN), harness side and **BTS C3238** Pin 3, circuit LR142 (YE/BU), harness side.



• N0133852

- For SuperCab vehicles, measure the resistance between **BTS C829** Pin 1, circuit VR210 (BU/BN), harness side and **BTS C829** Pin 3, circuit LR142 (YE/BU), harness side.



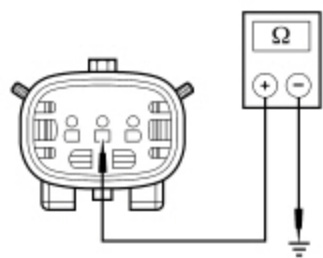
• N0133853

Is the resistance greater than 10,000 ohms?

Yes	GO to AA7 .
No	REPAIR the circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AA13 .

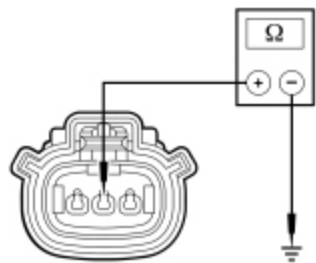
AA7 CHECK THE BTS GROUND CIRCUIT FOR AN OPEN

- For regular cab and Crew Cab vehicles, measure the resistance between ~~BTS~~ [C3238](#) Pin 2, circuit GD901 (BK), harness side and ground.



• N0101446

- For SuperCab vehicles, measure the resistance between ~~BTS~~ [C829](#) Pin 2, circuit GD174 (BK/WH), harness side and ground.



• N0109027

Is the resistance less than 3 ohms?

Yes	GO to AA10 .
No	REPAIR circuit GD901 (BK)/GD174 (BK/WH) or GD174 (BK/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AA13 .

AA8 CHECK THE BTS

NOTE: This pinpoint test step attempts to change the fault reported by the ~~QCSM~~ by inducing a different fault condition. If the fault reported changes, this indicates the ~~QCSM~~ is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: ~~BTS~~ C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Connect a fused jumper wire between ~~BTS~~ [C3238](#) Pin 1/[C829](#) Pin 1, circuit VR210 (BU/BN), harness side and [C3238](#) Pin 2/[C829](#) Pin 2, circuit GD901 (BK)/GD174 (BK/WH), harness side.
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

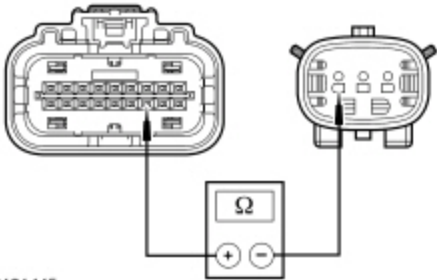
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — [OCSM](#) .
- **DIAGNOSTIC TIP:** When viewing DTCs with the [BTS](#) circuits shorted together, a short to ground fault is normally retrieved.

Did the on-demand DTC change from B0061:13 to B0061:11?

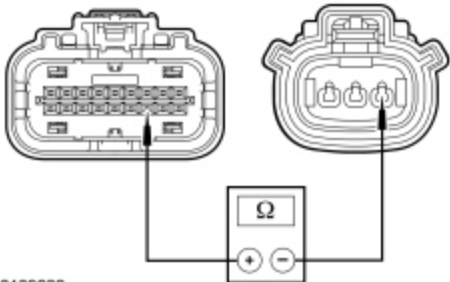
Yes	REMOVE the fused jumper wire and INSTALL a new passenger safety belt retractor pretensioner assembly (includes BTS). REFER to Section 501-20A. GO to AA13 .
No	REMOVE the fused jumper wire and GO to AA9 .

AA9 CHECK THE BTS SIGNAL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: [OCSM](#) C3159 .
- For regular cab and Crew Cab vehicles, measure the resistance between [BTS](#) [C3238](#) Pin 1, circuit VR210 (BU/BN), harness side and [OCSM](#) [C3159](#) Pin 12, circuit VR210 (BU/BN), harness side.



- N0101445
- For SuperCab vehicles, measure the resistance between [BTS](#) [C829](#) Pin 1, circuit VR210 (BU/BN), harness side and [OCSM](#) [C3159](#) Pin 12, circuit VR210 (BU/BN), harness side.

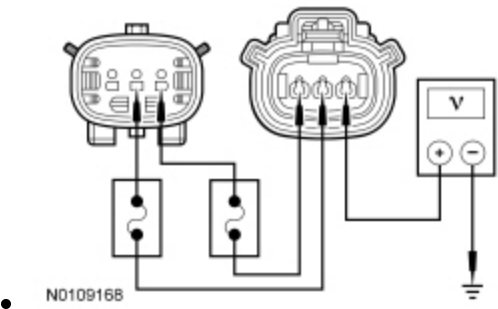


Are the resistances less than 0.5 ohm?

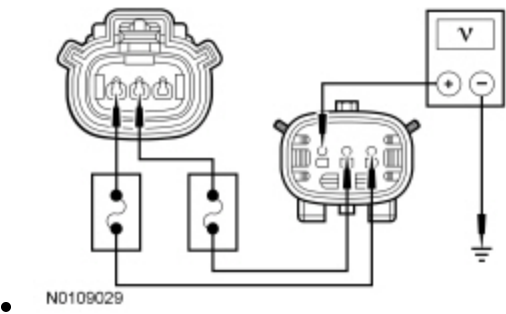
Yes	GO to AA10 .
No	REPAIR circuit VR210 (BU/BN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AA13 .

AA10 CHECK BTS VOLTAGE OUTPUT

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: BTS C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Connect a jumper lead between BTS C3238 Pin 3 (regular cab and Crew Cab)/C829 Pin 3 (SuperCab), circuit LR142 (YE/BU), harness side and C3238/C829 pin 3, circuit LR142 (YE/BU), component side.
- Connect a jumper lead between BTS C3238 Pin 2 (regular cab and Crew Cab)/C829 Pin 2 (SuperCab), circuit GD901 (BK), harness side and C3238/C829 pin 2, circuit GD901 (BK)/GD174 (BK/WH), component side.
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- For regular cab and Crew Cab vehicles, measure the voltage between BTS C3238 pin 1, component side and ground as the tension of the BTS varies.



- For SuperCab vehicles, measure the voltage between BTS C829 pin 1, component side and ground as the tension of the BTS varies.



Does the voltage vary from approximately 0.95 volt with no tension applied to the sensor to approximately 3.8 volts with full tension applied to the sensor?

Yes	REMOVE the fused jumper wires and GO to AA11 .
No	REMOVE the fused jumper wire and INSTALL a new passenger safety belt retractor pretensioner assembly (includes BTS). REFER to Section 501-20A. GO to AA13 .

AA11 CONFIRM THE OCSM FAULT

NOTE: Make sure all OCS components, restraint system sensor electrical connectors and the RCM electrical connectors are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Prior to reconnecting any previously disconnected SRS component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: OCSM C3159 (if disconnected previously) .
- Connect: BTS C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- If previously directed to depower the SRS , repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new <u>OCSM</u> . REFER to Occupant Classification System Module (OCSM) in this section. GO to AA13 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to AA12 .

AA12 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: BTS [C3238](#) (regular cab and Crew Cab)/[C829](#) (SuperCab) .
 - inspect connector(s) (including any in-line connectors) for loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: BTS [C3238](#) (regular cab and Crew Cab)/[C829](#) (SuperCab) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .

Was DTC B0061:11, B0061:12, B0061:13 or B0061:1C retrieved on-demand during self-test?

Yes	The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test. For DTC B0061:11 or B0061:12, GO to AA2 . For DTC B0061:13, GO to AA8 . For DTC B0061:1C, GO to AA10 .
No	CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. REPAIR any intermittent concerns found. GO to AA13 .

AA13 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~QCSM~~ .*

Are any ~~RCM~~ and/or ~~QCSM~~ DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM DTCs. PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AB: DTCs B00A0:54, B00A0:76, B00C2:49 and B00C3:49

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) monitors each Occupant Classification System (OCS) weight sensor bolt for correct calibration/weight and identification. When the ~~QCSM~~ detects a fault from either of the 2 sensors or if an ~~QCS~~ component has been installed new, it stores a DTC in memory and sends a message to the Restraints Control Module (RCM) .

When the ~~RCM~~ receives a fault message from the ~~QCS~~ , it stores DTC B00A0:09 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

DTC Description	Fault Trigger Condition
B00A0:54 — Occupant Classification System: Missing Calibration	A fault is indicated when the <u>OCSM</u> senses a calibration from one or both weight sensor bolt(s) that is not stored in memory. A new <u>OCS</u> component installed in the vehicle also sets this DTC.
B00A0:76 — Occupant Classification System: Wrong Mounting Position	A fault is indicated when the <u>OCSM</u> senses signals from weight sensor bolt(s) that are not the correct values for their position on the seat track. Sensors that are not connected correctly can set this DTC.
B00C2:49 — Passenger Seat Occupant Classification Sensor "C": Internal Electronic Failure	A fault is indicated when the <u>OCSM</u> measures sensor output to be between 4.375 volts and 4.85 volts (sensor is limited or clamped high) or when the measured sensor output voltage is between 0.15 to 0.625 volt (sensor is limited or clamped low). A faulted sensor, passenger seat not installed correctly or bent or binding passenger seat track also sets this DTC.
B00C3:49 — Passenger Seat Occupant Classification Sensor "D": Internal Electronic Failure	A fault is indicated when the <u>OCSM</u> measures sensor output to be between 4.375 volts and 4.85 volts (sensor is limited or clamped high) or when the measured sensor output voltage is between 0.15 to 0.625 volt (sensor is limited or clamped low). A faulted sensor, passenger seat not installed correctly or bent or binding passenger seat track also sets this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger seat track assembly

PINPOINT TEST AB : DTCS B00A0:54, B00A0:76, B00C2:49 AND B00C3:49

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: <i>Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.</i>
NOTE: <i>Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .</i>
NOTE: <i>Do not install any new <u>OCS</u> components until directed to do so. <u>OCS</u> components should only be installed when directed to do so in the pinpoint test.</i>
NOTE: <i>Always make sure the correct <u>SRS</u> component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect <u>SRS</u> component is installed, DTCs may set.</i>
NOTE: <i>The <u>SRS</u> must be fully operational and free of faults before releasing the vehicle to the customer.</i>
AB1 CHECK FOR ADDITIONAL OCS DTCS
- Ignition ON. - Enter the following diagnostic mode on the scan tool: Self Test — <u>OCSM</u> .

Were any DTCs other than B00A0:54, B00A0:76, B00C2:49 or B00C3:49 retrieved during the self-test?

Yes	REPAIR the DTCs before continuing with this test. REFER to the <u>QCSCM</u> DTC Chart in this section.
No	GO to AB2 .

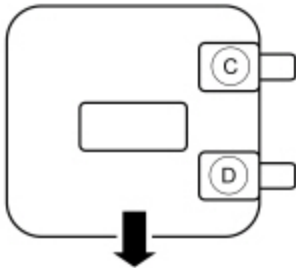
AB2 CHECK THE SEAT WIRING AND CONNECTORS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Carry out a thorough inspection of the QCSC wiring, terminals and connectors and the related seat wiring harness and body wiring harness terminals and connectors. Inspect the passenger seat wire harness for correct routing and retention. Refer to Seat Wire Harness Routing in the general procedures portion of Section 501-10.
- **NOTE:** *The QCSC weight sensor bolts do not have differently keyed connectors. Connecting the incorrect connector to a sensor causes DTCs to set.*

Use the wire colors for correct weight sensor bolt connection.
Refer to Wiring Diagrams Cell [46](#) for schematic and connector information.

Confirm that the sensors are connected correctly

- QCSC weight sensor bolt "C" is located at the rear of the inboard seat track
- QCSC weight sensor bolt "D" is located at the front of the inboard seat track



N0132320

Were any problems noted?

Yes	REPAIR the seat connectors and wiring as needed. Refer to Wiring Diagrams Cell 5 for schematic and connector information. REFER to Section 501-10 for wire harness routing information. CARRY OUT the Occupant Classification System (OCS) Reset. REFER to Occupant Classification Sensor (OCS) System Reset in this section. GO to AB4 .
No	GO to AB3 .

AB3 CHECK THE SEAT TRACK

NOTICE: Check for a bent or binding seat track assembly. Position the seat in all positions and check for correct operation after installation of the seat into the vehicle. Failure to follow this instruction may result in an unsuccessful system reset and incorrect operation of the Occupant Classification System (OCS) .

NOTICE: The passenger seat bolts must be tightened to the correct specifications and in the correct sequence. Refer to Section 501-10 Seat — Front in Removal and Installation for the correct sequence. Failure to follow

this instruction may result in an unsuccessful system reset and incorrect operation of the Occupant Classification System (OCS) .

- For vehicles with a manual seat, adjust the seat position to the full forward position of seat travel, then back to the mid-point of the seat track travel.
- Loosen the 4 seat track-to-floor mounting bolts.
- Tighten the seat track-to-floor bolts in the correct sequence. Refer to Section 501-10 Seat — Front in Removal and Installation for the torque sequence.
- Carry out the Occupant Classification System (OCS) Reset. Refer to Occupant Classification Sensor (OCS) System Reset in this section.
- If the System Reset successfully completed, carry out the Occupant Classification System (OCS) Zero Seat Weight Test. Refer to Occupant Classification Sensor (OCS) System Zero Seat Weight Test in this section.

Did the system reset and zero seat weight test pass?

Yes	Fault corrected. GO to AB4 .
No	INSTALL a new passenger seat track assembly with OCS weight sensor bolts. GO to AB4 .

AB4 CHECK FOR ADDITIONAL SRS DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~OCSM~~ .*

Are any ~~RCM~~ and/or ~~OCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AC: DTC B00A0:97

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) monitors each Occupant Classification System (OCS) weight sensor bolt and circuitry for faults. If the ~~OCSM~~ detects an upward force (obstruction) on the passenger seat for 60 seconds, it stores DTC B00A0:97 in memory and sends a message to the Restraints Control Module (RCM) , which stores DTC B00A0:09 or B00A0:68 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

- DTC B00A0:97 (Occupant Classification System: Component or System Operation Obstructed or Blocked) — This DTC sets if the ~~OCSM~~ detects a negative (upward) force on one or more ~~OCS~~ weight sensor bolt(s).

This pinpoint test is intended to diagnose the following:

- Foreign object placed under or contacting the passenger seat
- **OCSM**

PINPOINT TEST AC : DTC B00A0:97

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: *Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.*

NOTE: *Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .*

NOTE: *Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.*

NOTE: *The OCS weight sensor bolt(s) are 3 wire sensors and cannot be tested using a multi-meter.*

NOTE: *The SRS must be fully operational and free of faults before releasing the vehicle to the customer.*

AC1 RETRIEVE OCSM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **OCSM** .

Was DTC B00A0:97 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. GO to AC2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AC6 .

AC2 INSPECT UNDER THE PASSENGER SEAT FOR FOREIGN OBJECTS AND CORRECT WIRE HARNESS ROUTING

- Ignition OFF.
- Carry out a thorough inspection:
 - under the passenger seat for the presence of any foreign object(s) making contact with the seat cushion or track.
 - on the side of the passenger seat for foreign objects contacting the side of the seat (between console and seat).
 - of the passenger seat and body wire harness(es) at or near the **OCS** weight sensor bolts for correct routing. Refer to Seat Wire Harness Routing in the General Procedures portion of Section 501-10.
- For vehicles with a manual seat, adjust the seat position to the full forward position of seat travel, then back to the mid-point of the seat track travel.
- Remove any obstructions and/or repair any items that did not pass inspection.

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear All Continuous Memory Diagnostic Trouble Codes (CMDTCs) .
- Enter the following diagnostic mode on the scan tool: Self Test — **OCSM** .

Was DTC B00A0:97 retrieved again during the self-test?

Yes	GO to AC3 .
No	The fault is not present at this time and may have been caused by an obstruction, mis-aligned manual seat track, or incorrect wire routing. GO to AC7 .

AC3 CARRY OUT THE ZERO SEAT WEIGHT TEST

- Carry out the **OCS** Zero Seat Weight Test. Refer to Occupant Classification Sensor (OCS) System Zero Seat Weight Test in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear All Continuous Memory Diagnostic Trouble Codes (CMDTCs) .
- Enter the following diagnostic mode on the scan tool: Self Test — **OCSM** .

Was DTC B00A0:97 retrieved again during the self-test?

Yes	GO to AC4 .
No	The fault is not present at this time and may have been caused due to a mis-calibrated passenger seat OCSM . GO to AC7 .

AC4 CHECK THE OCS SENSOR STATUS PID (OCS_SENSTAT)

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Data Logger — **RCM** .
- Monitor the **RCM** PID (OCS_SENSTAT).

Does the PID display "Indeterminate"?

Yes	INSTALL a new OCS weight sensor assembly. REFER to Occupant Classification Sensor in this section. GO to AC7 .
No	GO to AC5 .

AC5 CONFIRM THE OCSM FAULT

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **OCSM** .

Was DTC B00A0:97 retrieved on-demand during self-test?

Yes	INSTALL a new OCSM . REFER to Occupant Classification System Module (OCSM) in this section. GO to AC7 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to AC6 .

AC6 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Carry out a thorough inspection:
 - under the passenger seat for the presence of any foreign object(s) making contact with the seat cushion or track.
 - on the side of the passenger seat for foreign objects contacting the side of the seat (between console and seat).
 - of the passenger seat and body wire harness(es) at or near the ~~OCS~~ weight sensor bolts for correct routing. Refer to Seat Wire Harness Routing in the General Procedures portion of Section 501-10.
- For vehicles with a manual seat, adjust the seat position to the full forward position of seat travel, then back to the mid-point of the seat track travel.
- Remove any obstructions and/or repair any items that did not pass inspection.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC B00A0:97 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. GO to AC2 .
No	CHECK for causes of the intermittent fault. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. REPAIR any intermittent concerns found. GO to AC7 .

AC7 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~OCSM~~ .

Are any RCM and/or OCSM DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AD: DTCs B00C2:12, B00C2:13 and B00C2:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) monitors the Occupant Classification System (OCS) weight sensor bolt "C" and circuitry for faults. If the OCSM detects a fault, it stores DTC B00C2:12, B00C2:13 or B00C2:1A in memory and sends a message to the Restraints Control Module (RCM) , which stores DTC B00A0:09 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

- DTC B00C2:12 (Passenger Seat Occupant Classification Sensor "C": Circuit Short To Battery) — A fault is indicated when the OCSM measures voltage on the OCS weight sensor bolt "C" signal circuit above 4.85 volts. An open sensor return circuit also sets this DTC.
- DTC B00C2:13 (Passenger Seat Occupant Classification Sensor "C": Circuit Open) — A fault is indicated when the OCSM measures resistance above a calibrated threshold on the OCS weight sensor bolt "C" signal circuit.
- DTC B00C2:1A (Passenger Seat Occupant Classification Sensor "C": Circuit Resistance Below Threshold) — A fault is indicated when the OCSM measures resistance below a calibrated threshold on the OCS weight sensor bolt "C" signal circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- OCS weight sensor bolt "C"
- OCSM

PINPOINT TEST AD : DTCS B00C2:12, B00C2:13 AND B00C2:1A

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The OCS weight sensor bolt(s) are 3 wire sensors and cannot be tested using a multi-meter.

NOTE: The *SRS* must be fully operational and free of faults before releasing the vehicle to the customer.

AD1 RETRIEVE OCSM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — *OCSM* .

Was DTC B00C2:12, B00C2:13 or B00C2:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. GO to AD2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AD12 .

AD2 CHECK THE SEAT WIRING AND CONNECTORS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Carry out a thorough inspection of the *OCS* wiring, terminals and connectors, weight sensor bolt connector and connection, wiring harness routing and the related seat wiring harness and body wiring harness terminals and connectors.

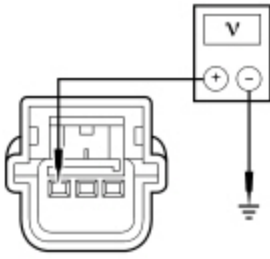
Were any problems noted?

Yes	DEPOWER the <i>SRS</i> . REFER to Supplemental Restraint System (SRS) Depowering and Repowering in this section. REPAIR the seat connectors and wiring as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AD13 .
No	For DTC B00C2:12, GO to AD3 . For DTC B00C2:13, GO to AD5 . For DTC B00C2:1A, GO to AD8 .

AD3 CHECK THE OCS WEIGHT SENSOR BOLT "C" SIGNAL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the *SRS* . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: *OCS* Weight Sensor Bolt "C" C3330 .
- Disconnect: *OCSM* C3159 .
- Repower the *SRS* . **Do not** prove out the *SRS* at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.

- Measure the voltage between OCS weight sensor bolt "C" [C3330](#) Pin 3, circuit VR226 (YE/GY), harness side and ground.



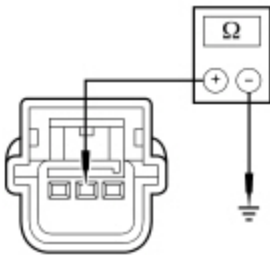
• N0066751

Is any voltage present?

Yes	REPAIR circuit VR226 (YE/GY). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AD13 .
No	GO to AD4 .

AD4 CHECK THE OCS WEIGHT SENSOR BOLT "C" RETURN CIRCUIT FOR AN OPEN

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Measure the resistance between OCS weight sensor bolt "C" [C3330](#) Pin 2, circuit RR153 (GY/BU), harness side and ground.



• N0066750

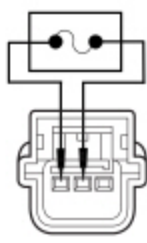
Is the resistance less than 3 ohms?

Yes	GO to AD11 .
No	REPAIR circuit RR153 (GY/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AD13 .

AD5 CHECK THE OCS WEIGHT SENSOR BOLT "C" FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the *QCSM* by inducing a different fault condition. If the fault reported changes, this indicates the *QCSM* is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: OCS Weight Sensor Bolt "C" C3330 .
- Connect a fused jumper wire between OCS weight sensor bolt "C" C3330 Pin 3, circuit VR226 (YE/GY), harness side and C3330 Pin 2, circuit RR153 (GY/BU), harness side.



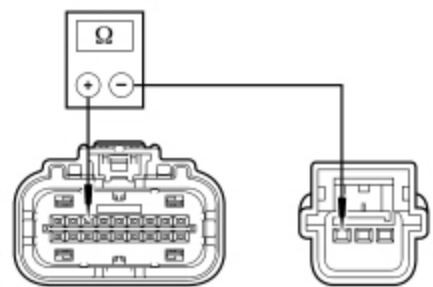
- N0114841
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the OCS weight sensor bolt "C" circuits shorted together, a low resistance circuit fault is normally retrieved.

Did the on-demand DTC change from B00C2:13 to B00C2:1A?

Yes	REMOVE the fused jumper wire and INSTALL a new seat track assembly with OCS weight sensor bolts. REFER to Section 501-10. GO to AD13.
No	REMOVE the fused jumper wire and GO to AD6.

AD6 CHECK THE OCS WEIGHT SENSOR BOLT "C" SIGNAL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: OCSM C3159 .
- Measure the resistance between OCSM C3159 Pin 7, circuit VR226 (YE/GY), harness side and OCS weight sensor bolt "C" C3330 Pin 3, circuit VR226 (YE/GY), harness side.



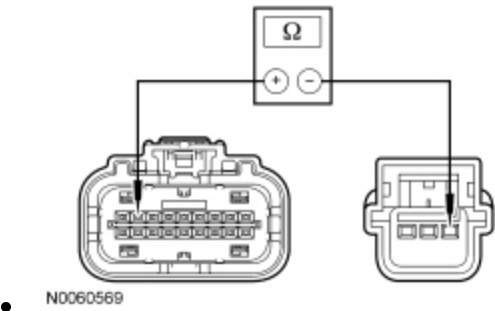
• N0091130

Is the resistance less than 0.5 ohm?

Yes	GO to AD7 .
No	REPAIR circuit VR226 (YE/GY). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AD13 .

AD7 CHECK THE OCS WEIGHT SENSOR BOLT "C" VREF CIRCUIT FOR AN OPEN

- Measure the resistance between OCS weight sensor bolt "C" [C3330](#) Pin 1, circuit CR153 (GN/WH), harness side and OCSM [C3159](#) Pin 8, circuit CR153 (GN/WH), harness side.



Is the resistance less than 3 ohms?

Yes	GO to AD11 .
No	REPAIR circuit CR153 (GN/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AD13 .

AD8 CHECK THE OCS WEIGHT SENSOR BOLT "C" FOR A FAULT STATUS CHANGE (RESISTANCE BELOW THRESHOLD INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the OCSM by inducing a different fault condition. If the fault reported changes, this indicates the OCSM is functioning correctly and is not the source of the fault.

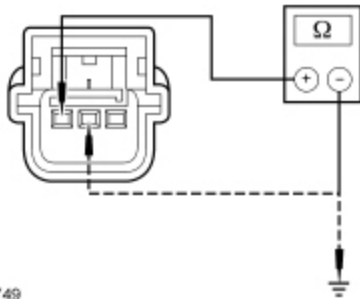
- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.
- Disconnect: OCS Weight Sensor Bolt "C" C3330 .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the OCS weight sensor bolt "C" disconnected, an open sensor circuit fault is normally retrieved.

Did the on-demand DTC change from B00C2:1A to B00C2:13?

Yes	INSTALL a new seat track assembly with <u>OCS</u> weight sensor bolts. REFER to Section 501-10. GO to AD13 .
No	GO to AD9 .

AD9 CHECK FOR A SHORT BETWEEN THE OCS WEIGHT SENSOR BOLT "C" CIRCUITS AND GROUND

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: OCSM C3159 .
- Measure the resistance between OCS weight sensor bolt "C":
 - [C3330](#) Pin 3, circuit VR226 (YE/GY), harness side and ground.
 - [C3330](#) Pin 3, circuit VR226 (YE/GY), harness side and [C3330](#) Pin 2, circuit RR153 (GY/BU), harness side.



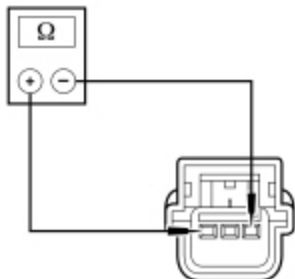
N0056749

Are the resistances greater than 10,000 ohms?

Yes	GO to AD10 .
No	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AD13 .

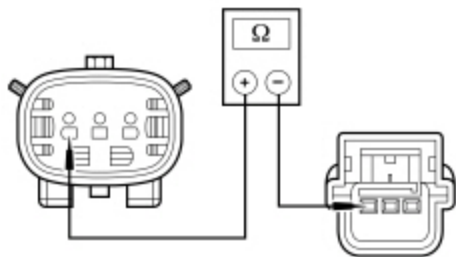
AD10 CHECK FOR A SHORT BETWEEN THE OCS WEIGHT SENSOR BOLT "C" SIGNAL CIRCUITS AND OTHER OCS SENSOR SIGNAL CIRCUITS

- Disconnect: OCS Weight Sensor Bolt "D" C3331 .
- Disconnect: BTS C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Measure the resistance between OCS weight sensor bolt "C" [C3330](#) Pin 3, circuit VR226 (YE/GY), harness side and OCS weight sensor bolt "D" [C3331](#) Pin 3, circuit VR227 (VT/GN), harness side.



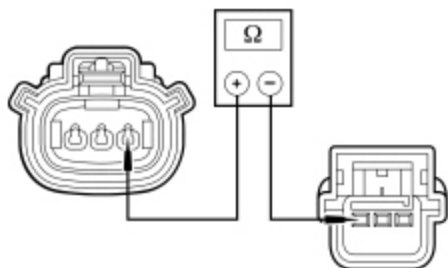
N0133854

- For regular cab and Crew Cab vehicles: Measure the resistance between OCS weight sensor bolt "C" [C3330](#) Pin 3, circuit VR226 (YE/GY), harness side and BTS [C3238](#) Pin 1, circuit VR210 (BU/BN), harness side.



• N0133855

- For SuperCab vehicles: Measure the resistance between OCS weight sensor bolt "C" [C3330](#) Pin 3, circuit VR226 (YE/GY), harness side and BTS [C829](#) Pin 1, circuit VR210 (BU/BN), harness side.



• N0133856

Are the resistances greater than 10,000 ohms?

Yes	GO to AD11 .
No	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AD13 .

AD11 CONFIRM THE OCSM FAULT

NOTE: Make sure all restraint system sensor electrical connectors, OCS components, and the OCSM electrical connector are all connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Prior to reconnecting any previously disconnected SRS component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: OCSM C3159 (if previously disconnected) .
- Connect: OCS Weight Sensor Bolt "C" C3330 .
- If previously directed to depower the SRS , repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new OCSM . REFER to Occupant Classification System Module (OCSM) in this section. GO to AD13 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to AD13 .

AD12 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: ~~OCS~~ Weight Sensor Bolt "C" [C3330](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC B00C2:12, B00C2:13 or B00C2:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. For DTC B00C2:12, GO to AD3 . For DTC B00C2:13, GO to AD5 . For DTC B00C2:1A, GO to AD8 .
No	The fault is not present and cannot be recreated at this time. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AD13 .

AD13 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).

- If previously directed to depower the **SRS** , repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the **RCM** and **OCSM** .*

Are any on-demand **RCM** and/or **OCSM** DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AE: DTCs B00C3:12, B00C3:13 and B00C3:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) monitors Occupant Classification System (OCS) weight sensor bolt "D" and circuitry for faults. If the **OCSM** detects a fault, it stores DTC B00C3:12, B00C3:13 or B00C3:1A in memory and sends a message to the Restraints Control Module (RCM) .

The **RCM** stores DTC B00A0:09 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

- DTC B00C3:12 (Passenger Seat Occupant Classification Sensor "D": Circuit Short To Battery) — A fault is indicated when the **OCSM** measures voltage on the **OCS** weight sensor bolt "D" signal circuit above 4.85 volts. An open sensor return circuit also sets this DTC.
- DTC B00C3:13 (Passenger Seat Occupant Classification Sensor "D": Circuit Open) — A fault is indicated when the **OCSM** measures resistance above a calibrated threshold on the **OCS** weight sensor bolt "D" signal circuit.
- DTC B00C3:1A (Passenger Seat Occupant Classification Sensor "D": Circuit Resistance Below Threshold) — A fault is indicated when the **OCSM** measures resistance below a calibrated threshold on the **OCS** weight sensor bolt "D" signal circuit.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- **OCS** weight sensor bolt "D"
- **OCSM**

PINPOINT TEST AE : DTCS B00C3:12, B00C3:13 AND B00C3:1A

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The OCS weight sensor bolt(s) are 3 wire sensors and cannot be tested using a multi-meter.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

AE1 RETRIEVE OCSM DTCs

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .

Was DTC B00C3:12, B00C3:13 or B00C3:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self-test. GO to AE2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AE12 .

AE2 CHECK THE SEAT WIRING AND CONNECTORS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Carry out a thorough inspection of the OCS wiring, terminals and connectors, weight sensor bolt connector and connection, wiring harness routing and the related seat wiring harness and body wiring harness terminals and connectors.

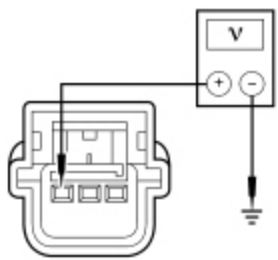
Were any problems noted?

Yes	DEPOWER the SRS . REFER to Supplemental Restraint System (SRS) Depowering and Repowering in this section. REPAIR the seat connectors and wiring as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AE13 .
No	For DTC B00C3:12, GO to AE3 . For DTC B00C3:13, GO to AE5 . For DTC B00C3:1A, GO to AE8 .

AE3 CHECK THE OCS WEIGHT SENSOR BOLT "D" SIGNAL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .

- Disconnect: OCS Weight Sensor Bolt "D" C3331 .
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between OCS weight sensor bolt "D" C3331 Pin 3, circuit VR227 (VT/GN), harness side and ground.



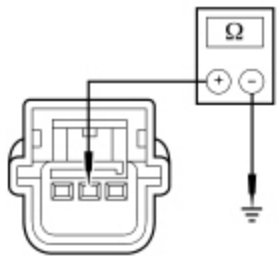
• N0056751

Is any voltage present?

Yes	REPAIR circuit VR227 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AE13.
No	GO to AE4.

AE4 CHECK THE OCS WEIGHT SENSOR BOLT "D" RETURN CIRCUIT FOR AN OPEN

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Measure the resistance between OCS weight sensor bolt "D" C3331 Pin 2, circuit RR154 (VT/WH), harness side and ground.



• N0056750

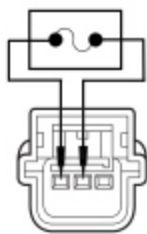
Is the resistance less than 3 ohms?

Yes	GO to AE11.
No	REPAIR circuit RR154 (VT/WH). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AE13.

AE5 CHECK THE OCS WEIGHT SENSOR BOLT "D" FOR A FAULT STATUS CHANGE (OPEN INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the OCSM by inducing a different fault condition. If the fault reported changes, this indicates the OCSM is functioning correctly and is not the source of the fault.

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag Module C339 .
- Disconnect: OCS Weight Sensor Bolt "D" C3331 .
- Connect a fused jumper wire between OCS weight sensor bolt "D" [C3331](#) Pin 3, circuit VR227 (VT/GN), harness side and [C3331](#) Pin 2, circuit RR154 (VT/WH), harness side.



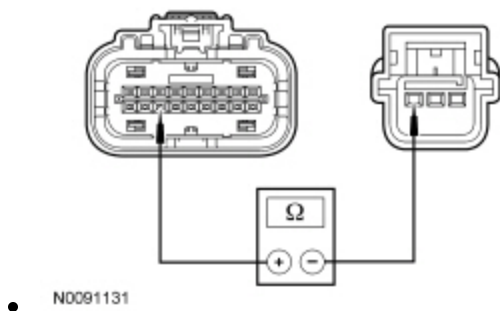
- NO114841
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the OCS weight sensor bolt "D" circuits shorted together, a low resistance circuit fault is normally retrieved.

Did the on-demand DTC change from B00C3:13 to B00C3:1A?

Yes	REMOVE the fused jumper wire and INSTALL a new seat track assembly with <u>OCS</u> weight sensor bolts. REFER to Section 501-10. GO to AE13 .
No	REMOVE the fused jumper wire and GO to AE6 .

AE6 CHECK THE OCS WEIGHT SENSOR BOLT "D" SIGNAL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Disconnect: OCSM C3159 .
- Measure the resistance between OCSM [C3159](#) Pin 16, circuit VR227 (VT/GN), harness side, and OCS weight sensor bolt "D" [C3331](#) Pin 3, circuit VR227 (VT/GN), harness side.

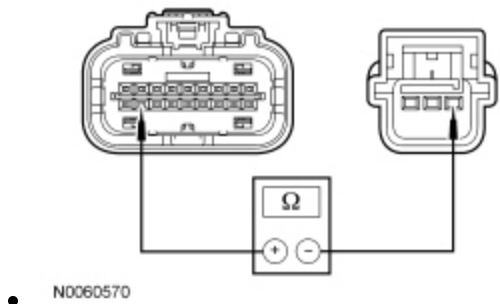


Is the resistance less than 0.5 ohm?

Yes	GO to AE7 .
No	REPAIR circuit VR227 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AE13 .

AE7 CHECK THE OCS WEIGHT SENSOR BOLT "D" VREF CIRCUIT FOR AN OPEN

- Measure the resistance between QCS weight sensor bolt "D" [C3331](#) Pin 1, circuit CR154 (VT/OG), harness side and QCSM [C3159](#) Pin 17, circuit CR154 (VT/OG), harness side.



Is the resistance less than 3 ohms?

Yes	GO to AE11 .
No	REPAIR circuit CR154 (VT/OG). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AE13 .

AE8 CHECK THE OCS WEIGHT SENSOR BOLT "D" FOR A FAULT STATUS CHANGE (RESISTANCE BELOW THRESHOLD INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the QCSM by inducing a different fault condition. If the fault reported changes, this indicates the QCSM is functioning correctly and is not the source of the fault.

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: QCS Weight Sensor Bolt "D" C3331 .
- Ignition ON.

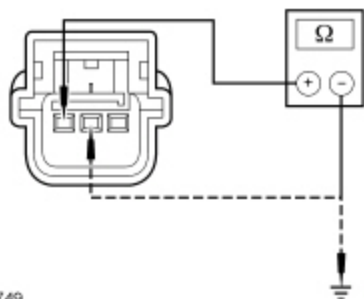
- Enter the following diagnostic mode on the scan tool: Self Test — [OCSM](#) .
- **DIAGNOSTIC TIP:** When viewing DTCs with the [OCS](#) weight sensor bolt "D" disconnected, an open circuit fault is normally retrieved.

Did the on-demand DTC change from B00C3:1A to B00C3:13?

Yes	INSTALL a seat track assembly with OCS weight sensor bolts. REFER to Section 501-10. GO to AE13 .
No	GO to AE9 .

AE9 CHECK FOR A SHORT BETWEEN OCS WEIGHT SENSOR BOLT "D" CIRCUITS AND GROUND

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: [OCSM](#) C3159 .
- Measure the resistance between [OCS](#) weight sensor bolt "D":
 - [C3331](#) Pin 3, circuit VR227 (VT/GN), harness side and ground.
 - [C3331](#) Pin 3, circuit VR227 (VT/GN), harness side and [C3331](#) Pin 2, circuit RR154 (VT/WH), harness side.



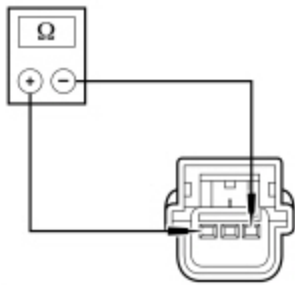
• N0056749

Are the resistances greater than 10,000 ohms?

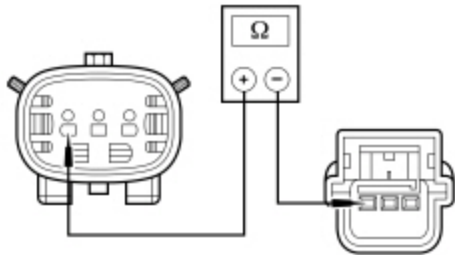
Yes	GO to AE10 .
No	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AE13 .

AE10 CHECK FOR A SHORT BETWEEN THE OCS WEIGHT SENSOR BOLT "D" SIGNAL CIRCUITS AND OTHER OCS SENSOR SIGNAL CIRCUITS

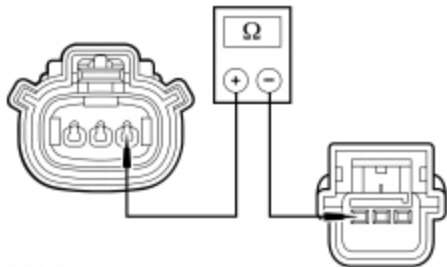
- Disconnect: [OCS](#) Weight Sensor Bolt "C" C3330 .
- Disconnect: [BTS](#) C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Measure the resistance between [OCS](#) weight sensor bolt "C" [C3330](#) Pin 3, circuit VR226 (YE/GY), harness side and [OCS](#) weight sensor bolt "D" [C3331](#) Pin 3, circuit VR227 (VT/GN), harness side.



- N0133854
- For regular cab and Crew Cab vehicles: Measure the resistance between OCS weight sensor bolt "D" [C3331](#) Pin 3, circuit VR226 (YE/GY), harness side and BTS [C3238](#) Pin 1, circuit VR210 (BU/BN), harness side.



- N0133855
- For SuperCab vehicles: Measure the resistance between OCS weight sensor bolt "D" [C3331](#) Pin 3, circuit VR226 (YE/GY), harness side and BTS [C829](#) Pin 1, circuit VR210 (BU/BN), harness side.



- N0133856

Are the resistances greater than 10,000 ohms?

Yes	GO to AE11 .
No	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AE13 .

AE11 CONFIRM THE OCSM FAULT

NOTE: Make sure all restraint system sensor electrical connectors, OCS components and the OCSM electrical connector are all connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.
- Prior to reconnecting any previously disconnected SRS component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.

- inspect wire harness for any damage, pinched, cut or pierced wires.
- repair any concerns found.

Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.

- Connect: ~~OCSM~~ C3159 (if previously disconnected) .
- Connect: ~~OCS~~ Weight Sensor Bolt "D" C3331 .
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new OCSM . REFER to Occupant Classification System Module (OCSM) in this section. GO to AE13 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to AE12 .

AE12 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: ~~OCS~~ Weight Sensor Bolt "D" [C3331](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.

Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC B00C3:12, B00C3:13 or B00C3:1A retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self test. For DTC B00C3:12, GO to AE3 . For DTC B00C3:13, GO to AE5 . For DTC B00C3:1A, GO to AE8 .
No	CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector

AE13 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~, repower the ~~SRS~~. **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~OCSM~~.*

Are any ~~RCM~~ and/or ~~OCSM~~ DTCs retrieved on-demand during self-test?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AF: DTCs B1A59:01 and B1A59:11

Refer to Wiring Diagrams Cell [46](#), Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) monitors each Occupant Classification System (OCS) weight sensor bolt and circuitry for faults. If the ~~OCSM~~ detects a fault, it stores DTC B1A59:01 in memory and sends a message to the Restraints Control Module (RCM).

The ~~RCM~~ stores DTC B00A0:09 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

- DTC B1A59:01 (Sensor 5 Volt Supply: General Electrical Failure) — A fault is indicated when the ~~OCSM~~ senses a short between the Belt Tension Sensor (BTS) VREF circuit and one of the ~~OCS~~ weight sensor bolt VREF circuits. DTC U200E:1A may also set when this DTC is present. The ~~OCSM~~ only checks for this fault when the module is first powered on, and does not monitor for this fault after the initial power-on self test.
- DTC B1A59:11 (Sensor 5 Volt Supply: Circuit Short To Ground) — A fault is indicated when the ~~OCSM~~ senses a short to ground on either of the ~~OCS~~ weight sensor bolt VREF circuits, or if the ~~BTS~~ VREF circuit is shorted to ground while an open circuit exists on the ~~BTS~~ VREF circuit (~~BTS~~ disconnected).

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- Passenger seat track assembly
- ~~OCSM~~

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.

WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.

NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.

NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.

NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS.

NOTE: Do not install any new OCS components until directed to do so. OCS components should only be installed when directed to do so in the pinpoint test.

NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.

AF1 RETRIEVE OCSM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — OCSM.

Was DTC B1A59:01 or B1A59:11 retrieved during self-test?

Yes	The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self test. GO to AF2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AF8 .

AF2 CHECK THE SEAT WIRING AND CONNECTORS

- Ignition OFF. **WARNING:** Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.
- Carry out a thorough inspection of the OCS wiring, terminals and connectors, jumper harness, weight sensor bolt connector and connection, wiring harness routing and the related seat wiring harness and body wiring harness terminals and connectors.

Were any problems noted?

Yes	DEPOWER the SRS. REFER to Supplemental Restraint System (SRS) Depowering and Repowering in this section. REPAIR the seat connectors and wiring as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AF9 .
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No	For DTC B1A59:11, GO to AF3 . For DTC B1A59:01, GO to AF6 .
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AF3 CHECK OCS WEIGHT SENSOR BOLTS FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the QCSM by inducing a different fault condition. If the fault reported changes, this indicates the QCSM is functioning correctly and is not the source of the fault.

- Disconnect: QCS Weight Sensor Bolts C3330 and C3331 .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear DTCs — QCSM .
- Enter the following diagnostic mode on the scan tool: Self Test — QCSM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the QCS weight sensor bolts disconnected, numerous sensor circuit faults would normally be retrieved.

Was DTC B1A59:01 retrieved during self-test?

Yes	GO to AF4 .
No	INSTALL a new passenger seat track assembly with <u>QCS</u> weight sensor bolts. REFER to Section 501-10. GO to AF9 .

AF4 CHECK BTS FOR A FAULT STATUS CHANGE (SHORT TO GROUND INDICATED)

NOTE: This pinpoint test step attempts to change the fault reported by the QCSM by inducing a different fault condition. If the fault reported changes, this indicates the QCSM is functioning correctly and is not the source of the fault.

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: BTS C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear DTCs — QCSM .
- Enter the following diagnostic mode on the scan tool: Self Test — QCSM .
- **DIAGNOSTIC TIP:** When viewing DTCs with the QCS weight sensor bolts and BTS disconnected, numerous sensor circuit faults are normally retrieved.

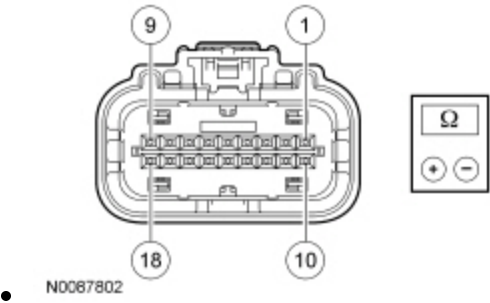
Was DTC B1A59:01 retrieved during self-test?

Yes	GO to AF5 .
No	INSTALL a new passenger safety belt retractor assembly (includes <u>BTS</u>). REFER to Section 501-20A. GO to AF9 .

AF5 CHECK OCSM VREF CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: ~~OCSM~~ C3159 .
- Measure resistance between a good ground and the following circuits using the following chart:

Connector	Circuit
C3159 Pin 8	CR153 (GN/WH)
C3159 Pin 11	LR142 (YE/BU)
C3159 Pin 17	CR154 (VT/OG)



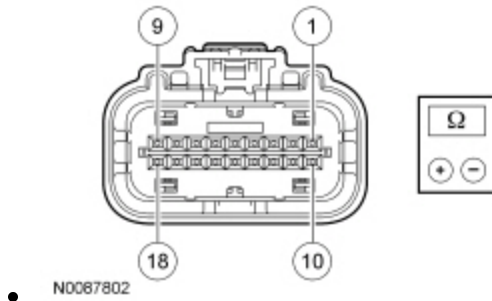
Are the resistances greater than 10,000 ohms?

Yes	GO to AF7 .
No	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AF9 .

AF6 CHECK THE OCSM

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: ~~BTS~~ C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Disconnect: ~~OCS~~ Weight Sensor Bolts C3330 and C3331 .
- Disconnect: ~~OCSM~~ C3159 .
- Measure resistance between [C3159](#) Pin 11, circuit LR142 (YE/BU), harness side and the following circuits using the chart below:

Connector	Circuit
C3159 Pin 8	CR153 (GN/WH)
C3159 Pin 17	CR154 (VT/OG)



Are the resistances greater than 10,000 ohms?

Yes	GO to AF7 .
No	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AF9 .

AF7 CONFIRM THE OCSM FAULT

NOTE: Make sure all restraint system sensor electrical connectors and the ~~RCM~~ electrical connector are connected before carrying out the self test. If not, DTCs will be recorded.

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: ~~OCSM~~ C3159 (if previously disconnected) .
- Connect: ~~OCS~~ Weight Sensor Bolts C3330 and C3331 .
- Connect: ~~BTS~~ C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear DTCs — ~~OCSM~~ .
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC B1A59:01 or B1A59:11 retrieved during self-test?

Yes	INSTALL a new OC SM . REFER to Occupant Classification System Module (OCSM) in this section. GO to AF9 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to AF9 .

AF8 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: **B**T~~S~~ [C3238](#) (regular cab and Crew Cab)/[C829](#) (SuperCab) .
- Disconnect: **OC**SM Weight Sensor Bolts [C3330](#) and [C3331](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear DTCs — **OC**SM .
- Enter the following diagnostic mode on the scan tool: Self Test — **OC**SM .

Was DTC B1A59:01 or B1A59:11 retrieved during self-test?

Yes	The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. GO to AF2 .
No	CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AF9 .

AF9 CHECK FOR ADDITIONAL DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all **S**R~~S~~ components (if previously disconnected).
- If previously directed to depower the **S**R~~S~~ , repower the **S**R~~S~~ . **Do not** prove out the **S**R~~S~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.

- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and QCSM .

Are any RCM and/or QCSM DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AG: DTC U200D:12

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) monitors each Occupant Classification System (OCS) weight sensor bolt and its circuitry for faults. If the QCSM detects unwanted voltage on any weight sensor VREF circuit, it stores DTC U200D:12 in memory and sends a message to the Restraints Control Module (RCM) .

The RCM stores DTC B00A0:09 in memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

- DTC U200D:12 (Control Module Output Power A: Circuit Short to Battery) — If the QCSM detects voltage above 5 volts on any weight sensor VREF circuit, it sets this DTC.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- QCSM

PINPOINT TEST AG : DTC U200D:12

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Do not install any new QCS components until directed to do so. QCS components should only be installed when directed to do so in the pinpoint test.
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.

NOTE: The ~~SRS~~ must be fully operational and free of faults before releasing the vehicle to the customer.

AG1 RETRIEVE OCSM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC U200D:12 retrieved on-demand during self-test?

Yes	The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self test. GO to AG2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AG5 .

AG2 CHECK THE SEAT WIRING AND CONNECTORS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Carry out a thorough inspection of the ~~OCS~~ wiring, terminals and connectors, jumper harness, weight sensor bolt connector and connection, wiring harness routing and the related seat wiring harness and body wiring harness terminals and connectors.

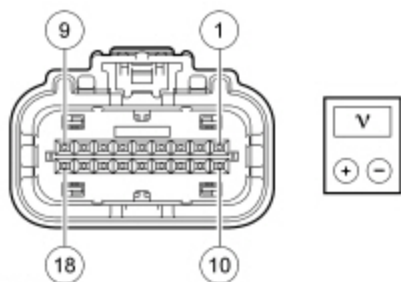
Were any problems noted?

Yes	DEPOWER the SRS . REFER to Supplemental Restraint System (SRS) Depowering and Repowering in this section. REPAIR the seat connectors and wiring as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AG6 .
No	GO to AG3 .

AG3 CHECK THE WEIGHT SENSOR VREF CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: Passenger Seat Side Air Bag C339 .
- Disconnect: ~~OCSM~~ C3159 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between ~~OCSM~~ C3159 and a good ground using the chart below:

Connector	Circuit
C3159 Pin 8	CR153 (GN/WH)
C3159 Pin 17	CR154 (VT/OG)



N0087854

Is voltage present on any circuit?

Yes	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AG6 .
No	GO to AG4 .

AG4 CONFIRM THE OCSM FAULT

NOTE: Make sure all restraint system sensor electrical connectors and the ~~RCM~~ electrical connector are connected before carrying out the self test. If not, DTCs will be recorded.

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Prior to reconnecting any previously disconnected ~~SRS~~ component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: ~~OCSM~~ C3159 .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC U200D:12 retrieved on-demand during self-test?

Yes	INSTALL a new OCSM . REFER to Occupant Classification System Module (OCSM) in this section. GO to AG6 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to AG6 .

AG5 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: ~~OCS~~ Weight Sensor Bolts [C3330](#) and [C3331](#).
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC U200D:12 retrieved on-demand during self-test?

Yes	The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. GO to AG2 .
No	CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AG6 .

AG6 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~OCSM~~ .*

Are any ~~RCM~~ and/or ~~OCSM~~ DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and OCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AH: DTCs U200E:11, U200E:12 and U200E:1A

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Occupant Classification System Module (OCSM) monitors the Belt Tension Sensor (BTS) VREF circuit for faults. If the OCSM detects a fault, it stores DTC U200E:11, U200E:12 or U200E:1A in memory and sends a message to the Restraints Control Module (RCM) .

The RCM stores DTC B00A0:09 memory and sends a message to the Instrument Panel Cluster (IPC) to illuminate the air bag warning indicator.

- DTC U200E:11 (Control Module Output Power B: Circuit Short to Ground) — A fault is indicated when the OCSM senses a short to a ground on the BTS VREF circuit. If an open circuit is present on the BTS signal circuit (such as if the BTS connector is disconnected) at the same time that the BTS VREF circuit is shorted to ground, only DTC B1A59:11 will be retrieved.
- DTC U200E:12 (Control Module Output Power B: Circuit Short to Battery) — A fault is indicated when the OCSM senses a short to voltage on the BTS VREF circuit.
- DTC U200E:1A (Control Module Output Power B: Circuit Resistance Below Threshold) — A fault is indicated when the OCSM senses that the BTS VREF circuit is shorted to the weight sensor VREF circuit(s). DTC B0061:12 and/or B1A59:01 may also set when this fault is present.

This pinpoint test is intended to diagnose the following:

- Wiring, terminals or connectors
- BTS (part of front passenger safety belt retractor)
- OCSM

PINPOINT TEST AH : DTCS U200E:11, U200E:12 AND U200E:1A

WARNING: Never probe the electrical connectors on airbag, Safety Canopy or side air curtain assemblies. Failure to follow this instruction may result in the accidental deployment of these assemblies, which increases the risk of serious personal injury or death.
WARNING: Never disassemble or tamper with seatbelt deployable components, including pretensioners, load limiters and inflators. Never back probe deployable device electrical connectors. Tampering or back probing may cause an accidental deployment and result in personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when making measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Do not install any new Occupant Classification System (OCS) components until directed to do so. OCS components should only be installed when directed to do so in the pinpoint test.
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the correct component is being installed. If an incorrect SRS component is installed, DTCs may set.
NOTE: The SRS must be fully operational and free of faults before releasing the vehicle to the customer.
AH1 RETRIEVE OCSM DTCS
• Ignition ON.

- Enter the following diagnostic mode on the scan tool: Self Test — **QCSM** .

Was DTC U200E:11, U200E:12 or U200E:1A retrieved during self-test?

Yes	The fault condition is still present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during self test. GO to AH2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AH8 .

AH2 CHECK THE SEAT WIRING AND CONNECTORS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Carry out a thorough inspection of the **QCS** wiring, terminals and connectors, jumper harness, **BTS** connector and connections, wiring harness routing and the related seat wiring harness and body wiring harness terminals and connectors.

Were any problems noted?

Yes	DEPOWER the SRS . REFER to Supplemental Restraint System (SRS) Depowering and Repowering in this section. REPAIR the seat connectors and wiring as needed. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AH9 .
No	GO to AH3 .

AH3 CHECK FOR CONTROL MODULE OUTPUT POWER B DTC FAULT STATUS CHANGE

NOTE: *This pinpoint test step attempts to change the fault reported by the **QCSM** by inducing a different fault condition. If the fault reported changes, this indicates the **QCSM** is functioning correctly and is not the source of the fault.*

- Disconnect: **BTS** C3238 (regular cab and Crew Cab)/C829 (SuperCab) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear DTCs — **QCSM** .
- Enter the following diagnostic mode on the scan tool: Self Test — **QCSM** .
- **DIAGNOSTIC TIP:** When viewing DTCs with the **BTS** disconnected, open sensor circuit faults are normally retrieved.

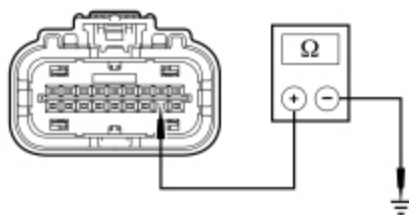
Was DTC U200E:11, B1A59:11, U200E:12 or U200E:1A retrieved during self-test?

Yes	For DTC U200E:11, GO to AH4 . For DTC U200E:12, GO to AH5 . For DTC U200E:1A, GO to AH6 .
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No	INSPECT the BTS harness, connectors and terminals for damage. If a concern is found, REPAIR the harness. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AH9. If a concern was not found, INSTALL a new passenger safety belt retractor assembly. REFER to Section 501-20A. GO to AH9.
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AH4 CHECK THE BTS VREF CIRCUIT FOR A SHORT TO GROUND

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: ~~QCSM~~ C3159 .
- Measure the resistance between ~~QCSM~~ [C3159](#) Pin 11, circuit LR142 (YE/BU), harness side and ground.



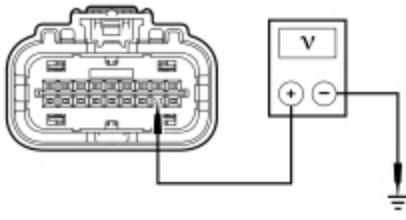
- N0133857

Is the resistance greater than 10,000 ohms?

Yes	GO to AH7 .
No	REPAIR circuit LR142 (YE/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AH9.

AH5 CHECK THE BTS VREF CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~QCSM~~ C3159 .
- Disconnect: Passenger Seat Side Air Bag C339 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between ~~QCSM~~ [C3159](#) Pin 11, circuit LR142 (YE/BU), harness side and ground.



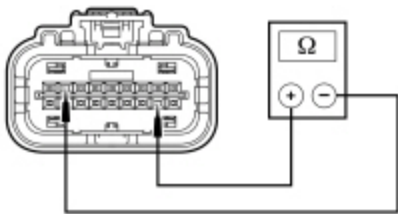
• N0133858

Is any voltage present?

Yes	REPAIR circuit LR142 (YE/BU). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AH9 .
No	GO to AH7 .

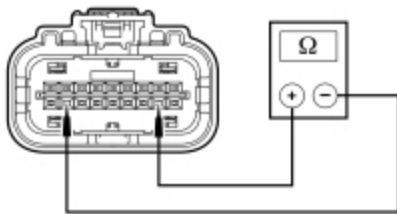
AH6 CHECK THE BTS VREF CIRCUIT FOR A SHORT TO THE WEIGHT SENSOR VREF CIRCUIT(S)

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~QCSM~~ C3159 .
- Disconnect: ~~QCS~~ Weight Sensor Bolts [C3330](#) and [C3331](#).
- Disconnect: Passenger Seat Side Air Bag C339 .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the resistance between ~~QCSM~~ [C3159](#) Pin 11, circuit LR142 (YE/BU), harness side and ~~QCSM~~ [C3159](#) Pin 8, circuit CR153 (GN/WH), harness side.



• N0133859

- Measure the resistance between ~~QCSM~~ [C3159](#) Pin 11, circuit LR142 (YE/BU), harness side and ~~QCSM~~ [C3159](#) Pin 17, circuit CR154 (VT/OG), harness side.



• N0133860

Is the resistance greater than 10,000 ohms?

Yes	REPAIR the affected circuits. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AH9 .
No	GO to AH7 .

AH7 CONFIRM THE OCSM FAULT

NOTE: Make sure all restraint system sensor electrical connectors and the **RCM** electrical connector are connected before carrying out the self test. If not, DTCs will be recorded.

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed-out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: **OCSM** C3159 .
- Connect: **BTS** C3238/C829 .
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear DTCs — **OCSM** .
- Enter the following diagnostic mode on the scan tool: Self Test — **OCSM** .

Was DTC U200E:11, U200E:12 or U200E:1A retrieved during self-test?

Yes	INSTALL a new OCSM . REFER to Occupant Classification System Module (OCSM) in this section. GO to AH9 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. GO to AH8 .

AH8 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF.
- Depower the ~~SRS~~ . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: ~~BTS~~ [C3238](#) (regular cab and Crew Cab)/[C829](#) (SuperCab) .
 - inspect connector(s) (including any in-line connectors) for corrosion, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- Repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Clear DTCs — ~~OCSM~~ .
- Enter the following diagnostic mode on the scan tool: Self Test — ~~OCSM~~ .

Was DTC U200E:11, U200E:12 or U200E:1A retrieved during self-test?

Yes	The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved during the on-demand self test. GO to AH2 .
No	CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition key frequently. ACTIVATE other systems in the same wire harness. Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. Refer to Wiring Diagrams Cell 5 , Connector Repair Procedures for schematic and connector information. GO to AH9 .

AH9 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Reconnect all ~~SRS~~ components (if previously disconnected).
- If previously directed to depower the ~~SRS~~ , repower the ~~SRS~~ . **Do not** prove out the ~~SRS~~ at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the ~~RCM~~ and ~~OCSM~~ .*

Are any ~~RCM~~ and/or ~~OCSM~~ DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM CMDTCs . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.

Pinpoint Test AI: DTCs B11D8:12 and B11D8:14

Refer to Wiring Diagrams Cell [46](#) , Supplemental Restraint System for schematic and connector information.

Normal Operation

The Restraints Control Module (RCM) transmits an event notification signal which communicates fuel cutoff status and Supplemental Restraint System (SRS) deployment status to the Fuel Pump Control Module (FPCM) . This allows the PCM to initiate fuel cut-off in the event of a crash.

The RCM monitors the event notification signal circuit for the following faults:

- Short to voltage
- Short to ground
- Open circuit

If a fault is detected, the RCM stores DTC B11D8:12 or B11D8:14 in memory. The airbag warning indicator will **not** be illuminated if B11D8:12 or B11D8:14 is set.

- DTC B11D8:12 (Restraint Event Notification: Circuit Short to Battery) — A fault is indicated when the RCM senses a short to voltage on the event notification signal circuit for more than 15 seconds.
- DTC B11D8:14 (Restraint Event Notification: Circuit Short to Ground or Open) — A fault is indicated when the RCM senses a short to ground or an open on the event notification signal circuit for more than 15 seconds. This fault can also set if there is high resistance or low voltage to the FPCM , or an open or high resistance on the FPCM ground circuit.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- FPCM
- RCM

PINPOINT TEST AI : DTCS B11D8:12 AND B11D8:14

WARNING: Do not handle, move or change the original horizontal mounting position of the restraints control module (RCM) while the RCM is connected and the ignition switch is ON. Failure to follow this instruction may result in the accidental deployment of the Safety Canopy and cause serious personal injury or death.
NOTICE: Use the correct probe adapter(s) from the Flex Probe Kit when taking measurements. Failure to use the correct probe adapter(s) may damage the connector.
NOTE: Most faults are due to connector and/or wiring concerns. Carry out a thorough inspection and verification before proceeding with the pinpoint test.
NOTE: Only disconnect or reconnect Supplemental Restraint System (SRS) components when instructed to do so within a pinpoint test step. Failure to follow this instruction may result in incorrect diagnosis of the SRS .
NOTE: Install new components only when directed to do so in the pinpoint test.
NOTE: Always make sure the correct SRS component is being installed. Parts released for other vehicles may not be compatible even if they appear physically similar. Check the part number listed in the parts catalog to make sure the

correct component is being installed. If an incorrect **SRS** component is installed, DTCs may set.

NOTE: The **SRS** must be fully operational and free of faults before releasing the vehicle to the customer.

AI1 RETRIEVE RCM DTCS

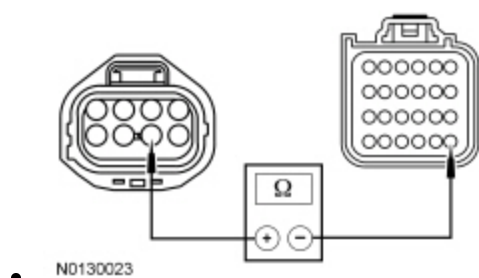
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B11D8:12 or B11D8:14 retrieved on-demand during self-test?

Yes	This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test. For DTC B11D8:12, GO to AI7 . For DTC B11D8:14, GO to AI2 .
No	This is an intermittent fault when present as a Continuous Memory Diagnostic Trouble Code (CMDTC) only. GO to AI10 .

AI2 CHECK THE EVENT NOTIFICATION SIGNAL CIRCUIT FOR AN OPEN

- Ignition OFF.
- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **RCM** [C310A](#) and [C310B](#).
- Disconnect: **FPCM** [C433](#) .
- Measure the resistance between **RCM** [C310A](#) Pin 19, circuit CR167 (YE/VT), harness side, and **FPCM** [C433](#) Pin 6, circuit CR167 (YE/VT), harness side.

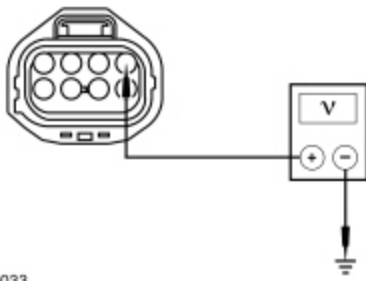


Is the resistance less than 3 ohms?

Yes	GO to AI3 .
No	REPAIR circuit CR167 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AI11 .

AI3 CHECK THE FPCM POWER CIRCUIT

- Measure the voltage between **FPCM** [C433](#) Pin 1, circuit CE608 (VT/GN), harness side, and ground.



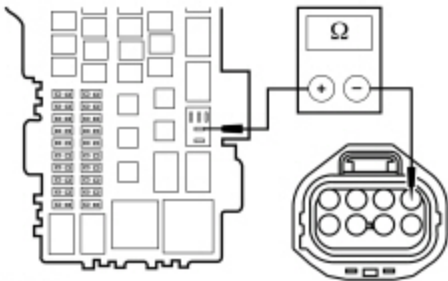
N0166033

Is the voltage greater than 10 volts?

Yes	GO to AI4 .
No	VERIFY BJB fuse 27 (20A) is OK. If OK, GO to AI4. If not OK, REFER to the Wiring Diagrams manual to identify the possible causes of the circuit short. REPAIR circuit CE608 (VT/GN). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AI11.

AI4 CHECK THE FPCM POWER CIRCUIT FOR AN OPEN OR HIGH RESISTANCE

- Remove the fuel pump relay from the **BJB** .
Refer to Wiring Diagrams Cell [11](#) for schematic and connector information.
- Measure the resistance between the fuel pump relay socket, pin 5, circuit CE608 (VT/GN) and **FPCM** [C433](#) Pin 1, circuit CE608 (VT/GN), harness side.



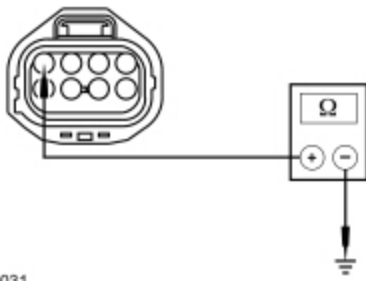
N0166032

Is the resistance less than 1 ohm?

Yes	GO to AI5 .
No	REPAIR circuit CE608 (VT/GN) for an open or high resistance. Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AI11.

AI5 CHECK THE FPCM GROUND CIRCUIT FOR AN OPEN OR HIGH RESISTANCE

- Measure the resistance between **FPCM** [C433](#) Pin 4, circuit GD117 (BK), harness side, and ground.



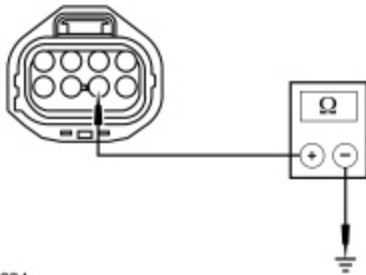
• N0166031

Is the resistance less than 1 ohm?

Yes	GO to AI6 .
No	REPAIR circuit GD117 (BK). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AI11 .

AI6 CHECK THE EVENT NOTIFICATION SIGNAL CIRCUIT FOR A SHORT TO GROUND

- Measure the resistance between FPCM [C433](#) Pin 6, circuit CR167 (YE/VT), harness side, and ground.



• N0130024

Is the resistance greater than 10,000 ohms?

Yes	INSTALL a new FPCM . REFER to Section 303-04A, Section 303-04B, Section 303-04C or Section 303-04D. CONNECT all RCM connectors. REPOWER the SRS . Do not prove out the SRS at this time. REFER to Supplemental Restraint System (SRS) Depowering and Repowering in this section. CLEAR the RCM DTCs and REPEAT the self-test. If DTC B11D8:14 returns, GO to AI9 . If no RCM DTCs are retrieved, GO to AI11 . If RCM DTCs other than B11D8:14 are retrieved, REFER to the Restraints Control Module (RCM) DTC Chart in this section to diagnose those DTCs.
No	REPAIR circuit CR167 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AI11 .

AI7 CHECK THE EVENT NOTIFICATION SIGNAL CIRCUIT DTC FOR A FAULT STATUS CHANGE

NOTE: This pinpoint test step attempts to change the fault reported by the RCM by inducing a different fault condition. If the reported fault changes, this indicates the RCM is functioning correctly and is not the source of the fault.

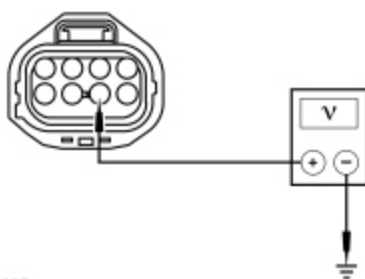
- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: FPCM [C433](#) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — RCM .

Was DTC B11D8:12 retrieved on-demand during self-test?

Yes	GO to AI8 .
No	INSTALL a new FPCM . REFER to Section 303-04A, Section 303-04B, Section 303-04C or Section 303-04D. GO to AI11 .

AI8 CHECK THE EVENT NOTIFICATION SIGNAL CIRCUIT FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Depower the SRS . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: RCM [C310A](#) and [C310B](#).
- Repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Measure the voltage between FPCM [C433](#) Pin 6, circuit CR167 (YE/VT), harness side, and ground.



Is any voltage present?

Yes	REPAIR circuit CR167 (YE/VT). Refer to Wiring Diagrams Cell 5 for schematic and connector information. GO to AI11 .
No	GO to AI9 .

AI9 CONFIRM THE RCM FAULT

NOTE: Make sure all SRS electrical connectors and the FPCM electrical connector are connected before carrying out the self-test. If not, DTCs will be recorded.

- Ignition OFF.

- Depower the **SRS** . Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Disconnect: **FPCM** [C433](#) .
- Disconnect: **RCM** [C310A](#) and [C310B](#).
- Prior to reconnecting any previously disconnected **SRS** component:
 - inspect connector(s) (including any in-line connectors) for pushed out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: **FPCM** [C433](#) .
- Connect: **RCM** [C310A](#) and [C310B](#).
- Repower the **SRS** . **Do not** prove out the **SRS** at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was the original DTC retrieved on-demand during self-test?

Yes	INSTALL a new RCM . REFER to Restraints Control Module (RCM) in this section. GO to AI11 .
No	In the process of diagnosing the fault, the fault condition has become intermittent. Do not install any new SRS components at this time. SRS components should only be installed when directed to do so in the pinpoint test. GO to AI10 .

AI10 CHECK FOR AN INTERMITTENT FAULT

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Disconnect: **FPCM** [C433](#) .
 - inspect connector(s) (including any in-line connectors) for pushed out, loose or spread terminals and loose or frayed wire connections at terminals.
 - inspect wire harness for any damage, pinched, cut or pierced wires.
 - repair any concerns found.
 Refer to Wiring Diagrams Cell [5](#) , Connector Repair Procedures for schematic and connector information.
- Connect: **FPCM** [C433](#) .
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — **RCM** .

Was DTC B11D8:12 or B11D8:14 retrieved on-demand during self-test?

Yes	The fault condition is now present. This fault cannot be cleared until it is corrected and the DTC is no longer retrieved on-demand during the self-test.
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For DTC B11D8:14, GO to [AI2](#).
For DTC B11D8:12, GO to [AI7](#).

No CHECK for causes of the intermittent fault. ATTEMPT to recreate the hard fault by flexing the wire harness and cycling the ignition frequently. ACTIVATE other systems in the same wire harness. **Do not install any new SRS components at this time. Install SRS components only when directed to do so in the pinpoint test. REPAIR any intermittent concerns found. GO to [AI11](#).**

AI11 CHECK FOR ADDITIONAL SRS DTCS

- Ignition OFF. **WARNING: Turn the ignition OFF and wait one minute to deplete the backup power supply. Failure to follow this instruction may result in serious personal injury or death in the event of an accidental deployment.**
- Connect: All Previously Disconnected Component(s)/Connector(s) .
- If previously directed to depower the SRS , repower the SRS . **Do not** prove out the SRS at this time. Refer to Supplemental Restraint System (SRS) Depowering and Repowering in this section.
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — Restraints . **NOTE:** *When selecting Restraints from the Self Test menu, DTCs are retrieved from the RCM and QCSM .*

Are any RCM and/or QCSM DTCs retrieved?

Yes	Do not clear any DTCs until all DTCs have been resolved. GO to the DTC Charts in this section for pinpoint test direction.
No	CLEAR all RCM and QCSM Continuous Memory Diagnostic Trouble Codes (CMDTCs) . PROVE OUT the SRS . The repair is complete. RETURN the vehicle to the customer.